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Review article

Metaheuristic algorithms for PID controller parameters tuning: review, approaches and open problems



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ABSTRACT

The simplicity, transparency, reliability, high efficiency and robust nature of PID controllers are some of the reasons for their high popularity and acceptance for control in process industries around the world today. Tuning of PID control parameters has been a field of active research and still is. The primary objectives of PID control parameters are to achieve minimal overshoot in steady state response and lesser settling time. With exception of two popular conventional tuning strategies (Ziegler Nichols closed loop oscillation and Cohen-Coon's process reaction curve) several other methods have been employed for tuning. This work accords a thorough review of state-of-the-art and classical strategies for PID controller parameters tuning using metaheuristic algorithms. Methods appraised are categorized into classical and metaheuristic optimization methods for PID parameters tuning purposes. Details of some metaheuristic algorithms, methods of application, equations and implementation flowcharts/algorithms are presented. Some open problems for future research are also presented. The major goal of this work is to proffer a comprehensive reference source for researchers and scholars working on PID controllers.

1. Introduction

Over the years, feedback control mechanisms have established a very strong impact in the fields of manufacturing, robotics, aviation and process control. Several efficient, robust and adaptive controllers (Brogliato and Lozano, 1994; Grimm et al., 2007; Kaloust and Qu, 1997; Lazar and Heemels, 2009; Rakovic et al., 2006) have been proposed aside proportional-integral-derivative (PID) controller. However, the acceptance and reputation of PID controllers in control systems is incomparable and supreme. PID control system can be implemented or applied in different modes (proportional only (P-mode), proportional and integral (PI mode), proportional and derivative (PD-mode) and proportional, integral and derivative (PID mode)) depending on process requirement (Lipták, 2002; Bharat et al., 2019). Several works have reported applications of typical PID controllers for different systems (Ramachandran et al., 2018; Elsrogy et al., 2013; Prasad et al., 2014; Wang, 2011).

PID controllers have been in use for many years in industries for process control applications. The use of PID controllers can be dated back to 1890s. Despite its long time of existence, PID controllers still remain the most popular controllers in both process and manufacturing industries today. Furthermore, research on PID controllers indicates that about ninety percent (90%) of process industries employ PID as controllers (Bansal et al., 2012; Abbasi and Naghavi, 2017). This popularity is as a result of their robustness, simplicity and ease of retuning control parameters (Åström and Hägglund, 1995; Abbasi and Naghavi, 2017). PID controller has been conventionally regarded as the best controller in the absence of fundamental process knowledge (Abbasi and Naghavi, 2017; Bennett, 1993). Despite these advantages of PID controllers, they are also faced with some shortcomings which include: unwanted speed overshoot and sluggish response as a result of unexpected variation in load torque and sensitivity to controller gains K_I and K_P as identified by Kushwah and Patra (2014). The success of PID controller bases on accuracy of system models and variables.

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综述 文章

元启发式 算法 用于 PID 控制器 参数 整定: 综述、方法 与 开放 问题



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摘要与关键词部分

PID控制器的简单性、透明性、可靠性、高效率 and 鲁棒性是其如今在全球过程工业控制中广受欢迎和接受的部分原因。PID控制参数的整定一直是一个活跃的研究领域, 至今仍是如此。PID控制参数的主要目标是实现稳态响应中的最小超调和更短的调节时间。除了两种流行的传统整定策略(齐格勒-尼科尔斯闭环振荡和科恩-库恩过程反应曲线)外, 还有其他几种方法被用于整定。本文对使用元启发式算法进行PID控制器参数整定的最新和经典策略进行了全面综述。评估的方法分为经典和元启发式优化方法, 用于PID参数整定。介绍了一些元启发式算法的细节、应用方法、方程和实现流程图/算法。还提出了一些未来研究的开放问题。这项工作的主要目标是从事PID控制器研究的学者和研究人员提供一个全面的参考来源。

1. 引言

多年来, 反馈控制机制在制造业、机器人学、航空和过程控制领域产生了非常深远的影响。除了比例-积分-微分(PID)控制器外, 还提出了几种高效、鲁棒和自适应的控制器(Brogliato and Lozano, 1994; Grimm et al., 2007; Kaloust and Qu, 1997; Lazar and Heemels, 2009; Rakovic et al., 2006)。然而, PID控制器在控制系统中的接受度和声誉是无与伦比且至高无上的。PID控制系统可以根据过程需求以不同模式(仅比例(P模式)、比例积分(PI模式)、比例微分(PD模式)和比例积分微分(PID模式))实现或应用(Lipták, 2002; Bharat et al., 2019)。多项工作报道了典型PID控制器在不同系统中的应用(Ramachandran et al., 2018; Elsrogy et al., 2013; Prasad et al., 2014; Wang, 2011)。

PID控制器已在工业中用于过程控制应用多年。PID控制器的使用可以追溯到19世纪90年代。尽管存在已久, 但PID控制器至今仍是过程工业和制造业中最受欢迎的控制器。此外, 对PID控制器的研究表明, 约百分之九十(90%)的过程工业采用PID作为控制器(Bansal等人, 2012; Abbasi和Naghavi, 2017)。这种普及性源于其鲁棒性、简单性和再整定控制参数的便利性(Åström和Hägglund, 1995; Abbasi和Naghavi, 2017)。在缺乏基本过程知识的情况下, PID控制器传统上被认为是最佳控制器(Abbasi和Naghavi, 2017; Bennett, 1993)。尽管PID控制器具有这些优点, 但它们也面临一些缺点, 包括: 因负载扭矩意外变化 and 控制器增益敏感性导致的不希望的速度超调和响应迟缓, 以及Kushwah和Patra(2014)指出的 K_I 和 K_P 。PID控制器的成功取决于系统模型和变量的准确性。

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2405-8440/© 2022 作者。由 Elsevier Ltd. 出版。这是一篇基于 CC BY-NC-ND 许可的开放获取文章 (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)。

Controller tuning is a long standing and important aspect of feed-back controllers. Hence, making the area of controller tuning a significant area of research both in academics and industries. The exploration and investigation of dynamic systems to create efficient, reliable and promising controllers are related to control engineering. After the development of PID control, numerous interests on tuning techniques that assure excellent performance of PID controllers sprang up. Furthermore, drawbacks of PID control strategy typically for nonlinear systems, include: challenges in choosing suitable and correct controller gains, and process termed tuning. The challenges commonly lie in choosing appropriate controller gains if set less than required, the control objective may never be achieved. While using excessively large controller gains may cause system instability, it is a viable option (Abbasi and Naghavi, 2017).

The first PID controller parameters tuning rules was introduced by Ziegler and Nichols (1942), several other tuning rules were proposed later. Though, some of these proposals focused on linear systems stabilization (Åström and Hägglund, 1995), which are not appropriate for nonlinear dynamic models. Conversely, some of these techniques encompass nonlinear systems (Lozano et al., 1999) with the goal of obtaining control variables that satisfy stability criteria. Conversely, the latter strategies do not assure definite response characteristics. Thus, one major difficulties and concern in control engineering area is appropriate controller parameters tuning. Controller variables are tuned to stabilize closed-loop control system and to achieve objectives which are related to stability, durability, performance tracking, performance measurement, noise reduction, disturbance rejection and robustness against environmental uncertainties.

Given a set of objectives, there are several types of tuning strategies for PID controller. Several authors and scholars presented different classifications of PID tuning strategies. Moradi (2003) and Johnson and Moradi (2005) classified PID tuning strategies relying on availability of a process model and model type. These classifications are grouped into four (4) subgroups namely:

- i. **Model free methods:** In these methods, there is no existence of a model or particular points of process identified. Hence, do not apply explicit identification important model points or parametric model as such.
- ii. **Non-parametric model methods:** These methods employ only a partial modeling information normally, steady state representation and critical frequency locations. The techniques are most appropriate for online application. They are typically used without needing extensive a priori plant information.
- iii. **In-depth data methods:** These techniques are midway amid parametric and non-parametric strategies. They use process data for tuning as used in subspace strategy or grey knowledge in fuzzy logic systems.
- iv. **Parametric model methods:** These methods require availability of model data, and can be obtained using standard offline or on-line identification process. They need a linear model of the system and its transfer function matrix/space model. These techniques are more appropriate for offline PID tuning. The parametric strategies can be classified into Non-Optimal Parametric, optimal (Restricted Structure control) and optimal (control signal matching) Methods.

A more comprehensive, detail discussion and application strategies of this classification can be obtained in Moradi (2003); Johnson and Moradi (2005).

Also, another classification of PID parameters tuning can be found in Bansal et al. (2012). These methods can be generally categorized into classical and computational techniques as presented in Bansal et al. (2012):

- i. **Classical Techniques:** These methods are centered on making some assumptions about plant model and required output. They try

to extract some system features analytically or graphically, which are then utilized to pick controller settings. These methods are simple to use, computationally efficient, and can be used as a first step in parameter adjustment. However, the controller settings often give desired results directly, this is as a result of assumptions made. Hence, further tuning is required. Some of these classical tuning methods include Ziegler and Nichols (1942, 1993) method and Cohen (1953) method.

- ii. **Computational or Optimization Techniques:** These techniques involve the use of data modeling and optimization of a cost function strategies to tune PID parameters. These methods of controller parameters depend on a cost function which they attempt to minimize. There are six (6) generally used cost functions for PID controller parameters tuning. These commonly used cost functions are:
 - (a) Integral Absolute Error (IAE) defined as

$$IAE = \int_0^{\tau} |e(t)|. \tag{1}$$

- (b) Integral Square Error (ISE) expressed by

$$ISE = \int_0^{\tau} |e(t)|^2. \tag{2}$$

- (c) Integral Time Absolute Error (ITAE) represented as

$$ITAE = \int_0^{\tau} t|e(t)|. \tag{3}$$

- (d) Integral Time Square Error (ITSE) denoted as

$$ITSE = \int_0^{\tau} t|e(t)|^2. \tag{4}$$

- (e) Mean Square Error (MSE) defined as

$$MSE = \frac{\int_0^{\tau} e(t)^2}{t}. \tag{5}$$

- (f) Integral Error (IE) given as

$$IE = \int_0^{\tau} e(t). \tag{6}$$

Some examples of computational or optimisation techniques are Artificial Immune System algorithm (Kim et al., 2002; Shatnawi and Quwaider, 2019; Quwaider and Shatnawi, 2020), Simulated annealing (Haber et al., 2007) and Particle Swarm Optimization (PSO) (Bassi et al., 2011). Computational strategies are employed for self tuning or automatic tuning of PID controller parameters. While a broader classification based on nature and usage can be classified in five (5) categories (the composition can be found in Rodríguez-Molina et al. (2020); Villarreal-Cervantes and Alvarez-Gallegos (2016); Ang et al. (2005)):

- i. **Analytical methods:** These methods involve analysis of closed loop model to attain stability. It involves computation of relationship between a given plant model and the set objective to determine appropriate PID parameters. Examples of these strategies may include Lyapunov (Li et al., 2009) and root locus approaches (Baker, 2011), where the latter is employed for linear systems and the former is used for nonlinear systems or frequency response. Other analytical methods include internal model control or lambda method.
- ii. **Heuristic methods:** Heuristic strategy of PID tuning is progressed with practical experience where controller variables are manually selected reliant on experimental knowledge of a professional designer who utilizes information of controlled variable estimates

控制器整定是反馈控制器中长期存在且重要的方面。因此，控制器整定领域成为学术界和工业界重要的研究领域。对动态系统的探索和研究，以创建高效、可靠和有前景的控制器，与控制工程相关。在PID控制发展之后，涌现出大量对保证PID控制器优异性能的调优技术的兴趣。此外，PID控制策略的缺点，特别是对于非线性系统，包括：选择合适且正确的控制器增益的挑战，以及称为整定的过程。挑战通常在于选择合适的控制器增益：如果设置得小于所需值，控制目标可能永远无法实现；而使用过大的控制器增益可能导致系统不稳定，这也是一种可行的选择 (Abbasi和Naghavi, 2017)。

第一个PID控制器参数整定规则由Ziegler和Nichols (1942)提出，之后又提出了许多其他整定规则。然而，其中一些方法侧重于线性系统的稳定性 (Åström和Hägglund, 1995)，这不适用于非线性动态模型。相反，一些技术涵盖了非线性系统 (Lozano等人, 1999)，旨在获得满足稳定性标准的控制变量。然而，后一种策略并不能保证明确的响应特性。因此，控制工程领域的一个主要难点和关注点是如何适当整定控制器参数。控制器变量经过整定以稳定闭环控制系统，并实现与稳定性、持久性、性能跟踪、性能测量、降噪、扰动抑制以及对环境不确定性的鲁棒性相关的目标。

给定一组目标，存在多种类型的PID控制器整定策略。许多作者和学者提出了不同的PID整定策略分类。Moradi (2003)和Johnson and Moradi (2005)根据过程模型的可用性和模型类型对PID整定策略进行了分类。这些分类分为四个子组，即：

- i. **无模型方法：**在这些方法中，不存在已识别的模型或特定过程点。因此，不需要应用明确的辨识重要模型点或参数模型本身。
- ii. **非参数模型方法：**这些方法通常仅利用部分建模信息，如稳态表示和关键频率点。这些技术最适合在线应用。它们通常在不需要大量先验对象信息的情况下使用。

iii. **深度数据方法：**这些技术介于参数化和非参数化策略之间。它们利用过程数据进行整定，如同在子空间策略或模糊逻辑系统中的灰色知识中那样。

iv. **参数模型方法：**这些方法需要模型数据的可用性，并且可以通过标准的离线或在线辨识过程获得。它们需要系统的线性模型及其传递函数矩阵/空间模型。这些技术更适用于离线PID整定。参数化策略可分为非最优参数方法、最优 (受限结构控制) 和最优 (控制信号匹配) 方法。

关于此分类的更全面、详细的讨论及应用策略可参见 Moradi (2003); Johnson and Moradi (2005)。

此外，另一类PID参数整定的分类可在 Bansal 等 (2012) 中找到。这些方法大致可分为经典方法和计算方法，如 Bansal 等人 (2012) 所述：

- i. **经典技术：**这些方法基于对被控对象模型和所需输出做出一些假设。它们尝试

通过分析或图形方式提取一些系统特征，然后利用这些特征选择控制器设置。这些方法使用简单、计算高效，可作为参数调整的第一步。然而，由于做出的假设，控制器设置通常能直接给出期望结果。因此，需要进一步整定。一些经典整定方法包括 Ziegler-Nichols 方法 (1942, 1993) 和 Cohen 方法 (1953)。

- ii. **计算或优化技术：**这些技术涉及使用数据建模和代价函数优化策略来整定PID参数。这些控制器参数方法依赖于一个它们试图最小化的代价函数。PID控制器参数整定通常使用六种代价函数。这些常用的代价函数是：(a) 积分绝对误差 (IAE)，定义为

$$IAE = \int_0^{\tau} |e(t)|. \tag{1}$$

- (b) 积分平方误差 (ISE) 表示为

$$ISE = \int_0^{\tau} |e(t)|^2. \tag{2}$$

- (c) 积分时间绝对误差 (ITAE) 表示为

$$ITAE = \int_0^{\tau} t|e(t)|. \tag{3}$$

- (d) 积分时间平方误差 (ITSE) 表示为

$$ITSE = \int_0^{\tau} t|e(t)|^2. \tag{4}$$

- (e) 均方误差 (MSE) 定义为

$$MSE = \frac{\int_0^{\tau} e(t)^2}{t}. \tag{5}$$

- (f) 积分误差 (IE) 表示为

$$IE = \int_0^{\tau} e(t). \tag{6}$$

一些计算或优化技术的例子包括人工免疫系统算法 (Kim 等, 2002; Shatnawi 和 Quwaider, 2019; Quwaider 和 Shatnawi, 2020)、模拟退火 (Haber 等, 2007) 和粒子群优化 (PSO) (Bassi 等, 2011)。计算策略用于 PID 控制器参数的自整定或自动整定。而基于性质和用途的更广泛分类可分为五 (5) 类 (组成可在 Rodríguez-Molina 等 (2020); Villarreal-Cervantes 和 Alvarez-Gallegos (2016); Ang 等 (2005) 中找到)：

- i. **解析方法：**这些方法涉及闭环模型的分析以实现稳定性。它需要计算给定植物模型与设定目标之间的关系，以确定合适的 PID 参数。这些策略的例子可能包括 Lyapunov (Li 等, 2009) 和根轨迹方法 (Baker, 2011)，其中后者用于线性系统，前者用于非线性系统或频率响应。其他解析方法包括内模控制或 λ 方法。

- ii. **启发式方法：**启发式策略的 PID 整定是基于实际经验发展起来的，其中控制器变量由手动选择，依赖于专业设计人员的实验知识，他们利用被控变量估计值的信息

- to set up correct variable-performance relationships. The heuristic method can also serve as a formula or a rule base for online tuning habitually with tradeoff in design goals or objectives. The most popular of these methods are the conventional Ziegler–Nichols and Cohen–Coon as depicted in (Rodríguez-Molina et al., 2020; Kumar et al., 2015; Ho et al., 1996).
- iii. **Frequency response methods:** These methods for PID parameter tuning are habitually offline and popular in the academic arena. The technique is based on the characteristics of target process or system such as loop shaping as reported in Ang et al. (2005). Natarajan and Gilbert (1997) and Wang et al. (1999) presented the use of frequency response for tuning PID controller parameters. Hagglund and Astrom presented adaptive techniques on frequency response to tune PID controller in Hägglund and Åström (1991).
- iv. **Optimization methods:** These strategies of PID controller tuning can be considered as an exceptional style of feedback optimal control system. PID variables are acquired on impromptu basis using an offline mathematical programming or numerical optimization technique for singular objective. The computation can also be implemented using computerized heuristics or evolutionary algorithms for multiple design objectives. The process involves stating a mathematical programming problem (MPP) for a targeted plant and employing optimization techniques to solve it. The main attribute of these methods is that control parameters are fixed and acquired by the solution of an offline numerical optimization method. Offline numerical optimization technique uses the derived state vector of plant by simulation or the real plant vectors. The obtained fixed control parameters are then added to a closed-loop system. This review fall to this class of tuning. Some recent optimization techniques for PID tuning may include; Firefly algorithm (dos Santos Coelho and Mariani, 2012), chaotic optimization (dos Santos Coelho, 2009), evolutionary multi-objective optimization (Nisi et al., 2019), PSO (Azar et al., 2019) and glow-worm optimization (Pilla et al., 2019).
- v. **Adaptive tuning methods:** The adaptive tuning strategies are on-line real-time tuning methods which involve the use of automated online mechanisms to tune PID controller parameters gains. The key feature of these techniques is that control parameters vary over preset time interval in the closed-loop system. Application of this technique can be found in (Pirabakaran and Becerra, 2001; Kim and Cho, 2005; Chang et al., 2003).

The complexity of plant dynamics depicts the effort or difficulty of tuning the controller parameters (Bernstein, 2002). Plant dynamics can be equipped with nonlinearity behavior, several tunable variables, and enormous numbers of input and output variables (Rodríguez-Molina et al., 2020). Furthermore, difficulties can be as a result of environmental and or operational uncertainties and perturbations. These features contribute to the difficulty of obtaining appropriate or suitable controller parameters geared towards achievement of set performance objectives. Additionally, existing applications require meeting numerous performance conditions/requirements at once, and are usually in conflict (Boyd et al., 1990). Some of these performance requirements may include: high performance accuracy, efficiency in energy consumption, low cost just to mention a few among others. Hence, tuning of controller parameters becomes a significant task since the level of satisfaction is determined by controller parameter settings.

Problems associated with several demand controller calibration can be handled as a multi-objective mathematical programming problem which can be solved using optimization or adaptive-optimization tuning approaches. This method permits the use of multi-objective optimization strategies to obtain a set of accurate controller settings with diverse trade-offs. These trade-offs are as a result of their compatibility with tuning performance conditions. As such operators, designers or decision makers need to select suitable configurations in accordance with the application demands depending on preferences.

The intricacy of multi-objective mathematical programming problems, otherwise termed as Multi-Objective Problem (MOP) with regards to controller parameters tuning function, has necessitated the employment of Computational Intelligence (CI) or soft computing approaches (Ruano et al., 2014). The implementation of CI for multi-objective optimization in control engineering is termed intelligent control (Rodríguez-Molina et al., 2020). Metaheuristic algorithms have been intensively employed to resolve several optimization problems in different fields of endeavor today. Some of these applications can be seen in Mathematics (Stojanović et al., 2017), robotics (Cruz-Bernal, 2013), agricultural engineering (Qi and Altınakar, 2011), structural design (Mirjalili and Lewis, 2016), greenhouse gas emission (Geem, 2011), medicine (Dey and Ashour, 2018), image processing (Alsmadi, 2018) and telecommunication (Kumar and Vidyarthi, 2016). The acceptance and popularity of metaheuristic algorithms can be attributed to their concurrent and quite simple implementation, their ability to handle large complex problems at a reduced computational cost, and their wide usage in different areas of endeavor (Gogna and Tayal, 2013; Talbi, 2009). These aforementioned qualities of metaheuristic algorithms make them appropriate substitutes for controller parameters tuning problem. Metaheuristic algorithms have a track record of outstanding performance and superiority in controller parameters tuning.

The diversity of various assortments of existing dynamic models and controller structures or architectures necessitates a wide range of distinct tuning MOP(s). These problems entail the application of diverse kinds of metaheuristics algorithms with specific search mechanisms to acquire the most appropriate or right controller settings. Thus, based on these assertions, this study seeks to give a comprehensive review on metaheuristic algorithms used for PID controller parameters tuning. Furthermore, in specific terms, the contribution of this work include:

- Comprehensive evolutionary background study of PID controller, its most important features, the evolution, developments and application strategies.
- Detailed background of metaheuristic algorithms and their classification, application flowcharts and implementation processes.
- Survey of PID controller tuning methods and their limitations.
- Presented an elaborate study of several techniques applied to PID tuning and a phenomenal review of literatures on PID controller parameters tuning using metaheuristic algorithm over a period of 2000–2020.
- Exposed researchers to some powerful metaheuristic algorithms that are not yet explored in PID parameters tuning.
- Identification of PID configuration problems.

The remainder of the paper is laid out as follows: Section 2 gives a succinct account of previous reviews, Section 3 is the background discussion, Section 4 describes the performance measures for evaluating the effectiveness of PID parameters tuning, Section 5 explains the metaheuristic algorithms that have found application in PID tuning, Section 6 is the comparative studies of existing tuning techniques used in PID tuning, Section 7 unveils open research problems in metaheuristic algorithms for PID tuning and future direction before concluding in Section 8.

To increase readability of manuscript and also enhance understanding of readers, the structure of this paper is depicted in Fig. 1.

2. Literature review

This section presents a brief history and background of PID controller, architecture (block diagram), equations, issues and application areas.

来建立正确的变量-性能关系。启发式方法还可以作为在线整定的公式或规则库，通常在设计目标或目标之间进行权衡。这些方法中最流行的是传统的齐格勒-尼科尔斯和科恩-库恩方法，如（Rodríguez-Molina et al., 2020; Kumar et al., 2015; Ho et al., 1996）所述。

iii. 频率响应法：这些用于PID参数整定的方法通常是离线的，并在学术界流行。该技术基于目标过程或系统的特性，如Ang等(2005)所述的循环成形。Natarajan和Gilbert (1997)以及Wang等(1999)提出了使用频率响应来整定PID控制器参数。Hagglund和Astrom在Hägglund和Åström (1991)中提出了频率响应上的自适应技术来整定PID控制器。

iv. 优化方法：这些PID控制器整定策略可以被视为一种特殊的反馈最优控制系统。PID变量是通过离线数学规划或数值优化技术针对单一目标即时获取的。该计算也可以通过计算机化启发式算法或进化算法来实现多个设计目标。该过程包括为指定被控对象陈述一个数学规划问题 (MPP)，并应用优化技术来解决它。这些方法的主要特点是控制参数是固定的，并通过求解离线数值优化方法获得。离线数值优化技术使用通过仿真或实际被控对象向量的导出状态向量。然后将获得的固定控制参数添加到闭环系统中。本综述属于此类整定。一些最近的PID整定优化技术包括：萤火虫算法 (dos Santos Coelho 和 Mariani, 2012)、混沌优化 (dos Santos Coelho, 2009)、进化多目标优化 (Nisi等, 2019)、粒子群优化 (Azar等, 2019) 和萤火虫优化 (Pilla等, 2019)。

v. 自适应整定方法：自适应整定策略是在线实时整定方法，涉及使用自动化在线机制来调整PID控制器参数增益。这些技术的关键特征是控制参数在闭环系统中的预定时间间隔内变化。该技术的应用可在（Pirabakaran and Becerra, 2001; Kim and Cho, 2005年; Chang et al., 2003）中找到。

被控对象动态特性的复杂性体现了整定控制器参数的难度（Bernstein, 2002）。被控对象动态特性可能具有非线性行为、多个可调变量以及大量的输入和输出变量（Rodríguez-Molina et al., 2020）。此外，困难可能源于环境和/或操作的不确定性与扰动。这些特征增加了获得旨在实现设定性能目标的合适控制器参数的难度。另外，现有应用需要同时满足多个性能条件/要求，且这些条件/要求通常相互冲突（Boyd et al., 1990）。其中一些性能要求可能包括：高精度性能、能效、低成本等。因此，控制器参数的整定成为一项重要任务，因为满意度水平由控制器参数设置决定。

与多个需求控制器校准相关的问题可以作为一个多目标数学规划问题来处理，该问题可以使用优化或自适应优化整定方法来解决。该方法允许使用多目标优化策略来获得一组具有不同权衡的准确控制器设置。这些权衡是由于它们与整定性能条件的兼容性而产生的。因此，操作员、设计人员或决策者需要根据应用需求选择适当的配置。

多目标数学规划问题的复杂性，就控制器参数整定功能而言也称为多目标问题，使得必须采用计算智能或软计算方法（Ruano 等人, 2014）。在控制工程中，将计算智能应用于多目标优化被称为智能控制（Rodríguez-Molina 等人, 2020）。元启发式算法已被广泛用于解决当今不同领域的多个优化问题。其中一些应用可以在数学（Stojanović 等人, 2017）、机器人学（Cruz-Bernal, 2013）、农业工程（Qi 和 Altınakar, 2011）、结构设计（Mirjalili 和 Lewis, 2016）、温室气体排放（Geem, 2011）、医学（Dey 和 Ashour, 2018）、图像处理（Alsmadi, 2018）和电信（Kumar 和 Vidyarthi, 2016）中看到。元启发式算法的接受度和普及性可归因于其并行且相当简单的实现、以较低计算成本处理大型复杂问题的能力，以及在不同领域的广泛应用（Gogna 和 Tayal, 2013; Talbi, 2009）。元启发式算法的这些前述品质使其成为控制器参数整定问题的合适替代方案。元启发式算法在控制器参数整定方面拥有出色的性能和优越性记录。

现有动态模型和控制器结构或架构的多样性需要各种不同的整定多目标问题。这些问题需要应用多种具有特定搜索机制的元启发式算法，以获取最合适或正确的控制器设置。因此，基于这些论断，本研究旨在对用于PID控制器参数整定的元启发式算法进行全面综述。此外，具体而言，本工作的贡献包括：

- PID控制器的全面进化背景研究，包括其最重要的特性、演变、发展及应用策略。
- 元启发式算法的详细背景及其分类、应用流程图和实现过程。
- PID控制器整定方法及其局限性的调查。
- 对应用于PID整定的几种技术进行了详尽研究，并对2000-2020年间使用元启发式算法进行PID控制器参数整定的文献进行了现象级综述。
- 向研究人员揭示了一些尚未在PID参数整定中探索的强大元启发式算法。
- 识别PID配置问题。

本文的其余部分安排如下：第2节简要介绍以往的综述，第3节提供背景讨论，第4节描述评估PID参数整定有效性的性能指标，第5节解释已应用于PID整定的元启发式算法，第6节对比PID整定中使用的现有调优技术，第7节揭示元启发式算法在PID整定中的开放性研究问题及未来方向，最后在第8节进行总结。

为了提高文稿的可读性并增强读者的理解，本文的结构如图1所示。

2. 文献综述

本节简要介绍了PID控制器的历史与背景，架构（框图）、方程、问题及应用领域。

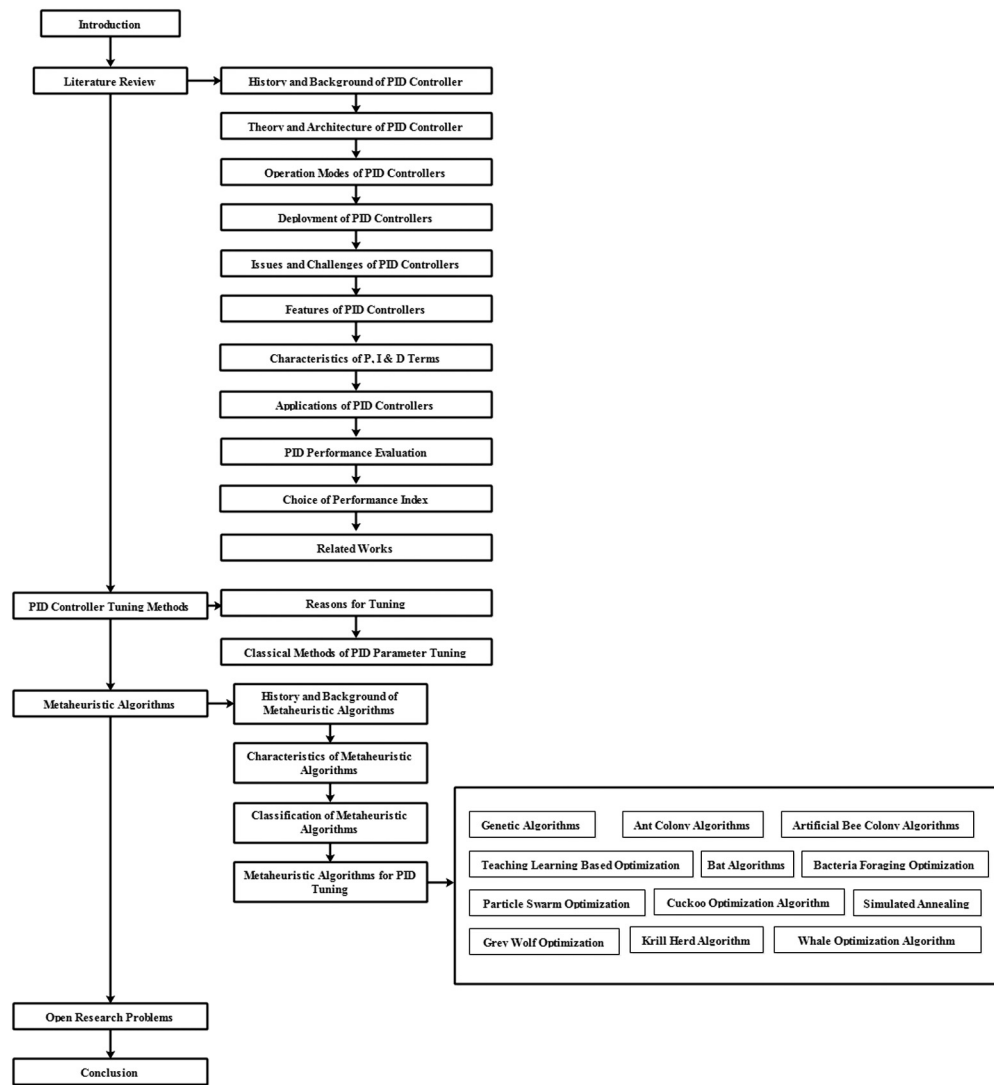


Fig. 1. Pictorial representation of paper structure.

2.1. Theory and architecture of PID controller

The PID controller is so called due to its output summation of three variables: proportional, integral and derivative terms which are denoted as K_P , K_I and K_D . These parameters K_P , K_I and K_D are reliant on error term e between input and output parameters. Correct tuning of these variables will enhance the dynamic reaction of a system, decrease overshoot, eradicate steady state error and increase stability of the process Saad et al. (2012). This relationship is described in Equation (7).

$$\text{output} = K_P \times e(t) + K_I \times \int_0^t e(t) dt + K_D \times \frac{de}{dt}, \quad (7)$$

where K_P , K_I and K_D are P , I and D parameters respectively. K_I and K_D can be represented by Equation (8).

$$K_I = K_P \times T_i, \quad K_D = K_P \times \frac{1}{T_d}, \quad (8)$$

where T_i and T_d are reset time and derivative time, respectively. The reset and derivative times are responsible for a systems response type. The controller variables P , I and D properties are briefly presented and discussed as follows:

- Proportional term (P):** The proportional parameter facilitates the reaction as closed loop time constant declines with proportional

parameter nonetheless ensures no alteration in system order as the output is just proportional to the input. The proportional parameter reduces nevertheless does not eradicate steady state error, or offset. The main task of the proportional element is to examine the magnitude of error and the PID controller response proportionally. In proportional term P is denoted by Equation (9) as

$$P_{out} = K_P e, \quad (9)$$

where

P_{out} : proportional part of controller output,

K_P : proportional gain,

e : error term.

- Integral term (I):** This parameter removes offset as it rises the nature and order of the system by 1. This parameter also rises the system reaction speed, but at the cost of continued oscillations. The integral control attempts to mitigate the issues of the proportional control. It studies the error over time and adjust (increase) a small error with time. Reset rate, which is a time factor, is used to denote adjustment of integral control.

$$I_{out} = \frac{1}{T_i} \int e dt = K_I \int e dt, \quad (10)$$

where

I_{out} : is the integral part of controller output,

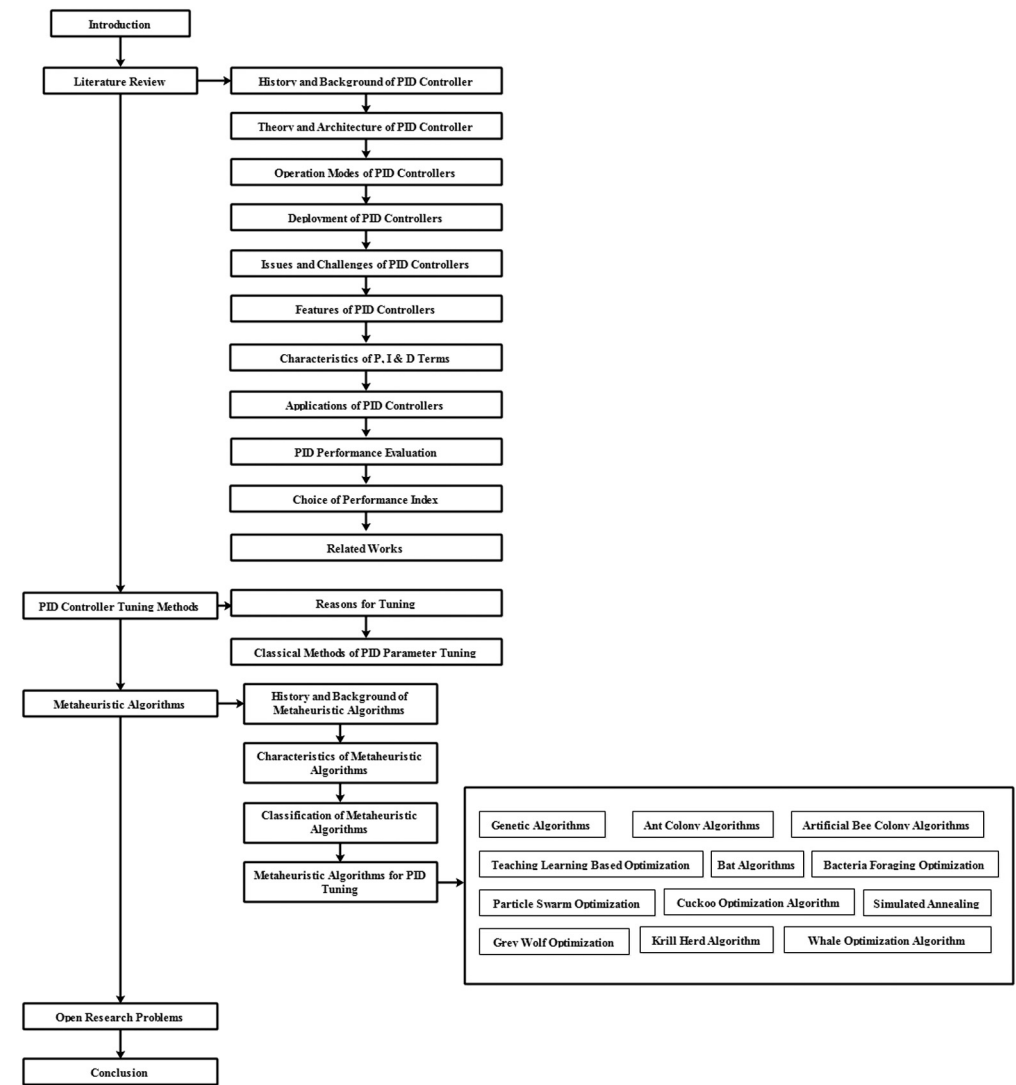


图 1. 论文结构的图示。

2.1. PID 控制器的理论与架构

PID控制器之所以得名，是因为其输出是三个变量的总和：比例项、积分项和微分项，分别表示为 K_P 、 K_I 和 K_D 。这些参数 K_P 、 K_I 和 K_D 依赖于输入与输出参数之间的误差项 e 。正确整定这些变量将增强系统的动态响应，减少超调，消除稳态误差，并提高过程的稳定性（Saad等人，2012）。该关系如方程(7)所示。

$$\text{output} = K_P \times e(t) + K_I \times \int_0^t e(t) dt + K_D \times \frac{de}{dt}, \quad (7)$$

其中 K_P , K_I 和 K_D 分别是 P , I 和 D 参数。 K_I 和 K_D 可以用公式表示 (8)。

$$K_I = K_P \times T_i, \quad K_D = K_P \times \frac{1}{T_d}, \quad (8)$$

其中 T_i 和 T_d 分别是复位时间和微分时间，分别。复位和微分时间负责系统的响应类型。控制器变量 P , I 和 D 属性简要介绍和讨论如下：

- 比例项 (P):** 比例参数有助于反应随着闭环时间常数下降而比例

参数确保系统阶数不变，因为输出仅与输入成正比。比例参数虽然减小了稳态误差或偏移，但并未完全消除。比例元素的主要任务是检查误差大小，并让PID控制器做出比例响应。比例项 P 由方程(9)表示：

$$P_{out} = K_P e, \quad (9)$$

其中

P_{out} : 控制器输出的比例部分

K_P : 比例增益

e : 误差项

- 积分项 (I):** 该参数通过将系统的性质和阶次提升1来消除偏移。该参数也提高了系统反应速度，但代价是持续振荡。积分控制试图缓解比例控制的问题。它随时间研究误差，并随时间调整（增加）小误差。复位速率是一个时间因子，用于表示积分控制的调整。

$$I_{out} = \frac{1}{T_i} \int e dt = K_I \int e dt, \quad (10)$$

其中

I_{out} : 是控制器输出的积分部分，

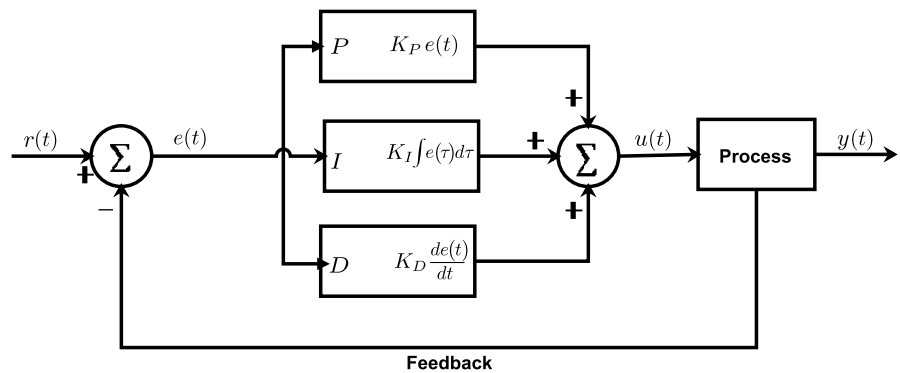


Fig. 2. Fundamental PID control structure.

T_i : reset time or integral time,
 K_I : integral gain, e : error term.
iii. **Derivative term (D):** This setting essentially reduces the system's oscillatory reaction. It has no effect on the offset, and has no effect on the system's nature and order. It investigates the error signal's rate of variation. A bigger system reaction to a rapid pace of change is caused by the derivative. The derivative term is adjusted with time. Too much of derivative term may cause an erratic control or overshoot. It is denoted by Equation (11).

$$D_{out} = T_d \frac{d}{dt} e = K_D \frac{d}{dt} e, \quad (11)$$

where
 D_{out} : is the derivative part of controller output,
 T_d : derivative time,
 K_D : derivative gain, e : error term.

Fig. 2 depicts the standard configuration of a PID control system. When the set value $r(t)$ changes, the error term $e(t)$ between the set value and the actual output is recalculated. The $e(t)$ error signal or term is utilized to construct the K_P , K_I , and K_D parameter actions, with the resulting signals weighted and summed to make the $u(t)$ control signal applied to the plant model. The output signal $y(t)$ is obtained as a result. The controller will receive this updated genuine signal, and the error signal will be recalculated. The plant receives the newly received control signal, $u(t)$. This process will run indefinitely until it encounters a steady-state mistake.

The PID control technique is a typical approach to control that uses a proportional factor (P), integrative action (I), and differential action (D) to convert error signals into input signals. The error signal is calculated by comparing the states to their corresponding references (set points). To improve the speed of the system's response to its defect, the error signal is amplified with proportional gain (K_P) and understood as current error term (Muliadi and Kusumoputro, 2018). The error signal is also integrated with the integral gain (K_I) as the accumulation of the past error term and differentiated with the derivative gain (K_D) as the prediction of future error term in parallel (Mange et al., 2015). These gains are fine-tuned using tuning methods based on a linear mathematical model of the system's dynamics. For systems with a modest amount of nonlinearity and perfect uncoupling between the modes of their dynamics, the PID gains can be simply modified.

2.2. Operation modes of PID controllers

The PID controller is a widely used industrial control structure with a common control loop feedback mechanism. PID controllers are the most generally used feedback controllers (Kushwah and Patra, 2014). It is worthy to note that majority of PID controllers are operated in either manual or automated state.

- Manual Mode:** Here, the output of controller is manipulated by an experienced who manipulates the controller parameters by pushing buttons which will either increase or decrease the controller output.
- Automatic Mode:** Controller parameters can be attuned during operation by the use of algorithms.

When variations in modes and terms are observed, it is imperative to stop changing transients. Meanwhile, PID controllers are dynamic systems, hence essential to ascertain state of the scheme is accurate before changing modes of operation. In automatic state, PID algorithm yields a control variable which may be distinct from manually created variable. Consequently, it is essential for operator(s) to ascertain both control terms match during switching. This process is referred to as bumpless transfer. Details of switching from one mode to another can be found in Anthony and Ho (2020).

2.3. PID performance evaluation

The acceptance and wide use of PID controllers has led to an explosion in methods of tuning its parameters. In the past decades, numerous proposals on PID parameters tuning have been published. Some of these proposals are focused on stable processes (Cohen, 1953; Zhuang and Atherton, 1993), integrating and unstable processes Tan et al. (1999) and multivariate processes (Luyben, 1986). Based on this, a natural question emanated: how can PID setting from different strategies be compared? In other words, what is the performance of a controller to be examined or measured? To answer this question, several researchers and authors have proposed and presented different measures to address the issue of performance measurements.

In the past decades, minimum variance (Huang and Shah, 1999; Harris, 1989), as a standard, have been used as criteria in process control to assess closed loop performance (Tan et al., 2004). A comprehensive review and assessment on minimum variance can be seen in Qin (1998); Huang et al. (1997).

Although minimum variance is a highly prized metric of system performance, it pays little heed to traditional metrics like setpoint tracking and disturbance rejection. Furthermore, another vital and important aspect of system performance, robustness, is not directly addressed. Obviously, a benchmark that can be used for single- and/or multi-loop processes that are stable, integrating, or unstable would be quite useful. The benchmark should include both time domain and frequency domain robustness requirements (Tan et al., 2004).

Tan et al. (2004) further suggested the use of integral error as a good measure for single loop process and gain-phase margin for robustness measure. However, these methods are not suitable for multi-loop processes. The use of integral error and gain-phase measure with different tuning methods can be seen in Huang and Shah (1999).

The most commonly used performance indices or measures employed in control system applications are Integral Absolute Error (IAE)

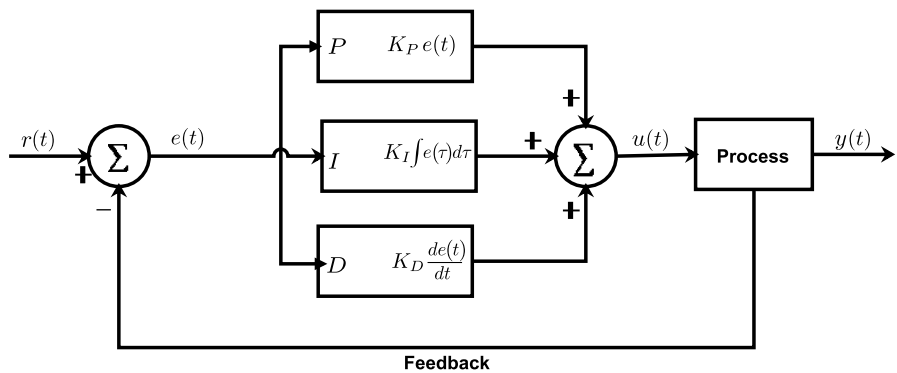


图 2.基本 PID 控制结构。

T_i : 复位 时间 或 积分 时间 ,
 K_I : 积分 增益, e : 误差 项。
iii. 微分项 (D): 该设置基本上减少了系统的振荡反应。它对偏移没有影响, 对系统的性质和阶次也没有影响。它研究误差信号的变化速率。较大的系统反应是由微分引起的快速变化率。微分项随时间调整。过多的微分项可能导致不稳定的控制或超调。它由方程 (11) 表示。

$$D_{out} = T_d \frac{d}{dt} e = K_D \frac{d}{dt} e, \quad (11)$$

其中
 D_{out} : 是 控制器输出 的 微分 部分 , T_d : 微分
时间 , : 微分 增益 , e : 误差 项。

图2展示了PID控制系统的标准配置。当设定值 $r(t)$ 变化时, 设定值与实际输出之间的误差项 $e(t)$ 会重新计算。 $e(t)$ 误差信号或误差项用于构造 K_P 、 K_I 和 K_D 参数作用, 所得信号经过加权求和后生成应用至被控对象模型的 $u(t)$ 控制信号。输出信号 $y(t)$ 作为结果获得。控制器将接收此更新后的真实信号, 并重新计算误差信号。被控对象接收到新接收的控制信号 $u(t)$ 。此过程将无限运行, 直到遇到稳态误差。

PID控制技术是一种典型的控制方法, 它利用比例因子(P)、积分作用(I)和微分作用(D)将误差信号转换为输入信号。误差信号通过将状态与其对应参考值(设定点)进行比较来计算。为了提高系统对其缺陷的响应速度, 误差信号使用比例增益(K_P)放大, 并作为当前误差项理解 (Muliadi和Kusumoputro, 2018)。误差信号还与积分增益(K_I)积分作为过去误差项的累积, 并与微分增益(K_D)微分作为未来误差项的预测并行处理 (Mange等, 2015)。这些增益使用基于系统动态线性数学模型的整定方法进行微调。对于非线性程度适中且动态模式之间完美解耦的系统, PID增益可以简单地调整。

2.2. PID控制器 的 操作模式

PID控制器是一种广泛使用的工业控制结构, 具有常见的控制回路反馈机制。PID控制器是最常用的反馈控制器 (Kushwah和Patra, 2014)。值得注意的是, 大多数PID控制器以手动或自动状态运行。

- 手动模式: 在此模式下, 由经验丰富的操作员通过按下增加或减少控制器输出的按钮来操纵控制器参数。
- 自动模式: 控制器参数可以在运行期间通过算法进行调整。

当观察到模式和项的变化时, 必须停止改变瞬态。同时, PID控制器是动态系统, 因此在改变操作模式之前, 必须确保方案的状态是准确的。在自动状态下, PID算法产生的控制变量可能与手动变量不同。因此, 操作员必须确保在切换时两个控制项匹配。这个过程称为无扰切换。关于从一个模式切换到另一个模式的详细信息, 请参阅 Anthony and Ho (2020)。

2.3. PID性能评估

PID控制器的接受和广泛使用导致了其参数整定方法的激增。在过去的几十年中, 已经发表了大量关于PID参数整定的建议。其中一些建议侧重于稳定过程 (Cohen, 1953; Zhuang and Atherton, 1993)、积分和不稳定过程 Tan et al. (1999) 以及多变量过程 (Luyben, 1986)。基于此, 一个自然的问题产生了: 如何比较不同策略下的PID设置? 换句话说, 如何检查或测量控制器的性能? 为了回答这个问题, 几位研究人员和作者提出并介绍了不同的措施来解决性能测量问题。

在过去几十年中, 最小方差 (Huang and Shah, 1999; Harris, 1989) 作为标准, 被用作过程控制中评估闭环性能的准则 (Tan et al., 2004)。关于最小方差的全面回顾和评估可见于Qin (1998); Huang et al. (1997)。

虽然最小方差是系统性能的一个高度珍视的度量标准, 但它很少关注传统的度量标准, 如设定点跟踪和扰动抑制。此外, 系统性能的另一个至关重要方面——鲁棒性, 也没有直接涉及。显然, 一个可用于稳定、积分或不稳定的单回路和/或多回路过程的基准将非常有用。该基准应包括时域和频域的鲁棒性要求 (Tan et al., 2004)。

Tan et al. (2004) 进一步建议使用积分误差作为单回路过程的一个良好度量, 并使用增益相位裕度作为鲁棒性度量。然而, 这些方法不适用于多回路过程。使用积分误差和增益相位度量结合不同整定方法的情况可见于Huang and Shah (1999)。

最常用 的 性能 指标 或 度量 方法 应用 于 控制 系统 领域 是 积分 绝对 误差 (IAE)

(Equation (1)), Integral Square Error (ISE) (Equation (2)), Integral Time Absolute Error (ITAE) (Equation (3)), Integral Time Square Error (ITSE) (Equation (4)), Mean Square Error (MSE) (Equation (5)) and Integral Error (Equation (6)). The application of ITAE for PID tuning can be seen in Martins (2005); da Silva et al. (2019). Jin et al. (2018) evaluated disturbance rejection performance for multiple objectives using IAE to estimate time response of process. Soni and Bhatt (2013) used ISE, IAE, IATE and MSE indices with BF-PSO to optimize PID controller. The performance and robustness trade-offs of PID control using IE and IAE can be seen in Garpinger et al. (2014). A comparison of ISE, IAE, ITAE and MSE using a heating tank was presented in Hussain et al. (2014).

2.4. Related works

There is still a rapid increase in the interest being shown by the academic and global research communities on PID parameter tuning. This segment presents related reviews that have been presented in literature on same subject area. This format is adopted to enhance articulation of issues (gaps) yet to be addressed or not fully addressed and to highlight differences with current review.

Bhawana and Neelu (2017) presented an appraisal of traditional as well as present methods employed for tuning PID controller parameters. The work presented different PID controller tuning procedures existing in literature and gave assessment of some methods. These methods can be typically grouped as classical methods and optimization methods. The methods appraised are classified into either traditional methods designed for tuning PID controller or heuristic methods of tuning. The methods proposed for tuning PID controller for processes like coupled tank system, trajectory tracking of Unmanned Aerial Vehicle (UAV), quadrotor stabilization are as well appraised.

Gholap et al. (2017) presented a review paper comparing different algorithms based on time domain analysis, acquired tuned parameters, iteration number and fitness function values. Two distinct process plants were considered based on ITAE and time domain analysis as objective functions. Variations in values of tuning parameters were considered using two systems and cost functions. Tuning parameters were obtained for both systems using Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Simulated Annealing (SA). Assessment for all the three algorithms was conducted with gotten tuning variables, time domain analysis, number of iterations, and value of cost functions. Closed loop responses of individual arrangements were plotted. Process with SA-PID variables settles down fast compared to PSO and GA, however its peak overshoot is higher. PSO-PID tuned system variables highest overshoot is least. PSO and SA out performed GA system with GA settling slowly with more overshoot and larger minimum cost function value. PSO algorithm converges fast as compared to SA, also cost minimization is better in PSO. Different algorithms may not yield the same results after implementing with same system and objective function.

In this paper (Srinivas et al., 2014), the results of the GA-PID controller for different tuning techniques employing three tank level process control systems were compared to the classic PID controller. Using computer simulation, this comparison analysis was performed for set point tracking of process plant procedures. Using Matlab, the optimal tuning variables of a PID controller were evaluated using six tuning methodologies (Z-N technique, C-C technique, ISE, IAE, IATE, and genetic algorithm centered PID methodology). The GA-PID controller produces a rapid reaction with less peak overshoot and integral square error, according to simulation results. Furthermore, this technique has a fine capacity to modify tuning variables for system dynamics variations. In conclusion, the GA-PID controller has been shown to be a reliable solution for controlling three tank levels. This method can also be used in a variety of nonlinear process control systems with long transportation lags.

Emmanuel and Inyama (2015) conducted a survey on robot manipulator controller design strategies that can function optimally in harsh environs. PID controller tuning approaches like the conventional Ziegler

Nichol's (ZN) software tool, fuzzy logic, GA, and Artificial Neural Network (ANN) approaches were studied. The study concluded that it is challenging to attain optimal performance using conventional and ZN technique. Fuzzy logic, GA and ANN approaches can be employed to attain anticipated optimal performance but they lack factors to calculate disturbance elimination settling time which aids to measure how fast the system can cancel or reject disturbance in harsh environments. Conversely, software tool scheme using PID tuner can be utilized to attain anticipated performance and good disturbance rejection settling time which satisfies the goal of design.

Fister et al. (2016) did a study on two reactive evolutionary algorithms (differential evolution and genetic algorithm), and four reactive, swarm intelligence-based algorithms (bat, hybrid bat, PSO and Cuckoo Search (CS)), were employed to tune PID controller in the comparative study. Only ten individuals and ten iterations (generations) were utilized in order to choose the most suitable optimization algorithm for fast tuning of controller parameters. The results were compared using statistical analysis and showed that PSO is the best option for such a task.

A survey on PSO and automated voltage regulators for synchronous generators was presented by Dixit and Pragati Joshi (2016). The generator is utilized to maintain electrical appliance control and stability. Automatic Voltage Regulators (AVR) employ a variety of ways. Diverse controllers are also utilized to improve robustness, overshoot, rise time, and voltage control, however the issue is with the AVR survey. According to several research paper analyses, assessment studies focusing on PID Controller are conducted to demonstrate rise time, overshoot, and undershoot.

Kim and Cho (2006) suggests a hybrid technique for adjusting a PID controller for an AVR that combines GAs and Bacterial Foraging (BF). The proposed method was first demonstrated using four test functions, and the algorithm's outcome was evaluated with a focus on mutation, crossover, step size variation, chemotactic steps, and bacteria life duration. Furthermore, the proposed algorithm was used to calibrate an AVR's PID controller. GA, PSO, GA-BF, and GA-PSO are among the algorithms compared.

Nagaraj et al. (2008) presents a survey on how PID controller variables are tuned for DC motor armature control. The performance of conventional procedures such as the continuous cycle approach and the ZN step response methodology was compared and analyzed with intelligent tuning strategies such as the GA, Evolutionary Programming (EP), and PSO. Tuning methods based on GA, EP, and PSO have demonstrated their superiority by achieving better results. They increased the performance indices and steady state characteristics.

Zulu and John (2014) reviewed some conventional control algorithms used on quadrotors in studies. Clearly, no single algorithm can provide the greatest of all essential features, as evidenced by the evaluation. It has been determined that achieving superior performance often demands hybrid control structures with the best combination of robustness, adaptability, optimality, simplicity, tracking ability, fast reaction, and disturbance rejection, among other things. However, such hybrid structures do not guarantee the optimal performance; consequently, for any control application, a compromise must be found as to which of the components is the most appropriate. The assessors are still undecided on which mathematical model will provide the best overall performance. PID, Linear Quadratic Regulator (LQR), Sliding mode, Backstepping, Feedback linearization, Adaptive, Robust, Optimal, L1, H_∞, Fuzzy logic, and ANN have all been appraised after a brief description of the system. The article concludes with a recommendation to examine hybrid systems, which integrate advantages from many control paradigms.

Divya and Nirmalkumar (2018) in their paper titled "A Survey on Tuning of PID Controller for Industrial Process Using Soft Computing Techniques" presented a review of some soft computing methods for PID controller parameters tuning. The review considered five (5) soft computing techniques, namely Neural Networks (NN), EP, GA, PSO and Ant Colony Optimization (ACO) methods. The survey gave some basic

(公式(1))、积分平方误差 (ISE) (公式(2))、积分时间绝对误差 (ITAE) (公式(3))、积分时间平方误差 (ITSE) (公式(4))、均方误差 (MSE) (公式(5)) 和积分误差 (公式(6))。ITAE在PID整定中的应用可见于 Martins (2005); da Silva 等 (2019)。Jin 等 (2018) 使用IAE评估了多目标下的扰动抑制性能，以估计过程的时间响应。Soni and Bhatt (2013) 使用ISE、IAE、IATE和MSE指标，结合BF-PSO优化PID控制器。使用IE和IAE的PID控制的性能与鲁棒性权衡可见于 Garpinger 等 (2014)。使用加热罐对ISE、IAE、ITAE和MSE的比较在 Hussain 等 (2014) 中提出。

2.4. 相关工作

学术界和全球研究界对PID参数整定的兴趣仍在迅速增加。本节介绍了文献中关于同一主题领域的相关综述。采用这种格式是为了更好地阐述尚未解决或未完全解决的问题（差距），并突出与当前综述的差异。

Bhawana and Neelu (2017) 对用于PID控制器参数整定的传统方法和现有方法进行了评估。该工作介绍了文献中存在的不同PID控制器整定程序，并对一些方法进行了评估。这些方法通常可分为经典方法和优化方法。被评估的方法分为用于整定PID控制器的传统方法或整定的启发式方法。针对耦合水箱系统、无人机轨迹跟踪、四旋翼无人机稳定等过程提出的PID控制器整定方法也得到了评估。

Gholap 等 (2017) 发表了一篇综述论文，比较了基于时域分析的不同算法，获取了整定参数、迭代次数和适应度函数值。基于 ITAE 和时域分析作为目标函数，考虑了两种不同的过程工厂。使用两个系统和代价函数考虑了整定参数值的变化。使用遗传算法 (GA)、粒子群优化 (PSO) 和模拟退火 (SA) 获得了两个系统的整定参数。对所有三种算法进行了评估，评估内容包括获取的整定变量、时域分析、迭代次数和代价函数值。绘制了各个配置的闭环响应。与 PSO 和 GA 相比，采用 SA-PID 变量的过程收敛更快，但其峰值超调更高。PSO-PID 整定系统的峰值超调最小。PSO 和 SA 优于 GA 系统，GA 收敛缓慢，超调更大，且最小代价函数值更大。与 SA 相比，PSO 算法收敛更快，且代价最小化效果更好。不同算法在相同系统和目标函数下实现后，可能不会产生相同的结果。

在本文中 (Srinivas 等人, 2014)，针对采用三种液位过程控制系统的不同调优技术的 GA-PID 控制器结果与经典 PID 控制器进行了比较。通过计算机仿真，该比较分析针对过程工厂程序的设定点跟踪进行。使用 Matlab，通过六种调优方法（Z-N 技术、C-C 技术、ISE、IAE、IATE 和基于遗传算法的 PID 方法）评估了 PID 控制器的最优调优变量。根据仿真结果，GA-PID 控制器产生快速响应，具有更小的峰值超调和积分平方误差。此外，该技术具有调整调优变量以适应系统动态变化的良好能力。总之，GA-PID 控制器已被证明是控制三个液位的可靠解决方案。该方法也可用于具有长传输滞后的各种非线性过程控制系统。

Emmanuel and Inyama (2015) 进行了一项关于机器人机械臂控制器设计策略的调查，这些策略能在恶劣环境中最优运行。PID 控制器整定方法，如传统的 Ziegler

Nichol's (ZN) 软件工具、模糊逻辑、遗传算法和人工神经网络 (ANN) 方法被研究。研究表明，使用传统和 ZN 技术难以实现最优性能。模糊逻辑、遗传算法和 ANN 方法可用于实现预期的最优性能，但它们缺乏计算扰动消除稳定时间的因素，这有助于衡量系统在恶劣环境中消除或抑制扰动的速度。相反，使用 PID 调谐器的软件工具方案可用于实现预期性能和良好的扰动抑制稳定时间，从而满足设计目标。

Fister 等人 (2016) 进行了一项研究，其中使用了两种反应式进化算法（差分进化和遗传算法）和四种反应式群体智能算法（蝙蝠算法、混合蝙蝠算法、粒子群优化和布谷鸟搜索 (CS)）来调优 PID 控制器进行比较研究。仅使用十个个体和十次迭代（代数）以选择最适合快速调优控制器参数的优化算法。通过统计分析比较结果，表明粒子群优化是此类任务的最佳选择。

Dixit 和 Pragati Joshi (2016) 对同步发电机的粒子群优化和自动电压调节器进行了综述。发电机用于维持电器控制和稳定性。自动电压调节器 (AVR) 采用多种方式。不同的控制器也被用于改善鲁棒性、超调、上升时间和电压控制，但问题在于 AVR 的综述。根据几篇研究论文的分析，开展了以 PID 控制器为重点的评估研究，以展示上升时间、超调和下冲。

Kim 和 Cho (2006) 提出了一种混合技术，用于调整 AVR 的 PID 控制器，该技术结合了遗传算法和细菌觅食 (BF)。首先使用四个测试函数演示了所提出的方法，并评估了算法的结果，重点关注突变、交叉、步长变化、趋化步骤和细菌寿命。此外，所提出的算法被用于校准 AVR 的 PID 控制器。比较的算法包括 GA、PSO、GA-BF 和 GA-PSO。

Nagaraj 等人 (2008) 提出了一项关于如何为直流电机枢控制整定 PID 控制器参数的综述。比较并分析了传统方法（如连续循环方法和 ZN 阶跃响应方法）与智能整定策略（如遗传算法、进化规划 (EP) 和粒子群优化）的性能。基于遗传算法、进化规划和粒子群优化的整定方法通过获得更好的结果证明了其优越性。它们提高了性能指标和稳态特性。

Zulu 和 John (2014) 回顾了研究中用于四旋翼飞行器的一些传统控制算法。显然，没有任何单一算法能提供所有基本特征的最佳表现，评估结果也证实了这一点。已确定实现卓越性能通常需要混合控制结构，以最佳组合鲁棒性、适应性、最优性、简单性、跟踪能力、快速反应和扰动抑制等特性。然而，这种混合结构并不能保证最优性能；因此，对于任何控制应用，必须权衡哪个组件最合适。评估者仍未决定哪种数学模型能提供最佳整体性能。在简要描述系统后，评估了 PID、线性二次调节器 (LQR)、滑模、反步、反馈线性化、自适应、鲁棒、最优、L1、H_∞、模糊逻辑和人工神经网络。文章最后建议研究混合系统，这些系统整合了多种控制范式的优势。

Divya 和 Nirmalkumar (2018) 在其题为“基于软计算技术的工业过程PID控制器整定综述”的论文中，综述了一些用于PID控制器参数整定的软计算方法。该综述考虑了五种软计算技术，即神经网络 (NN)、进化规划、遗传算法、粒子群优化和蚁群优化 (ACO) 方法。该调查给出了一些基本

highlights of selected algorithms and presented their flowcharts. Details of implementation of selected algorithms were not presented.

Bharat et al. (2019) presented a review of PID tuning method. The paper presented a survey on works related to PID controller tuning methods using fuzzy-logic and GA. Also, a comparison of conventional two methods (ZN and process reaction curve) and fuzzy-logic based tuning methods was presented. A brief review of tuning methods like Fuzzy Set-Point Weight (FSPW), GA, fuzzy logic, ZN, and PRC were presented.

A review on tuning techniques for PID controllers was presented in Bansal et al. (2012). The work presented a review on current and classical technique for PID tuning. The work gave a brief description of two main classifications of tuning strategies namely classical and computational or optimization techniques. On classical method, Ziegler and Nichols method, and Cohen Coon method were presented. Artificial Immune Systems (AIS), ACO, Bacteria Forage Technique (BFT), GA, Differential Evolution (DE), EP, ANN, SA, Support Vector Machine (SVM), PSO, Fuzzy logic, and Response Surface Method (RSM) were among the topics discussed. The work presented relevant equations, algorithms and flowcharts for every algorithm considered in the study. A good number of works implementing such algorithms for PID tuning was also presented.

Kushwah and Patra (2014) presented a review on tuning techniques using soft computing strategies. The study was focused on DC motor speed control using PID controller. The study gave a brief discussion on PID controller, its pictorial representations and parametric equations. The review was focused on GA, PSO, and fuzzy logic soft computing techniques for DC motor speed control. The paper only briefly highlighted on a few works that implemented the selected algorithms.

The study in Jain et al. (2012) titled “A Review of Soft Computing Techniques for PID Controller Tuning” presented a review of three soft computing algorithms (GA, ACO, and PSO) for tuning PID parameters. The study presented algorithms for each technique and also gave a little background information on the three algorithms chosen.

A review of how swarm intelligence computation algorithms can be used to tune PID controllers was presented in Ghosal et al. (2012). The study highlighted on different structures of PID controller and presented a survey on swam intelligence computation. The study further listed the advantages of intelligence, and also gave reviews on tuning PID parameters using ACO, PSO and Bacterial Foraging Optimization (BFOA). A comparative analysis of four (4) tuning methods (ZN, PSO, BFO and GA) using MATLAB was presented. Their analysis reported that BFOA gave a better result compared to others.

Despite the significance of metaheuristic algorithms in PID tuning, there is no comprehensive, complete and methodical review about their classification. Additionally, future problems and significant role of metaheuristic algorithms in PID tuning problem have not been itemized properly. In summary, reviewed papers have some shortcomings:

- Only a few literatures have been investigated.
- Details of algorithms implementation are not explicitly defined.
- Future studies and unanswered issues are not properly expressed.
- Few qualitative and recent papers were cited.
- In some cases, only a few metaheuristic algorithms were explored.
- Classification of studied techniques has not been adequately presented.
- Advantages and disadvantages of selected works not clearly presented.

Further readings on metaheuristic algorithms reviews and taxonomies on other subject matters like: scheduling techniques in cloud computing (Kalra and Singh, 2015), vehicle routing problem and its variants (Elshaer and Awad, 2020), optimizing 3D well-path designs (Biswas et al., 2020), reactive power planning problem (Shaheen et al., 2018), optimization of Service-Level Agreement (SLA) violation costs for dynamic execution of cloud services (Kumar and Bawa, 2020), multi-objective virtual machine placement mechanisms in cloud com-

puting (Donyagard Vahed et al., 2019), applications on PV cell parameter identification (Yang et al., 2020), solving grouping problems (Ramos-Figueroa et al., 2020) and hydropower energy optimization (Azad et al., 2020). A summary of related works is presented in Table 1.

3. PID controller tuning methods

A summary of tuning methods for PID controller is depicted in Fig. 3.

3.1. Classical methods of PID parameters tuning

This class of tuning strategy attempts to derive controller configurations for common models. To compute the dynamics of these systems, step response of systems is obtained (Bansal et al., 2012). This response is characterized by diverse equations, which leads to development of different classical approaches.

Before tuning PID parameters, it is crucial to obtain configuration using traditional approaches since the gains of controllers rely on transfer function variables. Although traditional PID tuning approaches are broadly used, oftentimes their behaviors do not match the set desired goal or objective (Reis et al., 2016).

- Ziegler-Nichols tuning method:** Ziegler and Nichols proposed the most fundamental acknowledged and popular tuning technique in 1942 (Åström and Hägglund, 2004). Ziegler and Nichols conducted multiple studies and proposed rules for calculating K_p , K_I , and K_D values based on a plant's transient step reaction. They presented two methods: closed-loop and open-loop. Process basic features such as process gain (K_p), process time constant (T_p), and process dead time (L_p) are obtained using the ZN tuning approach. Ziegler and Nichols suggested using the values of K_p , T_I and T_D for closed and open loops using different rules. The tables and equations can be seen in (Shahrokh and Zomorrodi, 2013; Sorensen, 2007; Hahn, 2006).
- Cohen-Coon tuning method:** Cohen and Coon (Cohen, 1953) proposed a technique with PID controller parameters obtained through First Order Plus Time Delay (FOLPD) model. The key design condition is elimination of load turbulence. The controller parameter configuration is presented in (Cohen, 1953), while some application examples can be seen in Hambali et al. (2014); Azman et al. (2017). Regardless of a superior model, results of Cohen and Coon technique are not much superior than Ziegler Nichols approach.
- Gain and Phase Method:** Åström and Hägglund (1984) suggested a test that, in terms of ease of use in automatic loop tuners, was a significant improvement over the ZN test. They proposed replacing the variable proportional gain with a nonlinear function, with relay nonlinearity as the function. Åström and Hägglund (1984) proposed a gain and phase strategy for getting specific spots on the Nyquist curve to aid in determining controller pretuning parameters to overcome the Z-N technique's trail by error limitation. The tuning method is based on the ability to move the critical point on the process Nyquist curve to a specific location. On the Nyquist plot, Åström and Hägglund (1984) proposed that this point be situated at unity gain and a phase of $\phi_m - 180$, where ϕ_m denotes the predicted phase margin and A_m represents the desired gain margin. A control system's phase and gain margins are a measure of how close the system's polar plot is to the $(-1 + j0)$ point.

There are other varieties or modifications of ZN tuning methods suggested by other researchers. Åström and Hägglund (2004) proposed an improved method of Ziegler and Nichols tuning with robust loop shaping strategy (McFarlane and Glover, 1992). Chien, Hrones, and Reswick (CHR) technique was created from original ZN Open Loop technique. It offers swiftest response without overshoot and also with 20% overshoot.

亮点, 选定的算法的流程图以及它们的展示。选定的算法的实现细节并未被呈现。

Bharat等人(2019)对PID整定方法进行了综述。该论文调查了使用模糊逻辑和遗传算法的PID控制器整定方法的相关工作。此外,还对两种传统方法(ZN和过程反应曲线)与基于模糊逻辑的整定方法进行了比较。简要介绍了诸如模糊设定点加权(FSPW)、遗传算法、模糊逻辑、ZN和PRC等整定方法。

Bansal等人(2012)对PID控制器的整定技术进行了综述。该工作回顾了当前和经典的PID整定技术。简要描述了整定策略的两大分类:经典方法以及计算或优化技术。经典方法中介绍了Ziegler和Nichols方法以及科恩-库恩方法。讨论的主题包括人工免疫系统(AIS)、ACO、细菌觅食技术(BFT)、遗传算法、差分进化(DE)、进化规划(EP)、人工神经网络、模拟退火(SA)、支持向量机(SVM)、粒子群优化(PSO)、模糊逻辑和响应面法(RSM)。该工作为研究中考考虑的每种算法提供了相关方程、算法和流程图。还介绍了大量使用此类算法进行PID整定的研究工作。

Kushwah和Patra(2014)提出了一篇关于使用软计算策略的调优技术的综述。该研究专注于使用PID控制器进行直流电机速度控制。研究简要讨论了PID控制器、其图形表示和参数方程。该综述侧重于遗传算法、粒子群优化和模糊逻辑软计算技术用于直流电机速度控制。该论文仅简要介绍了实现所选算法的一些工作。

Jain等(2012)的研究题为“用于PID控制器参数整定的软计算技术综述”,综述了三种软计算算法(遗传算法、ACO和粒子群优化)用于PID参数整定。该研究介绍了每种技术的算法,并提供了所选三种算法的一些背景信息。

Ghosal等(2012)提出了一篇关于如何使用群体智能计算算法来整定PID控制器的综述。该研究重点介绍了PID控制器的不同结构,并进行了群体智能计算的综述。该研究进一步列出了智能的优势,并且还综述了使用ACO、粒子群优化和细菌觅食优化(BFOA)来整定PID参数。使用MATLAB对四种整定方法(齐格勒-尼科尔斯方法、粒子群优化、BFO和遗传算法)进行了比较分析。他们的分析报告称,BFOA相比其他方法给出了更好的结果。

尽管元启发式算法在PID整定中具有重要意义,但目前尚无全面、完整且系统的关于其分类的综述。此外,元启发式算法在PID整定问题中的未来问题和重要作用也未得到充分列举。总之,已综述的文献存在一些不足:

- 仅研究了少量文献。
- 算法实现的细节未明确说明。
- 未来研究和未解决的问题未充分表达。
- 引用的定性和近期论文较少。
- 在某些情况下,仅探讨了少数元启发式算法。
- 所研究技术的分类未充分呈现。
- 所选工作的优缺点未清晰呈现。

关于元启发式算法在其他主题上的综述和分类的进一步阅读,包括:云计算中的调度技术(Kalra和Singh, 2015)、车辆路径问题及其变体(Elshaer和Awad, 2020)、优化三维井眼轨迹设计(Biswas等, 2020)、无功功率规划问题(Shaheen等, 2018)、优化云服务动态执行中的服务水平协议(SLA)违约成本(Kumar和Bawa, 2020)、云计算

算中的多目标虚拟机放置机制(Donyagard Vahed等, 2019)、光伏电池参数识别的应用(Yang等, 2020)、解决分组问题(Ramos-Figueroa等, 2020)以及水电能源优化(Azad等, 2020)。表1总结了相关工作。

3. PID 控制器 整定 方法

图 3 展示了 PID 控制器 整定 方法 的 概述 。

3.1. PID 参数 整定 的 经典 方法

这类整定策略试图为常见模型推导控制器配置。为计算这些系统的动态特性,需要获取系统的阶跃响应(Bansal等人, 2012)。该响应由多种方程描述,从而导致了不同经典方法的开发。

在整定PID参数之前,使用传统方法获取配置至关重要,因为控制器的增益依赖于传递函数变量。尽管传统PID整定方法被广泛使用,但其行为往往与设定的期望目标或目标不符(Reis等, 2016)。

- 齐格勒-尼科尔斯整定方法: Ziegler和Nichols于1942年提出了最基础、公认且流行的整定技术(Åström和Hägglund, 2004年)。Ziegler和Nichols进行了多项研究,并提出了基于被控对象瞬态阶跃响应计算 K_p 、 K_I 和 K_D 值的规则。他们提出了两种方法:闭环和开环。使用ZN整定方法可获得被控对象基本特征,如过程增益(K_p)、过程时间常数(T_p)和过程死区时间(L_p)。Ziegler和Nichols建议使用不同规则对闭环和开环的 K_p 、 T_I 和 T_D 值进行计算。相关表格和方程可参见(Shahrokh和Zomorrodi, 2013年; Sorensen, 2007年; Hahn, 2006年)。
- Cohen-Coon整定方法: Cohen和Coon (Cohen, 1953)提出了一种技术,通过一阶加时滞模型获得PID控制器参数。关键设计条件是消除负载扰动。控制器参数配置在(Cohen, 1953)中给出,而一些应用示例可见于Hambali等人(2014)和Azman等人(2017)。尽管模型更优,但Cohen和Coon技术的结果并不比Ziegler-Nichols方法优越多少。

iii. 增益相位法: Åström和Hägglund (1984)提出了一种测试,在自动回路整定器的便利性方面,比ZN测试有了显著改进。他们建议用非线性函数替代可变比例增益,并以继电器非线性作为该函数。

Åström和Hägglund (1984)提出了一种增益相位策略,用于获取奈奎斯特曲线上的特定点,以帮助确定控制器预整定参数,从而克服Z-N技术的试错限制参数。该整定方法基于将过程奈奎斯特曲线上的临界点移动到特定位置的能力。在奈奎斯特图上,Åström和Hägglund (1984)建议将该点置于单位增益和相位为 $\phi_m - 180$ 的位置,其中 ϕ_m 表示预期的相位裕度, A_m 表示期望的增益裕度。控制系统的相位和增益裕度是衡量系统极坐标图接近 $(-1 + j0)$ 点的程度。

其他研究人员提出了ZN整定方法的其他变体或修改。Åström和Hägglund (2004)提出了一种改进的Ziegler和Nichols整定方法,采用鲁棒环路整形策略(McFarlane和Glover, 1992)。Chien-Hrones-Reswick (CHR)技术源于原始的ZN开环技术。它提供无超调的快速响应以及具有20%超调的响应。

Table 1. Summary of previous related surveys.

S/No	References	No of study referenced	Focus of review	Number of algorithms reviewed	Taxonomy (classification)
1	Bhawana and Neelu (2017)	33	Classical and Optimisation techniques used for PID tuning	9	Yes
2	Gholap et al. (2017)	10	PID controller tuning parameters, time domain analysis, number of iterations, and value of cost functions using GA, PSO, and SA.	3	No
3	Srinivas et al. (2014)	16	Optimum tuning parameters of PID controller are estimated by six tuning methods (Z-N method, C-C method, ISE, IAE, IATE and GA based PID method) for three tank level process for using Matlab/Simulink	8	
4	Emmanuel and Inyiama (2015)	32	PID controller tuning methods such as the manual, Ziegler Nichols, software tool, fuzzy logic, Genetic Algorithm (GA), and the Artificial Neural Network (ANN)		
5	Fister et al. (2016)	32	PID controller tuning methods reactive using differential evolution, genetic algorithm, bat, hybrid bat, particle swarm optimization and cuckoo search	6	Yes
6	Dixit and Pragati Joshi (2016)	8	PID Controller tuning for Automatic Voltage Regulator using PSO and GA	2	No
7	Kim and Cho (2006)	29	PID controller of an AVR. using GA, PSO, GA-BF, GA-PSO	4	No
8	Nagaraj et al. (2008)	9	PID controller are tuned for controlling the armature controlled DC motor using Z-N (step response), Kappa-Tau Continuous cycling, EP, GA and PSO.	6	No
9	Zulu and John (2014)	34	PID control tuning for quadrotors using Linear Quadratic Regulator (LQR), Sliding mode, Backstepping, Feedback linearization, Adaptive, Robust, Optimal, L1, H ∞ , Fuzzy logic and Artificial neutral networks.	12	Yes
10	Divya and Nirmalkumar (2018)	13	Soft Computing Techniques for PID tuning using Neural Networks (NN), Evolutionary Programming (EP), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO).	5	No
11	Bharat et al. (2019)	15	Techniques for PID tuning using Fuzzy Logic (FL), Genetic Algorithm (GA).	2	No
12	Bansal et al. (2012)	64	Review of tuning Techniques for PID parameters classification of tuning strategies using 14 methods (Z-N method, C-C method, AIS,ACO, BFO, DE, EP, ANN, SA, SVM, PSO, FL and RSM based PID method)	14	No
13	Kushwah and Patra (2014)	43	Speed control of DC motor using for PID controller using three tuning strategies GA, PSO, and fuzzy logic algorithms	14	No
14	Ghosal et al. (2012)	51	Swam intelligence computation for PID controller using four tuning strategies ZN, BFO, PSO, and GA	4	No

This technique has capability for tuning the set point and disturbance. Details and application of CHR can be found in Sen et al. (2015). A comparison of several tuning strategies can be seen in Shahrokhi and Zomorrodi (2012).

4. Metaheuristic algorithms

Metaheuristic based strategies (Talbi, 2009) deals with problems by providing near optimal solutions within reasonable time. Due to its efficiency and effectiveness in solving vast and complicated issues, metaheuristics have exploded in popularity in recent years. Metaheuristics are a type of heuristic that performs better than simple heuristics. Meta implies “beyond” or “higher level”. Every metaheuristic algorithm employs a particular mix of local and global search. Different results are often realized through randomization. There is no standard or verifiable definition of heuristics and metaheuristics in the literature, regardless of whether metaheuristics are accepted. A number of academics and researchers utilize heuristics and metaheuristics interchangeably. Current developments, however, appear to apply to all stochastic algorithms that use randomization and global exploration as metaheuristics. This assessment will also adopt same convention.

4.1. Classification of metaheuristic algorithms

This section briefly explores and summarizes a comprehensive review of metaheuristic algorithms and their emergence. Metaheuristic algorithms can be categorized in several forms, depending on selected characteristics or features that distinguish them. Different authors and scholars classify metaheuristic algorithms considering different features or characteristics. The categorization could also be dependent or centered on objectives or goals of research. Examples of this category may be found in Blum and Roli (2003) which divides metaheuristics into two categories: nature-inspired and non-nature-inspired. Whether it is a population or a single point search, whether the objective function is dynamic or static, memory utilization or memory-less approaches algorithm and one or more neigh-

borhood structures, Mirjalili and Lewis in Mirjalili and Lewis (2016) classified metaheuristic algorithms into: evolutionary, physics-based, swarm-based and human-based algorithms. A further classification based on population is presented in Beheshti and Shamsuddin (2013); Nojhan (2007). Another classification can be seen in Jackson et al. (2018), metaheuristics can be divided into two types: single-solution and multiple-solution problems. Several other taxonomies of metaheuristic algorithms can be seen in Kumar and Bawa (2018); Shaheen et al. (2018); DrEO et al. (2007); Beheshti and Shamsuddin (2013); Gogna and Tayal (2013); Hussien et al. (2020); Kumar and Bawa (2020); Abdel-Basset et al. (2018). Fig. 4 presents a summary of meta-heuristics algorithms classification/taxonomy. Furthermore, it is worthy to note that metaheuristic algorithms can be conveniently implemented for global optimization (BoussaiD et al., 2013).

4.2. Metaheuristic algorithms for PID tuning

Though analytical classical PID tuning techniques are extensively used, occasionally, their performance does not match to the desired results (Ribeiro et al., 2017; Reis et al., 2016). When an optimization approach is used to tune controllers, it is feasible to achieve controller gains to reach as quick as possible responses with reduced overshoot, resulting in errors in the steady-state regime that are close to zero (Ribeiro et al., 2017). It is legal to say that these algorithms are heuristic techniques, which means that they produce acceptable results even if they are not the best. Many controller gains can be discovered in the same system by simply restarting the study. According to recent literature, heuristic algorithm-based optimization strategies have emerged as a potent tool for solving a variety of control engineering difficulties (Latha et al., 2013). Metaheuristic algorithms are extensively utilized in process control due to their architectural simplicity, good optimization capability, and fast response. When compared to traditional optimization approaches, metaheuristic algorithms are more efficient in solving higher-dimensional optimization problems. They can easily adapt to existing classical controller

表1 先前相关综述 的 摘要。

S/No	参考文献	无 研究 引用	综述 重点	已审查 算法 数量	分类法 (分类)
1	Bhawana和Neelu (2017)	33	经典 和 优 化 技术 用于 r PID 整定	9	Yes
2	Gholap等人 (2017)	10	PID 控制器 整定 参数、时域 分析、迭代 次数 以及 成本 函数 值 使 用 GA、PSO 和 SA。	3	No
3	Srinivas 等人 (2014)	16	最优 整定 参数 的 PID 控制器 通过 六种 整定 方法 进行 估计 (Z -N m 方法, C -C m 方法, ISE, IA E, IATE 和 遗传算法 基于 的 PID 方法 用于 三个 ta nk level 过程 f or us ing Matlab/S imulink	8	
4	Emmanuel 和 Inyiama (2015)	32	PID 控制 器 整定 m 方法 如 手动 l, Ziegler Nichol` ft- 件 工具 l, fuzzy logic, 遗传 ic 算法 (GA), 以及 人工 神经 l 网络 (ANN)		
5	Fister 等人 (2016)	32	PID 控制器 整定 方法 使用 差分进化、遗传算法、蝙蝠算法、混合蝙蝠算法、粒子群优化 和 布谷鸟 搜索	6	Yes
6	Dixit 和 Pragati Joshi (2016)	8	PID 控制 器 调整 用于 自动 电压 调节器 使用 p SO 和 GA	2	No
7	Kim和Cho (2006年)	29	PID 控制器 的一个 A VR. usi ng GA, PSO, GA-BF, GA-P SO	4	No
8	Nagaraj等人 (2008年)	9	PID 控制器 是 整定 用于 控制 电枢 控制 的 直流 电机 使用 Z-N (阶跃 响应)、 Kappa-Tau 连续 循环、进化规划、遗传算法 和 粒子群优化。	6	No
9	祖鲁和约翰 (2014)	34	PID 控制器 l tuni ng 用于 四旋翼飞行器 使用 线性 二次型 调节器 (LQR) , 滑模 、 反步、反馈 线性 rization, 自适应、鲁棒、最优、 L1, H ∞ , 模糊 逻辑 和 人工 神经 l n 络。	12	Yes
10	Divya和Nirmalkumar (2018)	13	软 计算 技术 用于 P ID tuni ng Neural Networ k s (NN), 进化型 程 序设计 (EP), 遗传 算 l 法 hm (GA), 粒 子 群 优化 (P SO), 蚁 群 优化 (A CO).	5	No
11	Bharat等人 (2019)	15	使用 模糊 逻辑 (FL)、遗传 算法 (GA) 进行 PID 整定 的技术	2	No
12	Bansal等人 (2012)	64	回顾 PID 参数 整定 技术 的 分类 的 整定 策略 使用 14 种 方法 (Z -N m 方法, C -C 方法, AIS, ACO, BFO, DE, EP, ANN, SA, SVM, PSO, 模糊逻辑 和 IS M 基于 PID m 方法)	14	No
13	Kushwah和Patra (2014)	43	速度 控制 o f DC 电机 使用 f or PID 控制 器 使用 三 h 种 tuning 整定策略: 遗传算法、模糊 SO, 拟退火 和 模糊 逻辑算法	14	No
14	Ghosal 等 (2012)	51	群智能 计算 用于 PID 控制器 采用 四种 整定 策略 : 齐格勒-尼科尔斯 方法 、 BFO 、 粒子群优化和遗传算法	4	No

该技术能够整定设定点和干扰。有关CHR的详细信息和应用可在Sen等 (2015)中找到。几种整定策略的比较可见于Shahrokhi和 Zomorrodi(2012)。

4. 元启发式 算法

基于元启发式的策略 (Talbi, 2009年) 通过在合理时间内提供接近最优的解决方案来处理问题。由于其解决庞大而复杂问题的效率和有效性，元启发式近年来广受欢迎。元启发式是一种比简单启发式表现更好的启发式算法。“元”意味着“超越”或“更高层次”。每种元启发式算法都采用局部搜索和全局搜索的特定组合。不同的结果通常通过随机化实现。无论元启发式是否被接受，文献中并没有关于启发式和元启发式的标准或可验证的定义。许多学者和研究人员互换使用启发式和元启发式这两个术语。然而，当前的发展似乎将所有使用随机化和全局探索的随机算法视为元启发式。本次评估也将采用相同的惯例。

4.1. 元启发式 算法 的 分类

本节简要探索并总结了元启发式算法及其出现的全面回顾。元启发式算法可以根据区分它们的选定特征或特性以多种形式进行分类。不同的作者和学者根据不同的特征或特性对元启发式算法进行分类。分类也可能取决于或集中于研究的目标或目的。

该类别的示例可见于Blum和Roli(2003), 他们将元启发式分为两类：自然启发型和非自然启发型。无论是群体搜索还是单点搜索，目标函数是动态的还是静态的，使用内存还是无记忆方法算法，以及一种或多种邻域结构，

Mirjalili和Lewis在Mirjalili和Lewis(2016)中将元启发式算法分类为：进化型、物理型、群体型和人类类型算法。基于群体的进一步分类见于Beheshti和Shamsuddin(2013); Nojhan(2007)。另一种分类可见于Jackson等(2018)，元启发式可分为两种类型：单解问题 and 多解问题。

元启发式算法的其他几种分类可见于Kumar and Bawa (2018); Shaheen等 (2018); DrEO等 (2007); Beheshti和Shamsuddin(2013); Gogna and Tayal (2013); Hussien等 (2020); Kumar and Bawa (2020); Abdel-Basset等 (2018)。图4总结了元启发式算法的分类/分类法。此外，值得注意的是，元启发式算法可以方便地用于全局优化 (BoussaiD等, 2013) 。

4.2. 元启发式 算法 用于 PID 整定

尽管解析经典PID整定技术被广泛使用，但有时其性能并不符合预期结果 (Ribeiro等, 2017; Reis等, 2016)。当使用优化方法来整定控制器时，可以实现控制器增益，以尽快获得响应并减少超调，从而使稳态误差接近于零 (Ribeiro等, 2017)。可以合理地说，这些算法是启发式技术，这意味着即使它们不是最优的，也能产生可接受的结果。只需重新启动研究，就可以在同一个系统中发现许多控制器增益。

根据近期文献，基于启发式算法的优化策略已成为解决控制工程中各种难题的有力工具 (Latha等, 2013)。元启发式算法因其架构简单、优化能力好和快速响应而在过程控制中得到广泛应用。与传统优化方法相比，元启发式算法在求解高维优化问题时效率更高。它们能轻松适应现有的经典控制器

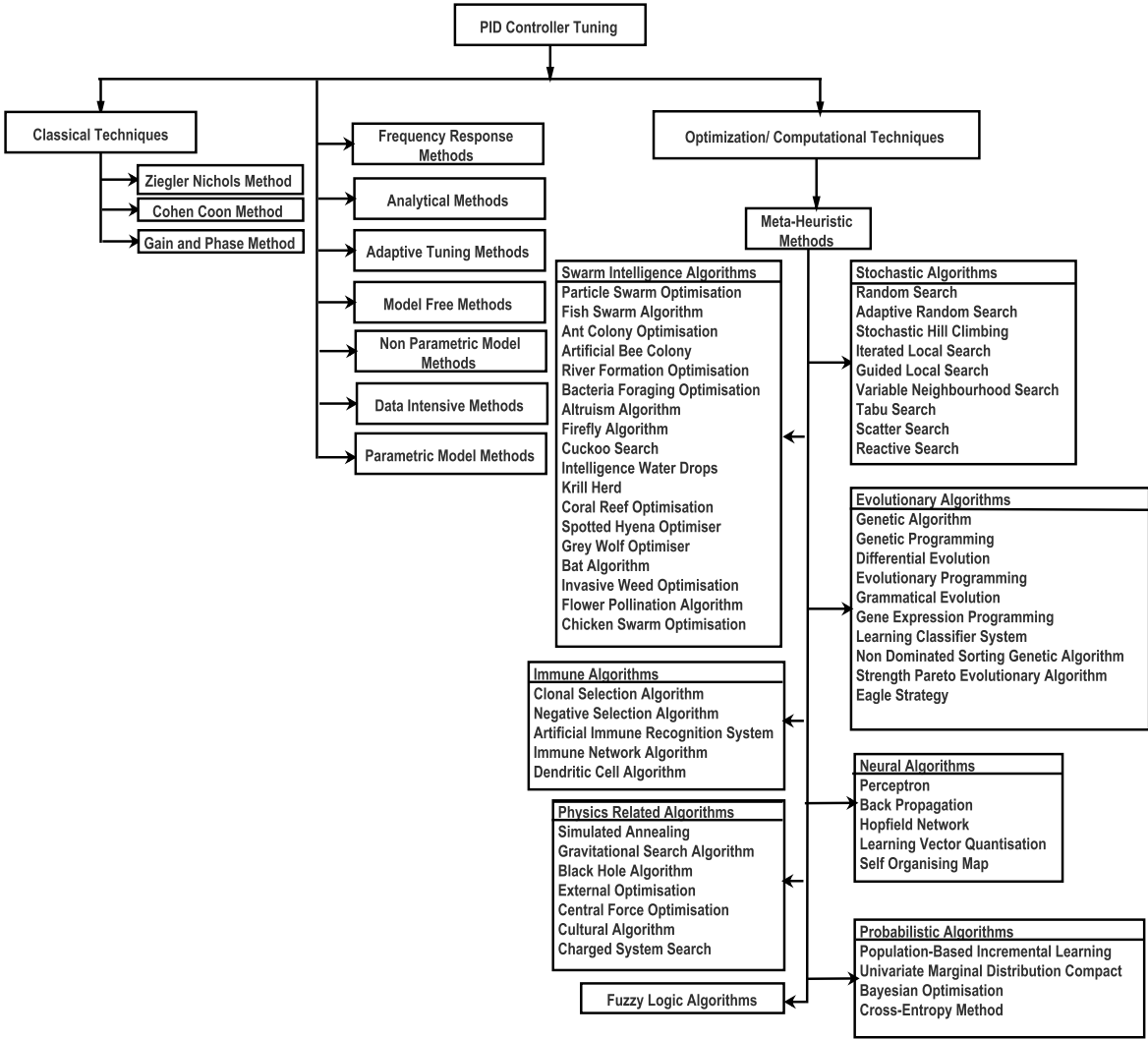


Fig. 3. Summary of PID tuning methods.

design methodologies because of their resiliency. Metaheuristic algorithms, regardless of model order, can be used as a basic mechanism for creating traditional and improved structured controllers for a group of unstable operational models.

In order to produce effective engineering applications, controllers are mechanisms that are utilized to govern the performance of dynamic systems. The main purpose of controllers is to ensure that the feedback of a dynamic system is balanced. Despite the fact that controllers are designed to create stability in principle, the way the system stabilizes in practice is determined by a collection of variables. These variables can be adjusted to satisfy a variety of performance requirements, such as shortening the stabilization time (Lequin et al., 1999), permitting recovery of stability under unpredictability or perturbation (Freeman et al., 1998), smoothing transition to stability (Hang et al., 1991).

Controller tuning, or the precise synchronization of controller variables, is a difficult process that is highly reliant on application needs. The common controller tuning challenge entails identifying a collection of controller variables that stabilizes dynamic system reactivity under a defined performance condition that meets the specified parameters (performance criteria). Engineering applications are becoming increasingly difficult with time, in the sense that they demand dynamic systems to meet a variety of competing needs (i.e., one requirement cannot be satisfied completely devoid of harming the achievement of others). As a result, fine-tuning controllers for complex and dynamic systems be-

come a challenging effort or requirement. A GA for controller tuning involves:

- Evaluate plant characteristics:** This is done to obtain a model for simulation, such as the number of inputs and outputs, nature of performance (linear or nonlinear), and appropriate sort of job it accomplishes.
- Choose controller structure:** It is feasible to select an optimal controller configuration that successfully balances plant operation based on the derived plant model.
- Identify tunable controller variables:** It has a collection of variables that directly compromise the controller process and plant reaction, regardless of the specified controller configuration. The designer must determine which of these variables are used for tuning and which are set to a fixed value.
- Establish needed performance goals:** Designers must describe conflicting performance settings or criteria for controller tuning based on application requirements. These criteria are concerned with the minimization or maximization of quantitative indicators connected to system responsiveness, which officially define a multi-objective optimization issue.
- Analyze various sets of controller variables:** Due to the difficulties of obtaining analytical answers for engineering optimization problems like controller tuning, various solutions (candidate group of controller variables) must be recommended and tested in simu-

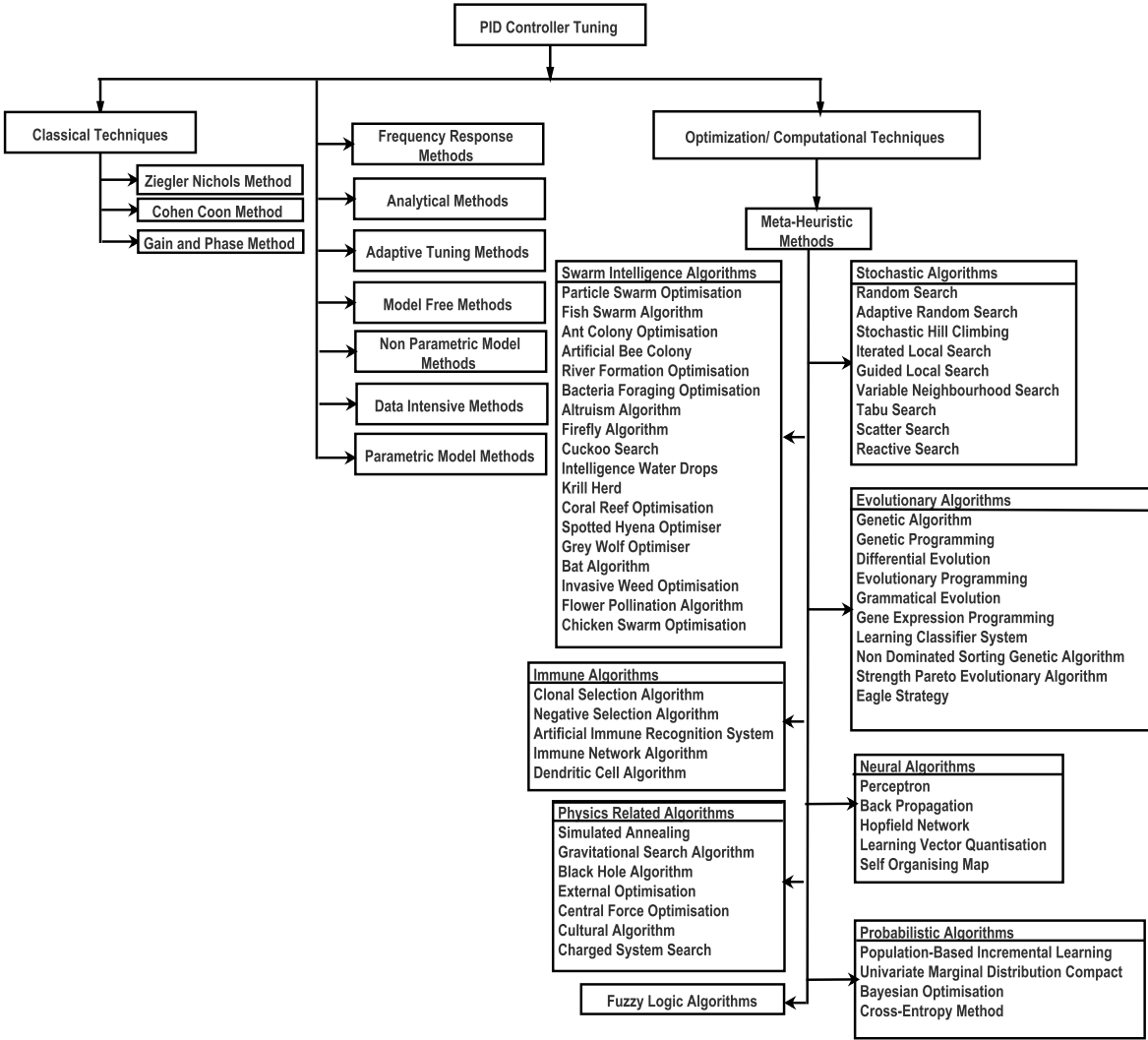


图 3. PID 整定方法总结。

设计方法，因为其弹性。元启发式算法无论模型阶次如何，都可作为基本机制来创建传统和改进结构的控制器，用于一组不稳定操作模型。

为了产生有效的工程应用，控制器是用来管理动态系统性能的机制。控制器的主要目的是确保动态系统的反馈平衡。尽管控制器在原理上旨在创造稳定性，但系统实际稳定化的方式由一组变量决定。这些变量可以调整以满足各种性能要求，例如缩短稳定时间 (Lequin 等人, 1999)、允许在不可预测性或扰动下恢复稳定性 (Freeman 等人, 1998)、平滑过渡到稳定 (Hang 等人, 1991)。

控制器整定，或控制器变量的精确同步，是一个高度依赖于应用需求的困难过程。常见的控制器整定挑战在于识别一组控制器变量，这些变量能在满足指定参数 (性能标准) 的特定性能条件下稳定动态系统的反应性。随着时间的推移，工程应用变得越来越困难，因为它们要求动态系统满足多种相互竞争的需求 (即，一个需求无法在不损害其他需求实现的情况下完全满足)。因此，为复杂和动态系统微调控制器变

得一项具有挑战性的工作或要求。遗传算法用于控制器整定涉及：

- 评估被控对象特性：这是为了获得用于仿真的模型，例如输入和输出的数量、性能性质 (线性或非线性) 以及其完成的适当任务类型。
- 选择控制器结构：基于推导出的被控对象模型，可以选择一种最优的控制器配置，从而成功平衡被控对象的运行。
- 识别可调控制器变量：无论指定的控制器配置如何，它都有一组变量，这些变量直接影响到控制器过程和被控对象响应。设计人员必须确定哪些变量用于整定，哪些被设为固定值。
- 建立所需的性能目标：设计人员必须根据应用需求，描述相互矛盾的性能设置或控制器整定标准。这些标准涉及系统响应性相关的定量指标的最小化或最大化，从而正式定义一个多目标优化问题。
- 分析多组控制器变量：由于在工程优化问题 (如控制器整定) 中难以获得解析解，必须推荐并在仿真中测试各种解决方案 (候选的控制器变量组)。

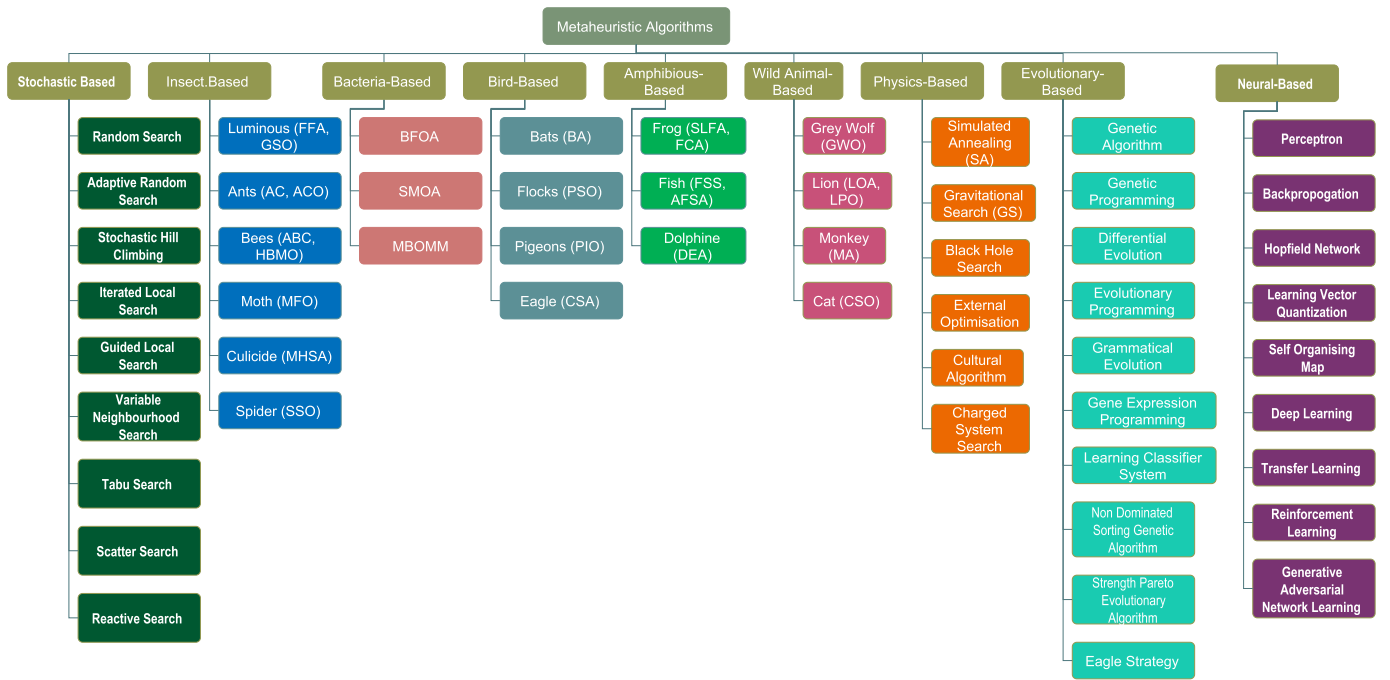


Fig. 4. Summary of metaheuristic classification based on operational procedure.

lation to determine their degree of compliance with the application requirements.

vi. **Choose most appropriate set of controller variables:** A collection of solutions with outstanding (non-dominated) performance trade-offs is discovered using the degree of compliance of each solution. Then, from among the options, one is chosen to be implemented in the final application.

Automatic PID tuning algorithms have received lots of attention because of their importance in the industry. This has been particularly true in the last two decades (Fraga-Gonzalez et al., 2017). Some methods applied include metaheuristic techniques such as GA, PSO and SA, among others. A comprehensive list of some popular metaheuristic algorithms are presented in Tables 2 and 3. In the following paragraphs, an overview of some of these algorithms and applications for PID tuning will be presented.

4.3. Genetic algorithms

Genetic Algorithms are search and optimization approaches motivated by two biological philosophies termed as procedure of natural selection and mechanism of natural genetics (Krishnakumar and Goldberg, 1992). GA (Sreekanth and Hari, 2016; Man et al., 1996) is a stochastic global search approach that mimics the natural evolution process or procedure. GA is a class of probabilistic optimization techniques that uses natural selection and genetic inheritance to solve problems. GA avoids local minima and converging to suboptimal solutions by starting at multiple distinct points and searching in parallel (Wang et al., 2003; Skogestad, 2001). As a result, GA has been shown to be effective in locating high performance space in complicated domains while avoiding the issues associated with high dimensionality that other approaches, such as gradient decent techniques that rely on derivative information, may encounter (Goldenberg, 1989).

The stages of GA implementation are summarized in a flowchart as shown in Fig. 5. Details of GA implementation for PID tuning can be seen in Meena and Devanshu (2017).

4.3.1. Genetic algorithm for PID tuning

Currently, GA has been getting a good deal of attention and further studies have been done to research its implementation. Implementa-

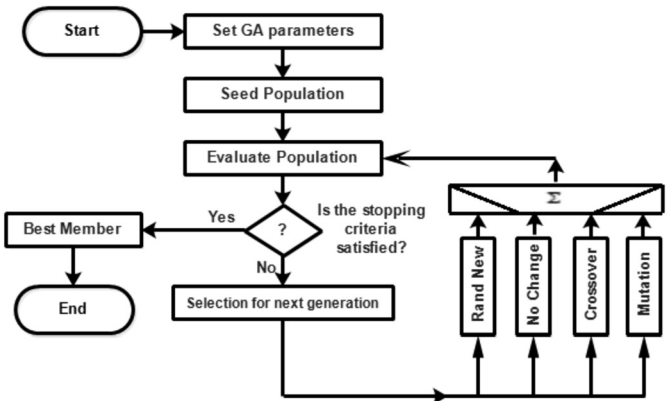


Fig. 5. Genetic algorithm process flowchart.

tion in Control Engineering has advanced greatly. Despite the fact that performance, system stability, static and dynamic index, and system robustness must all be considered while designing a control system, PID parameters have been fine-tuned using GAs for various processes or plants. El-Deen et al. (2015) and Joseph and Emmanuel (2018) discussed the best way to tune PID settings for DC motor speed control. The GA controlled PID controller is designed for synchronous generators to promote damping and maintain power system stability, as presented in Mohammed et al. (2014). GA based design of PID controller for cascade control process is presented in Patil and Lakhekar (2017). An investigation on applicability of genetic algorithms for automatic tuning of PID controller parameters is presented in Amaral et al. (2018).

4.4. Ant colony optimization

The ant colony optimization (ACO) algorithm is a distinct method inspired by insect swarm behavior and was designed for combinatorial issues at first. ACO is a stochastic based metaheuristic method for solving combinatorial optimization issues that employs artificial ants. ACO's goal is to find shorter routes from their nests to food sources. Ants leave behind a chemical compound called a pheromone that allows them to

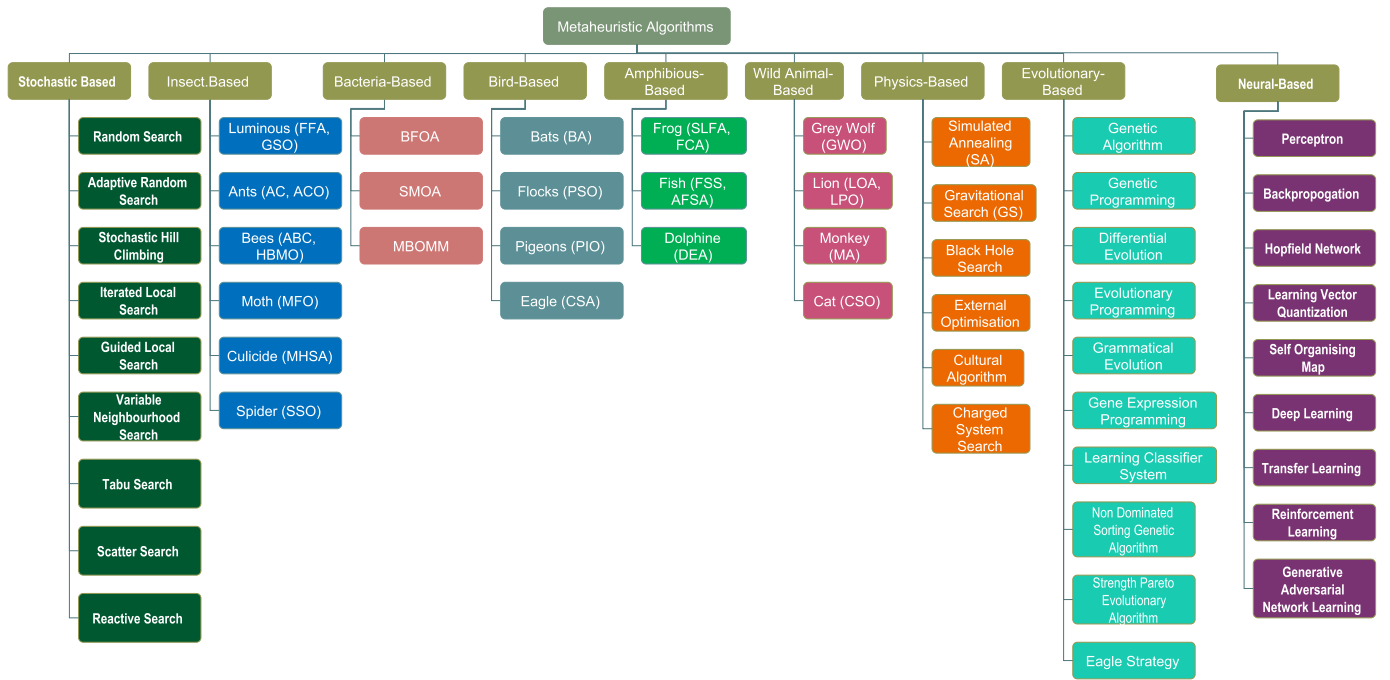


图4. 基于操作流程的元启发式分类总结。

lation 以 确定 其 对 应用需求 的 符合度 程度。

vi. 选择最合适的控制器变量集：根据每个解的符合度，发现一组具有出色（非支配）性能权衡的解。然后，从这些选项中选出一个用于最终应用。

自动PID整定算法因其在工业中的重要性而备受关注。在过去二十年中尤其如此（Fraga-Gonzalez 等人，2017）。应用的一些方法包括元启发式技术，如遗传算法、粒子群优化和模拟退火等。表2和表3列出了部分流行的元启发式算法的全面列表。以下段落将概述其中一些用于PID整定的算法及其应用。

4.3. 遗传 算法

遗传算法是一种搜索和优化方法，受两种生物学原理启发：自然选择过程和自然遗传学机制（Krishnakumar和Goldenberg, 1992）。GA（Sreekanth和Hari, 2016; Man等, 1996）是一种随机全局搜索方法，模仿自然进化过程。GA是一类概率优化技术，利用自然选择和遗传继承来解决问题。GA通过从多个不同点开始并行搜索，避免陷入局部极小值和收敛到次优解（Wang等, 2003; Skogestad, 2001）。因此，GA在复杂领域中定位高性能空间方面表现出色，同时避免了其他方法（如依赖导数信息的梯度下降技术）可能遇到的高维性问题（Goldenberg, 1989）。

GA实施的阶段总结在图5的流程图中。关于GA用于PID整定的详细信息，可参见Meena和Devanshu（2017）。

4.3.1. 遗传 算法 用于 PID 整定

目前，遗传算法已经得到了广泛关注，并且有更多的研究已经完成，以探索其应用。实现

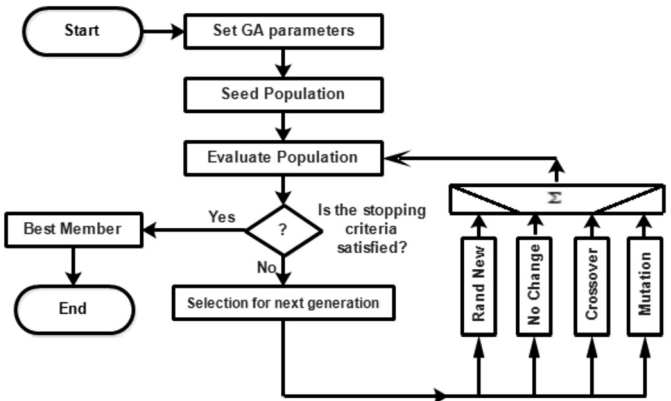


图5. 遗传算法流程图。

在控制工程中的应用已经取得了很大进展。尽管在设计控制系统时必须考虑性能、系统稳定性、静态和动态指标以及系统鲁棒性，但使用遗传算法对PID参数进行了针对各种过程或设备的微调。El-Deen等人（2015）以及Joseph和Emmanuel（2018）讨论了优化直流电机速度控制中PID设置的最佳方法。如Mohammed等人（2014）所述，针对同步发电机设计了由遗传算法控制的PID控制器，以促进阻尼并维持电力系统稳定性。Patil和Lakhekar（2017）提出了基于遗传算法的级联控制过程PID控制器设计。Amaral等人（2018）提出了关于遗传算法在PID控制器参数自动整定中适用性的研究。

4.4. 蚁群 优化

蚁群优化（ACO）算法是一种受昆虫群体行为启发的独特方法，最初设计用于解决组合问题。ACO是一种基于随机性的元启发式方法，用于解决组合优化问题，它采用人工蚂蚁。ACO的目标是从蚁巢到食物源找到更短的路径。蚂蚁会留下一种称为信息素的化学物质，使它们能够

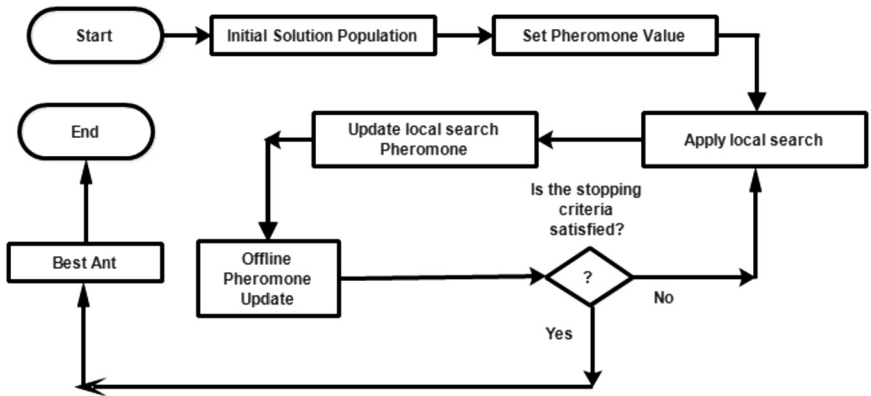


图 6. 蚁群优化搜索算法流程图

communicate with other ants. When an ant moves, it leaves a consistent amount of pheromone behind that other ants can detect. Every ant travels in its own unique way, but when it comes across a pheromone trail, it must decide whether or not to follow it. If one ant follows the trail, its own pheromone strengthens it, and an increase in pheromone enhances the likelihood that the next ant will follow it. As a result, the more ants that go along a path, the more appealing it becomes to succeeding ants. In addition, an ant who takes a shorter route to a food source will return to the nest sooner.

Dorigo suggested the fundamental ant structure, an earlier version of ACO, in his PhD thesis in 1992. (Dorigo, 1992). Dorigo devised a difficult optimization algorithm based on both simple and complex foraging ant features, which sparked widespread interest in the 1990s and beyond. Dorigo and others introduced a number of alterations and hybrids. Offline pheromone updates and pheromone evaporation were demonstrated in particular, which led to more widespread of ACO in 1999 (Cooper and Smeresky, 2020; Dorigo et al., 2006). In Adubi and Misra (2014), an overview of the many types of ACO algorithms, their applications, and a comparative study of a few selected versions are presented. The navigation of ant is represented by the following probability Equation (12) in Sundareswaran and Nayak (2012):

$$P_j^{(k)} = \frac{\tau_j^{(k)}}{\sum_{j=1}^n \tau_j^{(k)}}. \quad (12)$$

A high level algorithmic concept of ACO is presented in Fig. 6.

4.4.1. ACO for PID tuning

The parameters of a PID controller for DC motor of a robotic arm was tuned using ACO in Sandoval et al. (2015). The findings were compared to typical tuning techniques, and ACO-PID performed better than ZN tuned PID. As demonstrated in Priyambodo et al. (2015), a PID controller is utilized for quadrotor stabilization utilizing the ACO method. An application of optimization method based on artificial ants colonies (ACO) for the synthesis of an optimal PID regulator for human heart control is presented in Aabid et al. (2017). Load Frequency Control (LFC) of a single area re-heat thermal power system is one of the other PID tuning applications that uses ACO (Kaliannan et al., 2016), Autonomous Underwater Vehicle (AUV) control system (Herlambang et al., 2019), Single Machine Infinite Bus control (Yunus and Djalal, 2019), position control operation of a DC motor (Şimşek et al., 2019) and multi objective ACO for PID can be found in Chiha et al. (2012).

4.5. Artificial bee colony optimization

Karaboga proposed the Artificial Bee Colony (ABC) metaheuristic optimization technique in 2005. Honey bees' clever foraging habit served as inspiration. The method is based on the model for honey

bee colony foraging behavior presented by Karaboga (2005). Working and unemployed foraging bees, as well as food sources, are the three main components of the concept. The employed and unemployed foraging bees in the first two components look for ample food sources close to their hive, which is the third component. Forager recruitment to rich food sources, which results in positive feedback, and forager desertion of low food sources, which results in negative feedback, are two fundamental behaviors required for self-organization and collective intelligence, according to the concept. A comprehensive discussion and application strategy of ABC can be seen in Karaboga (2010).

The ABC optimization algorithm, like the ant colony optimization algorithm, was created to solve a combinatorial problem in which several solutions can be discretized into an array of building blocks that can be rearranged and mutated to find the best answer. The ABC approach was also utilized to optimize the trajectory (Murrieta-Mendoza et al., 2018), parameter optimization (Zhong et al., 2014; Ding et al., 2015), and remote sensing applications (Jayanth et al., 2015; Fu et al., 2020). Applications, overview and relevant equations can be seen in Xu et al. (2013); Kumar et al. (2017); Abu-Mouti and El-Hawary (2012). An illustration of ABC application flowchart is presented in Fig. 7.

4.5.1. ABC for PID tuning

The ABC method outperformed earlier population-based algorithms in numerical tests, with the extra bonus of fewer control settings (Liao et al., 2014). Apart from the usual population-based control parameters like population size or colony size (SN) and maximum generation number or maximum cycle number, the ABC algorithm's basic version only has one control parameter called limit (MCN). Because of its simplicity and ease of implementation, the ABC technique has sparked a lot of attention, and it has been utilized to solve a variety of real optimization problems. One of these practical challenges is the setting of PID controller parameters.

ABC is used to tune PID settings to perfection. Some benchmark functions are used to test the algorithm. For the computation of PID parameters, the test functions were used. On the basis of overshoot, settling time, and minimal error, the performance is compared to that of other approaches (Abachizadeh et al., 2010). ABC algorithm was proposed in El-Telbany (2013) for tuning the regulation parameters of PID controller for DC motor. The optimum parameters of PID controller were obtained using ABC algorithm for higher order oscillatory systems in Bagis and Senberber (2017). Other plants controlled using PID controllers and tuned by ABC algorithm include: fractional order systems (Senberber and Bagis, 2017), hub motion and end point vibration suppression of double-link flexible robotic manipulator (Annisa et al., 2018), bench scaled nonlinear dynamical system (NLDS) (Valluru and Singh, 2017) and single phase inverter (Zhi, 2019).

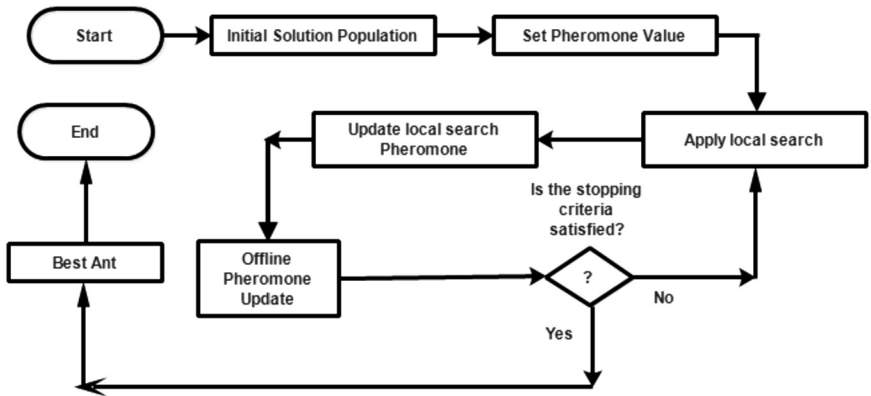


图 6. 蚁群优化搜索算法流程图

与其他蚂蚁通信。当蚂蚁移动时，它会留下恒定量的信息素，其他蚂蚁可以检测到。每只蚂蚁以自己独特的方式行进，但当遇到信息素轨迹时，它必须决定是否跟随。如果一只蚂蚁跟随轨迹，其自身的信息素会加强它，信息素的增加提高了下一只蚂蚁跟随它的可能性。因此，沿着一条路径走的蚂蚁越多，它对后续蚂蚁的吸引力就越大。此外，走更短路径到达食物源的蚂蚁会更早返回蚁巢。

Dorigo 于 1992 年在他的博士论文中提出了基本的蚂蚁结构，即 ACO 的早期版本 (Dorigo, 1992)。Dorigo 基于简单和复杂的觅食蚂蚁特征设计了一种困难的优化算法，这在 20 世纪 90 年代及以后引起了广泛兴趣。Dorigo 等人引入了许多修改和混合算法。特别是，离线信息素更新和信息素蒸发得到了展示，这导致 ACO 在 1999 年更加普及 (Cooper and Smeresky, 2020; Dorigo et al., 2006)。在 Adubi and Misra (2014) 中，介绍了多种 ACO 算法、它们的应用以及对少数选定版本的比较研究。蚂蚁的导航由 Sundareswaran and Nayak (2012) 中的以下概率方程 (12) 表示：

$$P_j^{(k)} = \frac{\tau_j^{(k)}}{\sum_{j=1}^n \tau_j^{(k)}}. \quad (12)$$

高层次 算法 概念 ACO 的 高层次 算法 概念 如图 6 所示。

4.4.1. ACO 用于 PID 整定

Sandoval等人 (2015) 使用ACO对机械臂直流电机的PID控制器参数进行了整定。研究结果与典型调优技术进行了比较，ACO-PID表现优于Z-N整定PID。如Priyambodo等人 (2015) 所示，采用ACO方法的PID控制器用于四旋翼无人机稳定。Aabid等人 (2017) 提出了一种基于人工蚁群 (ACO) 的优化方法，用于合成人类心脏控制的最优PID调节器。单区域再热火力发电系统的负荷频率控制 (LFC) 是另一个使用ACO的PID整定应用 (Kaliannan等人, 2016)，还有自主水下航行器 (AUV) 控制系统 (Herlambang等人, 2019)、单机无穷大母线控制 (Yunus and Djalal, 2019)、直流电机位置控制操作 (Simsek等人, 2019)，以及Chiha等人 (2012) 提出的多目标ACO用于PID。

4.5. 人工蜂群 优化

Karaboga 在 2005 年 提出了 人工蜂群 算法 (ABC) 元启发式 优化 技术。蜜蜂 的 聪明 觅食 习性 是 其 灵感 来源。该 方法 基于 蜜蜂

蜂群觅食行为模型，由Karaboga (2005年) 提出。雇佣蜂、非雇佣蜂以及食物源是该概念的三个主要组成部分。前两个组成部分中的雇佣蜂和非雇佣蜂寻找靠近蜂巢的充足食物源，蜂巢是第三个组成部分。根据该概念，招募觅食者到丰富食物源 (产生正反馈) 和放弃低质量食物源 (产生负反馈) 是自组织和集体智慧所需的两个基本行为。关于ABC的全面讨论和应用策略可参见Karaboga (2010年)。

人工蜂群优化算法，如同蚁群优化算法，最初是为了解决组合优化问题而创建的，在该问题中，多个解可以被离散化为一个可重新排列和突变的块状数组，以找到最佳答案。人工蜂群方法也被用于优化轨迹 (Murrieta-Mendoza等, 2018)、参数优化 (Zhong等, 2014; Ding等, 2015) 以及遥感应用 (Jayanth等, 2015; Fu等, 2020)。应用、概述及相关方程可参见Xu等 (2013)；Kumar等 (2017)；Abu-Mouti和El-Hawary (2012)。人工蜂群算法应用流程图的图示见图7。

4.5.1. 人工蜂群算法 用于 PID 整定

人工蜂群方法在数值测试中优于先前的基于种群的算法，且额外优势是控制设置更少 (Liao等, 2014)。除了常规的基于种群的控制参数 (如种群大小或蜂群大小 (SN) 以及最大代数或最大循环次数) 外，人工蜂群算法的基本版本只有一个称为限制参数的控制参数 (MCN)。由于其简单性和便利性，人工蜂群技术引起了广泛关注，并已被用于解决多种实际优化问题。其中一个实际挑战便是PID控制器参数的整定。

人工蜂群算法用于将PID参数整定至完美。使用一些基准函数来测试该算法。对于PID参数的计算，使用了测试函数。基于超调、调节时间和最小误差，将其性能与其他方法进行比较 (Abachizadeh等人, 2010)。El-Telbany (2013) 提出了人工蜂群算法用于整定直流电机PID控制器的调节参数。Bagis和Senberber (2017) 使用人工蜂群算法获得了高阶振荡系统的PID控制器最优参数。其他使用PID控制器并通过人工蜂群算法整定的受控对象包括：分数阶系统 (Senberber和Bagis, 2017)、双连杆柔性机器人机械臂的轮毂运动和端点振动抑制 (Annisa等人, 2018)、台架缩比非线性动力学系统 (NLDS) (Valluru和Singh, 2017) 以及单相逆变器 (Zhi, 2019)。

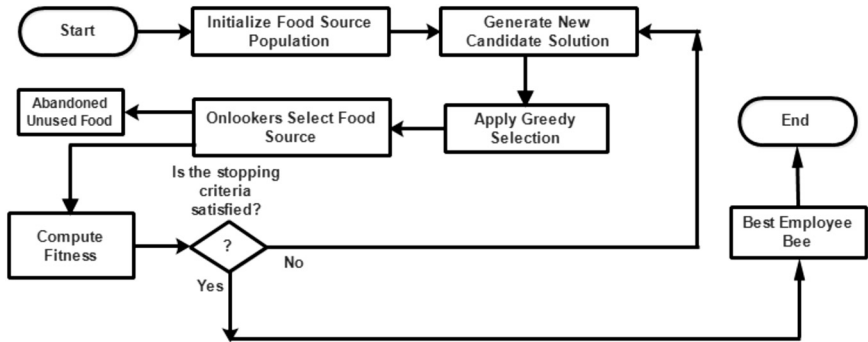


Fig. 7. Phases of artificial bee colony algorithm flowchart.

4.6. Teaching learning based optimization

This technique was suggested by Rao et al. (2012) and Rao and Kalyankar (2013) to find the worldwide solution to mechanical design difficulties using large-scale non-linear optimization problems TLBO (Teaching-Learning-Based Optimization) is a population-based meta-heuristic search algorithm that mimics a typical class’s teaching and learning methods Rao et al. (2012). A group of students is called a solution in TLBO, and the fitness of solutions is called results or grades. The TLBO algorithm adjusts each learner’s or student’s grade in class by learning from the teacher and learning from student engagement.

The algorithm procedure is divided into two stages or actions: the instructor stage and the learner stage. In the teacher stage, the instructor is seen as the best solution in the entire group, and the teacher shares his or her knowledge with the students in order to improve the class’s mean outcome. The equations and details of implementation can be seen in Rao et al. (2012). The application flowchart for TLBO is presented in Fig. 8.

4.6.1. TLBO for PID tuning

TLBO has gained a lot of attention, and has been adjusted to capture limited data due to its appealing properties such as simple concept, lack of exact algorithm variables, simple implementation, and fast convergence (Yu et al., 2016b), multiobjective (Patel and Savsani, 2016; Rao and Patel, 2014), large-scale (Biswas et al., 2012), and dynamic optimization problems (Chen et al., 2018a). TLBO has also been effectively applied to a variety of scientific and engineering sectors (Chen et al., 2018b).

While in the case of PID parameter tuning, TLBO has been employed to adjust the variables of PID for different processes. Some of these include: automatic voltage regulator (AVR) processes, Automatic Generation Control (AGC) in power systems (Sahu et al., 2016), automatic voltage regulator (AVR) (Chatterjee and Mukherjee, 2016; Priyambada et al., 2014), improve performance of the magnetic levitation system (Yadav et al., 2018).

With the help of TLBO, a fuzzy PID controller is proposed for automatic generation control (AGC) of a thermal system (Sahu et al., 2015). There are two degrees of freedom: one and two. Using a teaching learning based approach, a PID controller design is used on an AVR device (Rajinikanth and Satapathy, 2015). The fuzzy-PID controller was proposed by Sahu et al. (2015) for the AGC of a two unequal area interconnected thermal system. For the first time in this sector, the TLBO technique is employed to obtain parameters for the proposed fuzzy-PID controller. The design challenge is formulated as an optimization problem, and the fuzzy-PID controller’s parameters are optimized via TLBO. When the proposed method’s results are compared to those of several recently published approaches, such as the Lozi Map-based Chaotic Optimization Algorithm (LCOA), GA, Pattern Search (PS), and Simulated Algorithm (SA) based PID controller, the proposed method clearly wins for the same system and objective function. The TLBO optimized fuzzy-PID controller has superior dynamic performance in terms of settling

time, frequency overshoot and undershoot, and tie-line power deviation when compared to LCOA, GA, PS, and SA based PID controllers.

4.7. Bat algorithm (BA)

The Bat Algorithm (BA) is a metaheuristic algorithm derived from natural bats’ sound waves activity to find prey and differentiate between various kinds of insects even in dark environment (Yang, 2010). BA was recently introduced by Yang (2010), which was influenced by the sonar of microbats. Each bat in the original population updates its position in a correlated fashion using echolocation. Bat echolocation uses a sequence of powerful ultra-sound pulses to generate echoes, which are then received with latencies and different sound frequencies, qualifying bats to identify the exact prey. In natural world, echolocation can last only a few thousandths of a second (up to around 8–10 ms) and also have a varying amplitude in the range of 25–150 kHz, which corresponds to air wavelengths of 2–14 mm (Gandomi and Yang, 2014).

Microbats typically use sound waves to locate prey. Microbats transmit small signals when traveling; nonetheless, when a possible prey is nearby, their transmitted pulse rates increase and the frequency is tuned up. The addition of frequency, also known as frequency tuning, in combination with the speed of pulse emission will reduce the wavelength of echolocations and hence improve detection precision (Gandomi and Yang, 2014). An application illustration flowchart for BA is presented in Fig. 9.

Further details, readings or consultation on BAs can be seen in Yang (2012a); Chaib et al. (2017); Gandomi and Yang (2014); Yang and Gandomi (2012); Singh et al. (2015); Al-Betar et al. (2018).

4.7.1. BAT for PID tuning

BA was used to solve realistic real-world problems because it is a suitable metaheuristic algorithm with unique ability to handle hard optimization problems. A few of these classifications are presented in Jayabarathi et al. (2018): structural Optimization, classification and feature selection, electrical power systems, aerospace engineering, petroleum engineering, and nuclear engineering. Furthermore, BA has been implemented in control engineering where PID tuning is an aspect.

The BA for optimizing PID parameters has been used in a variety of process plants with encouraging results. In Katal et al. (2014), BA was utilized to fine-tune the PID controller parameters for a connected tank liquid level control system, which is widely used in the mineral oil, food manufacturing, and water purification companies. BA in Kotteeswaran and Sivakumar (2013) was used to fine-tune the settings of a centrally controlled PI controller for a coal gas turbine, which is a non-linear multidimensional process with complex relationships across control loops. Singh et al. (2015) used BA for tuning the parameters of PID controller of a servo motor. Other implementations of BA for PID tuning include, speed control of brushless direct current drive (Premkumar and Manikandan, 2016), bio-inspired robot manipulator (Rahmani et al., 2016) and control of microelectromechanical systems (MEMS) gyroscopes (Rahmani et al., 2018).

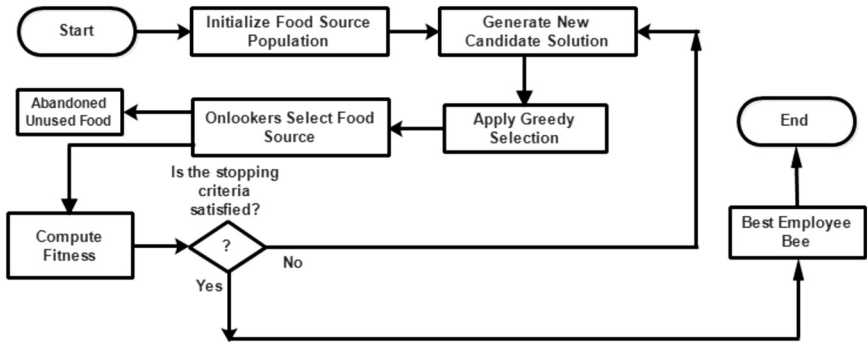


图 7. 人工蜂群算法流程图阶段。

4.6. 教学基优化

该技术由Rao等人 (2012) 以及Rao和Kalyankar (2013) 提出, 用于通过大规模非线性优化问题寻找机械设计难题的全局解。TLBO (教与学优化) 是一种基于群体的元启发式搜索算法, 它模仿了典型班级的教学和学习方法 (Rao等人, 2012)。在TLBO中, 一组学生被称为一个解, 而解的适应度被称为结果或成绩。TLBO算法通过向教师学习和通过学生互动学习来调整每个学习者或学生在班级中的成绩。

该算法过程分为两个阶段或操作: 教师阶段和学习者阶段。在教师阶段, 教师被视为整个群体中的最佳解决方案, 教师与学生分享知识以提高班级的平均成绩。方程和实施细节可参见 Rao 等人 (2012)。TLBO 的应用流程图如图 8 所示。

4.6.1. TLBO 用于 PID 整定

TLBO 因其简单的概念、无需精确算法变量、易于实现以及快速收敛等吸引人的特性, 已获得广泛关注, 并被调整用于捕捉有限数据 (Yu 等, 2016b), 以及多目标 (Patel 和 Savsani, 2016; Rao 和 Patel, 2014)、大规模 (Biswas 等, 2012) 和动态优化问题 (Chen 等, 2018a)。TLBO 也已有效应用于各种科学和工程领域 (Chen 等, 2018b)。

在 PID 参数整定方面, TLBO 已被用于调整不同过程的 PID 变量。其中一些包括: 自动电压调节器 (AVR) 过程、电力系统中的自动发电控制 (AGC) (Sahu 等, 2016)、自动电压调节器 (AVR) (Chatterjee 和 Mukherjee, 2016; Priyambada 等, 2014), 以及提高磁悬浮系统的性能 (Yadav 等, 2018)。

借助TLBO, 提出了一种用于热力系统自动发电控制(AGC)的模糊PID控制器(Sahu等人, 2015)。存在两个自由度: 一个和两个。使用基于教与学的方法, 在AVR设备上设计了PID控制器(Rajinikanth和Satapathy, 2015)。Sahu等人(2015)针对两个不等面积互联热力系统的AGC提出了模糊PID控制器。在该领域首次采用TLBO技术来获取所提模糊PID控制器的参数。设计挑战被表述为优化问题, 并通过TLBO优化模糊PID控制器的参数。当将所提方法的结果与几种最新发表的方法(例如基于Lozi映射的混沌优化算法(LCOA)、遗传算法(GA)、模式搜索(PS)和模拟算法(SA)的PID控制器)进行比较时, 所提方法在相同系统和目标函数下明显胜出。TLBO优化的模糊PID控制器在调节

时间、频率超调和下冲、以及联络线功率偏差方面与基于LCOA、GA、PS、和SA的PID控制器相比具有更优的动态性能。

4.7. 蝙蝠算法 (BA)

蝙蝠算法 (BA) 是一种元启发式算法, 源自自然蝙蝠利用声波活动在黑暗环境中寻找猎物并区分不同种类昆虫的原理 (Yang, 2010)。BA 由 Yang (2010) 最近提出, 受到微型蝙蝠声纳的启发。初始群体中的每只蝙蝠利用回声定位以相关的方式更新其位置。蝙蝠回声定位使用一系列强大的超声波脉冲产生回声, 然后通过延迟和不同的声音频率接收, 使蝙蝠能够识别精确的猎物。在自然世界中, 回声定位仅持续千分之几秒 (最多约8-10毫秒), 并且振幅在25-150 kHz范围内变化, 对应空气波长为2-14毫米 (Gandomi 和 Yang, 2014)。

微型蝙蝠通常使用声波定位猎物。微型蝙蝠在移动时发射小信号; 然而, 当可能的猎物靠近时, 其发射的脉冲率增加, 频率调高。频率的增加, 也称为频率调谐, 结合脉冲发射速度, 会减少回声定位的波长, 从而提高检测精度 (Gandomi 和 Yang, 2014)。BA 的应用流程图如图9所示。

关于蝙蝠算法 (BA) 的更多细节、阅读或咨询, 可参见Yang (2012a); Chaib et al. (2017); Gandomi and Yang (2014); Yang和Gandomi (2012); Singh et al. (2015); Al-Betar et al. (2018)。

4.7.1. 蝙蝠算法 用于 PID 整定

BA被用于解决现实世界中的实际问题, 因为它是一种合适的元启发式算法, 具有处理困难优化问题的独特能力。以下分类来自 Jayabarathi et al. (2018): 结构优化、分类与特征选择、电力系统、航空航天工程、石油工程和核工程。此外, BA已在控制工程中得到应用, 其中PID整定是一个方面。

BA用于优化PID参数已在多种过程工厂中得到应用, 并取得了令人鼓舞的结果。在Katal等人(2014)的研究中, BA被用于精细调整连通罐液位控制系统的PID控制器参数, 该系统广泛应用于矿物油、食品制造和水净化公司。在Kot-teeswaran和Sivakumar(2013)的研究中, BA被用于精细调整燃气轮机的集中控制PI控制器的设置, 这是一个具有跨控制回路复杂关系的非线性多维过程。Singh等人(2015)使用BA调整伺服电机的PID控制器参数。BA用于PID整定的其他应用包括: 无刷直流驱动器的速度控制(Premku-mar和Manikandan, 2016)、仿生机器人操作臂 (Rahmani等人, 2016)以及微机电系统陀螺仪的控制(Rahmani等人, 2018)。

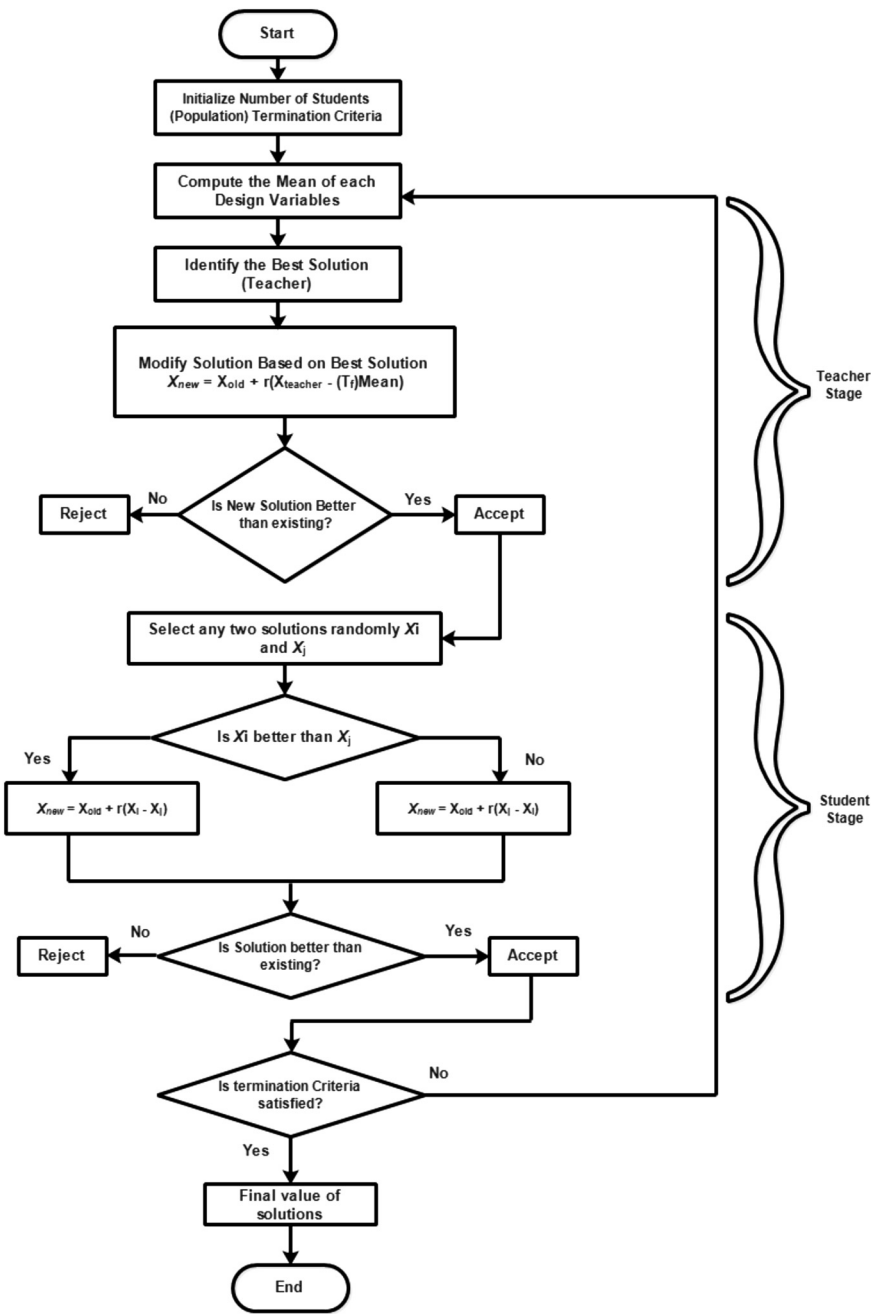


Fig. 8. Descriptive flow chart for Teaching-Learning-Based Optimization (TLBO).

4.8. Bacterial foraging optimization (BFO)

Passino (2002) presented the Bacteria Foraging Optimization (BFO) algorithm, which is a member of the family of nature-inspired optimization algorithms. BFO imitates a swarm of Escherichia coli bacteria's searching strategy. Bacteria look for nourishment in a way that maximizes the amount of energy they get per unit of time. By sending signals, each bacterium communicates with each other. After comparing various prior factors, a bacterium makes foraging decisions. Chemotaxis is the mechanism by which a bacterium moves by taking little steps while looking for nutrition, and a key principle of BFO is to simulate chemotactic motion of artificial bacteria in the problem search space. Specifics on these processes, as well as the BFO implementation steps, can be found in Das et al. (2009); Datta and Nandakumar (2017); Jhankal and Adhyaru (2011). A flowchart showing the flow of BFO algorithm is depicted in Fig. 10. The BFO variables to be defined for

applications include: number of bacteria S , swimming length N_n , number of chemotaxis steps N_c , number of reproduction steps N_r , number of elimination and dispersal events N_e , and the probability of elimination/dispersal P_{ed} . Step size of swimming C for bacteria is a vector $M \times 1$ corresponding to vector θ . Initial values of $\theta(i), i = 1, 2, \dots, S$ which are boundaries of parameters randomly created. The functions $J(i)$ and $F(i)$ are defined to evaluate cost and fitness of i^{th} bacterium respectively. Details of application flow diagram can be seen in Huynh et al. (2020). For further readings, variants and applications of BFO can be seen in Agrawal et al. (2012); Niu et al. (2010); Tang and Wu (2009); Rani and Kumar (2015). BFO has piqued the interest of scholars from several disciplines of study since its debut, owing to its biological purpose and beautiful shape. Researchers are attempting to combine BFO with a variety of other algorithms in order to investigate both its local and global search

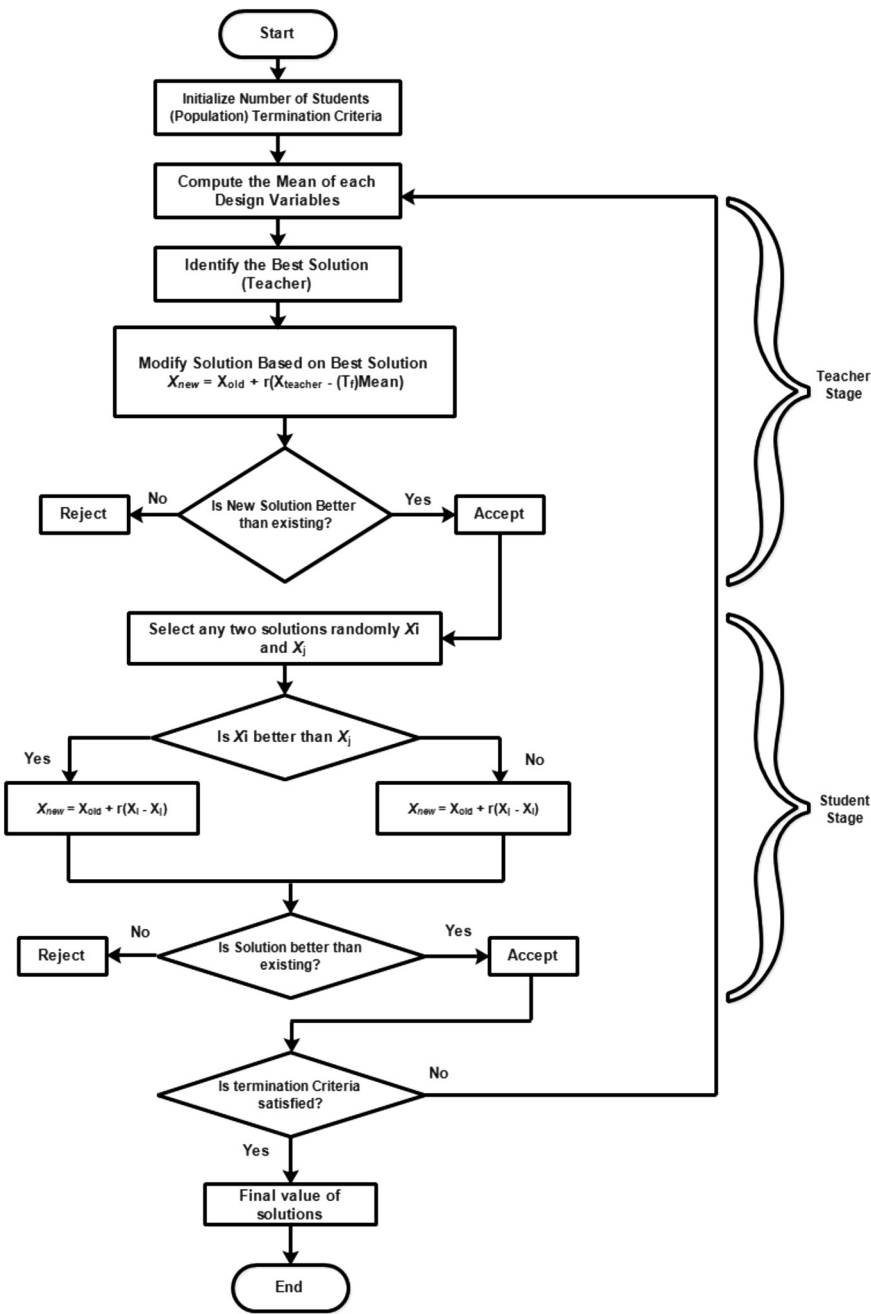


图8. 教与学优化 (TLBO) 的描述性流程图。

4.8. 细菌觅食优化 (BFO)

Passino (2002)提出了细菌觅食优化(BFO)算法, 该算法属于自然启发型优化算法家族。BFO模仿大肠杆菌群体的搜索策略。细菌以最大化单位时间所获能量的方式寻找营养。通过发送信号, 每个细菌与其他细菌进行通信。在比较各种先前的因素后, 细菌做出觅食决策。趋化性是细菌通过小步移动寻找营养的机制, BFO的一个关键原理是在问题搜索空间中模拟人工细菌的趋化运动。关于这些过程的具体内容以及 BFO 实现步骤, 可在 Das 等 (2009年)、Datta 和 Nandakumar (2017)、Jhankal 和 Adhyaru (2011) 中找到。图 10 展示了 BFO 算法的流程图。BFO 算法需要定义的变量包括:

细菌数量 S 、游泳长度 N_n 、趋化性步骤数量 N_c 、繁殖步骤数量 N_r 、消除和扩散事件数量 N_e , 以及消除/扩散概率 P_{ed} 。细菌的游泳步长 C 是一个向量 $M \times 1$, 对应于向量 θ 。 $\theta(i), i = 1, 2, \dots, S$ 的初始值是随机创建的参数边界。函数 $J(i)$ 和 $F(i)$ 分别用于评估第 i^{th} 个细菌的成本和适应度。应用流程图的详细信息可参见 Huynh 等 (2020)。关于BFO的进一步阅读、变体及应用可参见Agrawal等 (2012年); Niu等 (2010年); Tang和Wu (2009年); Rani和Kumar (2015年)。BFO自问世以来, 因其生物学用途和优美的形态引起了多个学科领域学者的兴趣。研究人员正试图将BFO与多种其他算法相结合, 以探究其局部和全局搜索

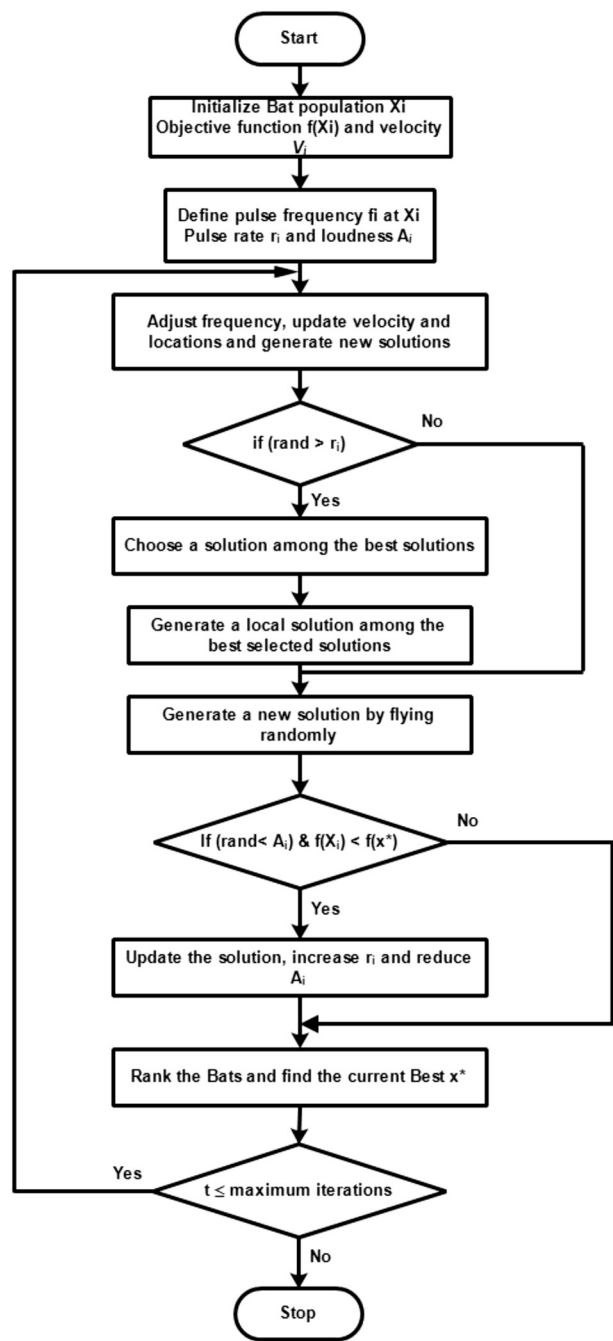


Fig. 9. Generic flowchart for Bat algorithm.

features. It has previously been used to solve a variety of real-world problems, and has been found to be effective over a wide range of GA and PSO types (Jhankal and Adhyaru, 2011).

4.8.1. BFO for PID tuning

The BFO algorithm is a new swarm intelligence system inspired by the foraging behavior of Escherichia coli (E.coli) in human intestines. After significant research over the previous few years, BFO has demonstrated good performance in a range of application domains (Niu et al., 2010). Lately, BFO has gotten a lot of interest, and it offers a lot of promising industrial applications and information processing. Few models that simulate bacterial foraging behavior have been devised and used to solve actual challenges (Bayoumi and Salem, 2012). The BFO method, like other SI algorithms, has been used to solve problems in a wide range of fields. Communication systems, communications engi-

neering, intelligent control engineering systems (Niu et al., 2010) are some of the application sectors in which PID is used.

BFO has been employed for suitable tuning of PID controller settings to overcome difficulties that numerous industrial machinery are regularly troubled with, such as high order and uncertainties. The BF-PID controller was optimized and tuned using BFO in Niu et al. (2006). BF-PID has a faster settling time, little or no overshoot, and greater resilience than GA-PID. Compared with GA-PID, BF-PID have a quicker settling time, little or without overshoot and higher robustness. Similarly, dos Santos Coelho and da Costa Silveira (2006) provided an improved BFO for PID optimization in a robotic manipulator system. To adjust the C constant in basic BFO, the new technique was tested with a chemotactic step parameter (C) variable speed based on Uniform, Gaussian, and Cauchy probability distributions. The BFO approach was used in Ali and Majhi (2006) to improve PID parameters by reducing the integral of time error squared (ISTE) condition. Also, addressing the challenge of PID controller tuning, Korani et al. (2009) coupled bacteria foraging actions with PSO. Rajinikanth and Latha have suggested utilizing BFO to tune PID controller parameters for unstable process models in Rajinikanth and Latha (2012). The results demonstrate that utilizing the bacterium foraging algorithm to tune PID parameters is a viable option.

Monitoring a moving target along an unpredictable route by a mobile robot is another application of BFO for tuning PID controllers (Benbouabdallah and Zhu, 2013; Huynh et al., 2020), speed controller of a permanent magnet synchronous motor (Agarwal et al., 2017), position control of Direct Current (DC) servo motor (Jasim, 2018) and tuning PID parameters for Multi-variable Input and Multi-variable Output (MIMO) systems (Sivakumar et al., 2016).

4.9. Particle swarm optimization (PSO)

Kennedy and Eberhart (1995) proposed the PSO algorithm, which is a metaheuristic algorithm based on the principle of swarm intelligence and capable of solving complicated mathematics problems encountered in engineering (de Almeida and Leite, 2019). The social activities of birds, fish shoals, and even humans inspired the PSO algorithm (Ribeiro et al., 2017). In PSO, it is assumed that each individual has their own experience and can assess its quality. Because the algorithm is based on social behavior, each individual is aware of what their neighbors are going through.

The PSO algorithm is based on these two perspectives (Ribeiro et al., 2017). The individual's actions are influenced by its past performance and its surroundings. A culture's adjustment process can be divided into three categories: assess, compare, and mimic. Every individual have the ability to assess their surroundings and predict their behavior, as well as compare their outcomes to those of their peers. It can detect flaws and advantages in each action and duplicate the important events produced by their neighbors. It indicates that mental talents are acquired and maintained (Ribeiro et al., 2017).

Detail information about the evaluation of the implementation can be found in de Almeida and Leite (2019); Kennedy and Eberhart (1995). The flow diagram for basic PSO algorithm is shown in Fig. 11.

4.9.1. PSO for PID tuning

PSO has recently received increased interest from researchers because of its suitability for realization and promising optimization capability in a variety of applications (Song and Gu, 2004). PSO's application to various domains of effort is based on two factors. The first is to increase PSO performance by changing parameters, boosting population diversity, and combining it with other optimization methods. The second is PSO's use in various domains such as multi-objective optimization, electronics, neural network training, network security and mute selection, medicine, and emergent system identification, to name a few.

Because of its intelligible performance and ease of use, PSO has been extensively used to a variety of domains of endeavor

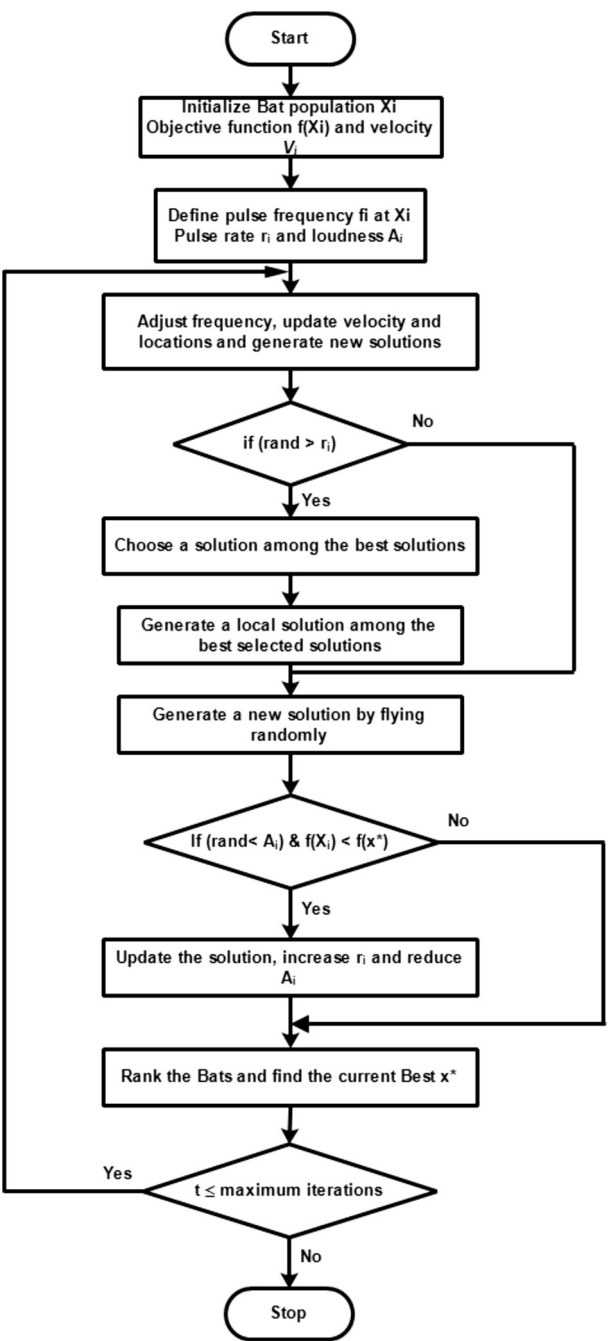


图9. 蝙蝠算法通用流程图。

特性。它已被用于解决多种实际问题, 并被发现在广泛的遗传算法和粒子群优化类型中有效 (Jhankal和Adhyaru, 2011)。

4.8.1. BFO用于 PID 整定

细菌觅食优化算法是一种受人类肠道中大肠杆菌觅食行为启发的新型群体智能系统。经过过去几年的深入研究, BFO在多个应用领域表现出良好的性能 (Niu 等人, 2010)。近年来, BFO引起了广泛关注, 并提供了许多有前景的工业应用和信息处理方案。一些模拟细菌觅食行为的模型已被设计并用于解决实际挑战 (Bayoumi 和 Salem, 2012)。与其他群体智能算法一样, BFO方法已被用于解决广泛领域的问题。通信系统、通信工

程智能控制系统 (Niu 等, 2010) 是一些应用领域, 其中 PID 被使用。

BFO已被用于PID控制器设定的适当整定, 以克服许多工业机械常遇到的问题, 如高阶和不确定性。Niu等(2006年)使用BFO优化和整定了BF-PID控制器。BF-PID比GA-PID具有更快的调节时间、很小或没有超调以及更强的鲁棒性。与GA-PID相比, BF-PID具有更快的调节时间、很少或没有超调以及更高的鲁棒性。类似地, dos Santos Coelho和da Costa Silveira (2006年)在机器人操纵器系统中提供了一种改进的BFO用于PID优化。为了调整基本BFO中的 C 常数, 新技术使用基于均匀分布、高斯分布和柯西概率分布的趋化步长参数(C)变速进行了测试。BFO方法在Ali和Majhi (2006年)中被用于通过降低时间误差平方积分(ISTE)条件来改进PID参数。此外, 为解决PID控制器整定的挑战, Korani等(2009年)将细菌觅食行为与PSO相结合。Rajinikanth和Latha在Rajinikanth and Latha (2012年)中提出使用BFO整定不稳定过程模型的PID控制器参数。结果表明, 使用细菌觅食算法整定PID参数是一种可行的选择。

通过移动机器人沿不可预测路径监控移动目标是BFO用于整定PID控制器的另一应用 (Benbouabdallah 和 Zhu, 2013; Huynh 等人, 2020)、永磁同步电机的速度控制器 (Agarwal 等人, 2017)、直流伺服电机的位置控制 (Jasim, 2018) 以及多输入多输出 (MIMO) 系统的PID参数整定 (Sivakumar 等人, 2016)。

4.9. 粒子群优化 (PSO)

Kennedy和Eberhart (1995) 提出了PSO算法, 这是一种基于群体智能原理的元启发式算法, 能够解决工程中遇到的复杂数学问题 (de Almeida和Leite, 2019)。鸟类、鱼群甚至人类的社会活动启发了PSO算法 (Ribeiro等, 2017)。在PSO中, 假设每个个体都有自己的经验并能够评估其质量。由于该算法基于社会行为, 每个个体都知道其邻居正在经历什么。

PSO算法基于这两个观点 (Ribeiro等, 2017)。个体的行为受其过去表现和环境的影响。文化适应过程可分为三类: 评估、比较和模仿。每个个体都有能力评估其环境并预测其行为, 以及将自己的结果与同伴的结果进行比较。它能检测每个动作中的缺陷和优势, 并模仿其邻居产生的重要事件。这表明心理才能是习得并维持的 (Ribeiro等, 2017)。

关于实现评估的详细信息可在de Almeida和Leite (2019); Kennedy和Eberhart (1995) 中找到。基本PSO算法的流程图如图11所示。

4.9.1. PSO 用于 PID 整定

近年来, PSO因其易于实现和在多种应用中具有良好优化能力而受到研究人员的更多关注 (Song和Gu, 2004)。PSO在不同领域的应用基于两个因素。第一是通过改变参数、增加种群多样性以及与其他优化方法结合来提高PSO性能。第二是PSO在多个领域的应用, 例如多目标优化、电子学、神经网络训练、网络安全和静音选择、医学以及涌现系统辨识等。

由于其卓越的性能和使用的便利性, 粒子群优化已被广泛应用于各种不同的工作领域中。

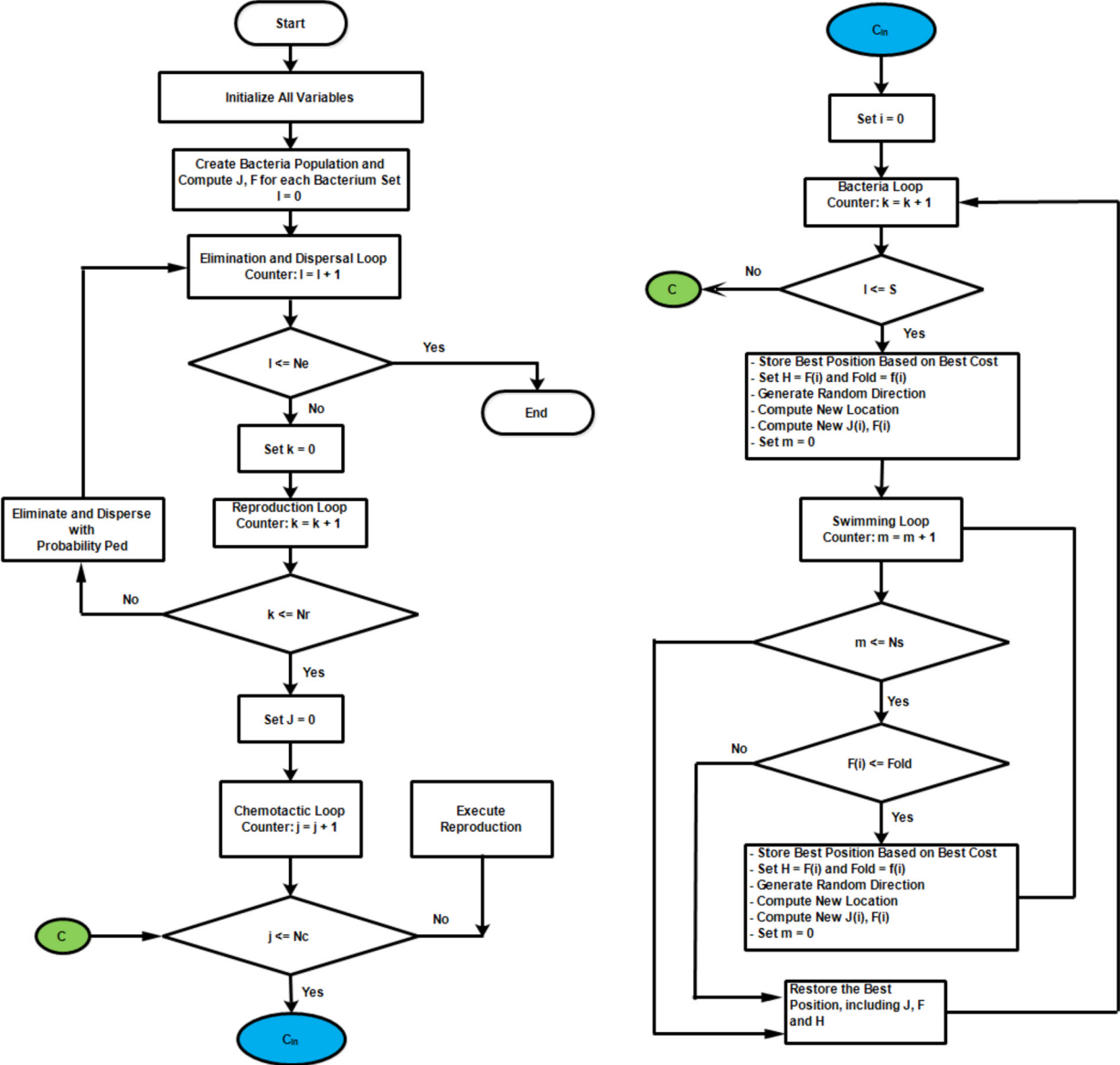


Fig. 10. Generic flow diagram for BFO algorithm.

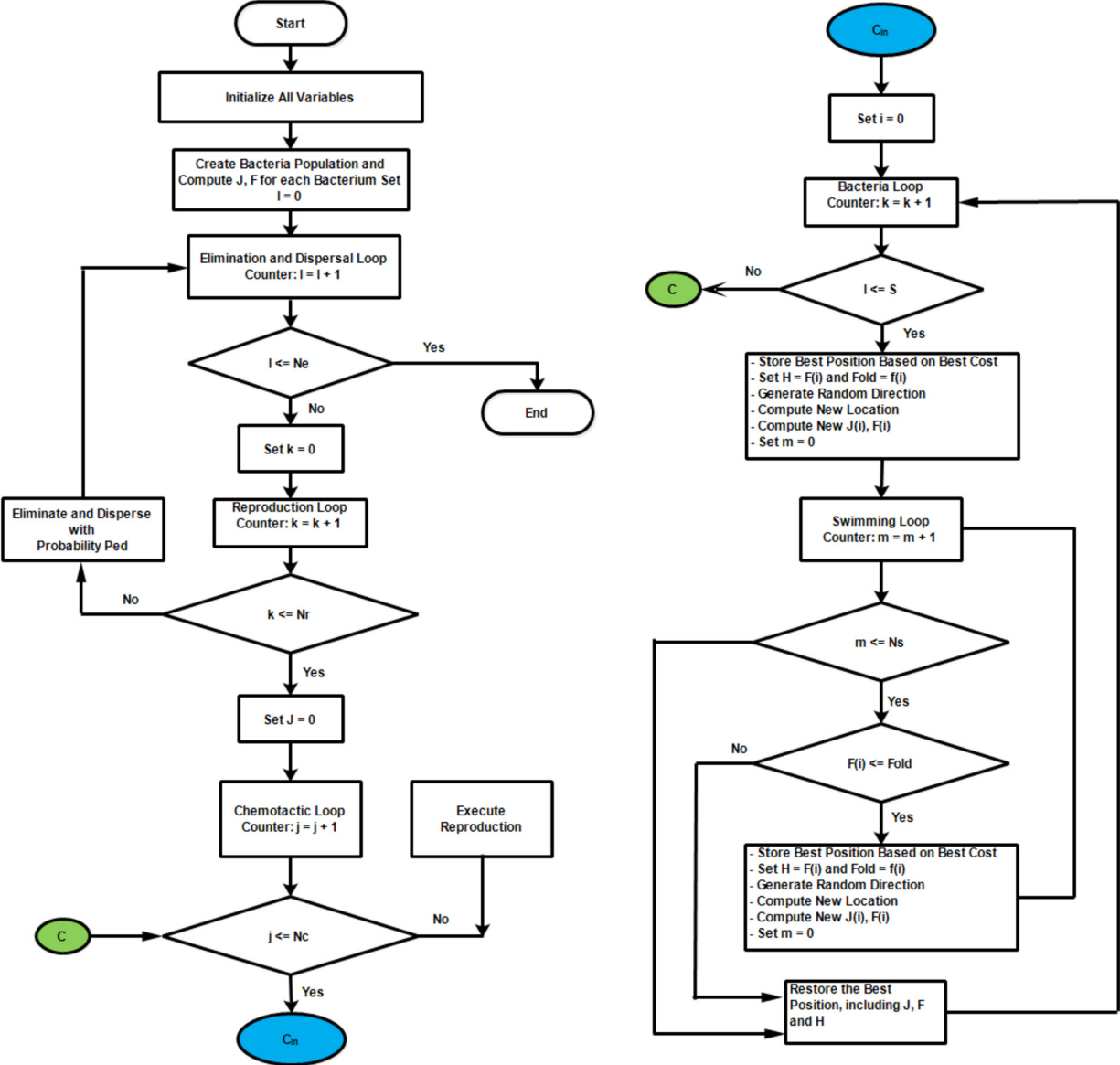


图10. 细菌觅食优化算法的通用流程图。

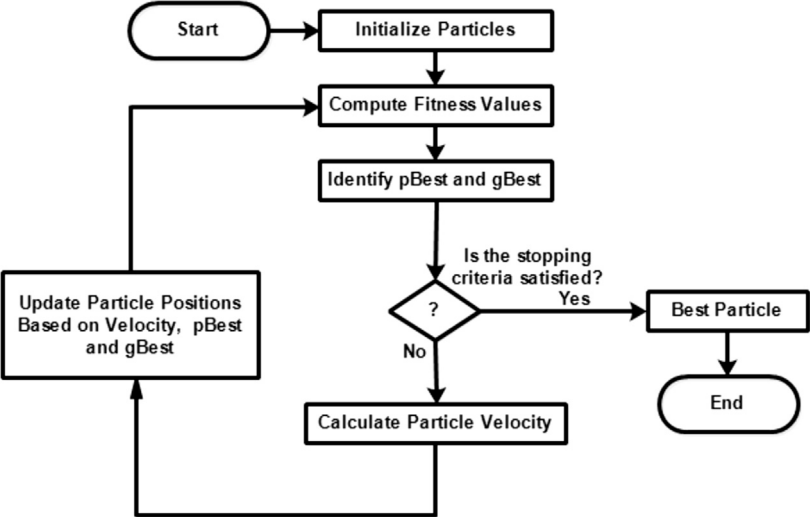


Fig. 11. Flow diagram for basic PSO algorithm.

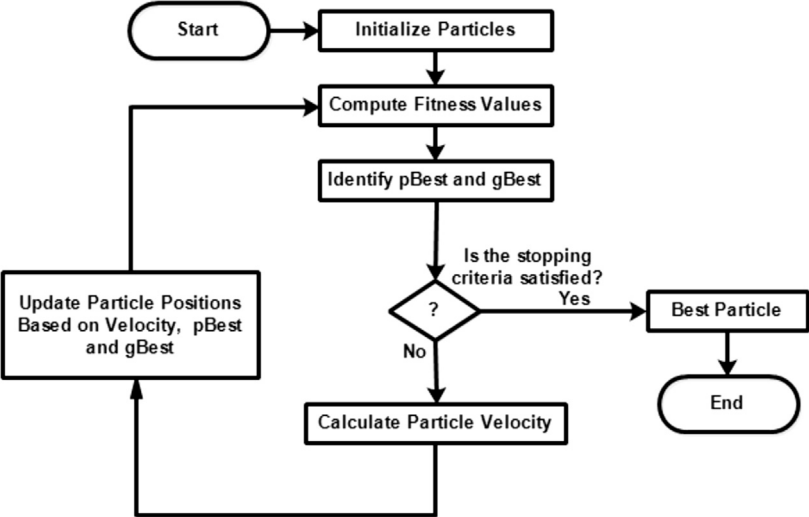


图11.基本PSO算法的流程图。

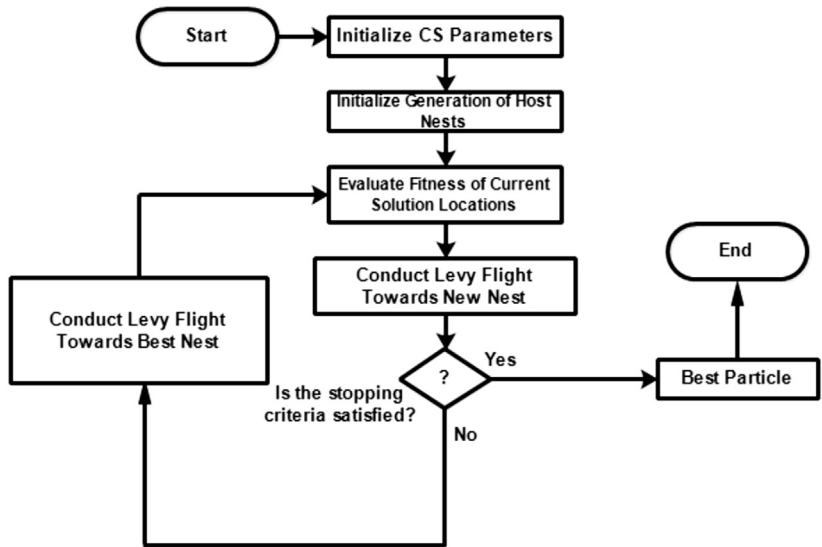


图 12. 布谷鸟搜索算法的应用流程图。

(Hajihassani et al., 2018). PSO is a popular optimization technique, and because of its versatility, it has a wide range of applications in fields such as medicine (Eshtay et al., 2018), financial (Shie et al., 2012), economics (Lee and Park, 2006), security and military (Cho et al., 2010; Cheng et al., 2008), biological, system identification etc. (Khare and Rangnekar, 2013).

PSO was used by many academics to tackle difficulties in the disciplines of structural, hydrological, and geotechnical civil engineering (Hajihassani et al., 2018). PSO has recently been used to solve problems in operations research (Yu et al., 2016a; Xia and Li, 2017). Some applications are also closely linked to mechanical engineering (Chen et al., 2013), and PSO algorithms have been used to solve problems in operations research (Che et al., 2012). Academic researchers have also used PSO to address optimization problems in the field of electronics and communication engineering (Ganguly et al., 2013). Liu et al. (2013), Liu et al. created a job scheduling model for cloud computing based on the PSO algorithm to reduce energy usage and enhance profit. Zhang et al. (2015) contains a detailed overview of the implementation of PSO techniques. Swarm robotics flocking motion is another domain where PSO has been successfully utilized (Dada and Ramlan, 2015; Dada and Ramlana, 2013).

PSO has been used in various applications of automatic control systems that heavily rely on PID controllers. In Štimac et al. (2014), PSO was used to tune the settings of a PID controller employing an active magnetic bearing that suspends a rotating shaft and keeps it suspended using regulated electromagnetic forces on the rotor in radial and axial directions. Chang and Chen (2014) designed a PID control system for MIMO activities. In MIMO control systems, an improved version of the PSO algorithm was utilized to tune PID control gains. Particle swarm optimization approach was used in Rajesh and Ananda (2015) to optimize settings of PID controllers in order to modify the location of the camera in an Unmanned Aerial Vehicle (UAV). The findings were compared to those of a traditional PID controller. PSO approach was used to calibrate PID controllers for various transitory circumstances of Permanent Magnet DC motor in Medewar and Munje (2015). The proposed approach increased both noise immunity and steady state detection accuracy, according to the results of experiments. PSO has also been used to tune PID control variables for the control of an inverted pendulum in the past (Joseph and Dada, 2018), and speed control of a DC motor in Bassi et al. (2011).

4.10. Cuckoo optimization algorithm

Yang and Deb (2009) developed the Cuckoo Search (CS) metaheuristic evolutionary optimization algorithm in 2009. The cuckoo is a bird species that inspired the concept of CS. Cuckoos are appealing birds, not only because of their wonderful songs, but rather because of their proactive breeding approach. Cuckoos lay their eggs in the nests of other host birds or species when they are grown. Compel brood parasitism is the term for this type of parasitism. The ability of a single cuckoo to lay an egg that may duplicate, or resemble, host nest eggs in such a way that the host nest mother bird cannot distinguish the cuckoo egg from her own is crucial. The cuckoo's egg will hatch before the host egg, confirming the cuckoo's control in the nest and so extending their lifespan.

The concept of Cuckoo Optimization Algorithm (COA) is depicted in Fig. 12. The pseudocode, equations and application details for CS can be seen in Yang and Deb (2009); Bhandari et al. (2014); Shair et al. (2015); Mamizadeh et al. (2018); Rajabioun (2011).

4.10.1. Cuckoo search algorithm for PID tuning

Recently, CS has been applied in almost every area and domain of function optimization (Fister et al., 2014), engineering optimization (Yang and Deb, 2010), image processing (Brajevic and Tuba, 2014), scheduling (Burnwal and Deb, 2013), planning (Akbarzadeh and Shadkam, 2015), feature selection (Abd El Aziz and Hassanien, 2018), forecasting (Wu et al., 2019), and real-world applications (Expósito-Izquierdo and Expósito-Márquez, 2017). Engineering optimization appears to be just one of the many uses. CS and its derivatives have been used in nearly every field of research, technology, and manufacturing.

CS techniques have been used to tune PID controller parameters in many studies. Kumar et al. (2014) presents PID parameter tuning for inverted pendulum monitoring, experimental validation of Van der Pol oscillator, and stability study of capsizing problem as an example of nonlinear systems employing cuckoo optimization approach. Bingul and Karahan (2018) presents a new tuning approach for PID parameters in the AVR system utilizing the CS algorithm and a new frequency-domain performance benchmark. In their paper, a new way for designing PID controllers utilizing the CS optimization algorithm was provided in order to improve the performance of buck-boost converters in LED driver circuits (Verma et al., 2016; Nimisha and Senthilkumar, 2018). The CS optimization approach was recently used to optimize the PID parameters for various different industrial plants. Maximum Power Point Tracking (MPPT) of PV systems (Mosaad et al., 2019b), Speed of induction motor regulation (Peram et al., 2018), and optimization of

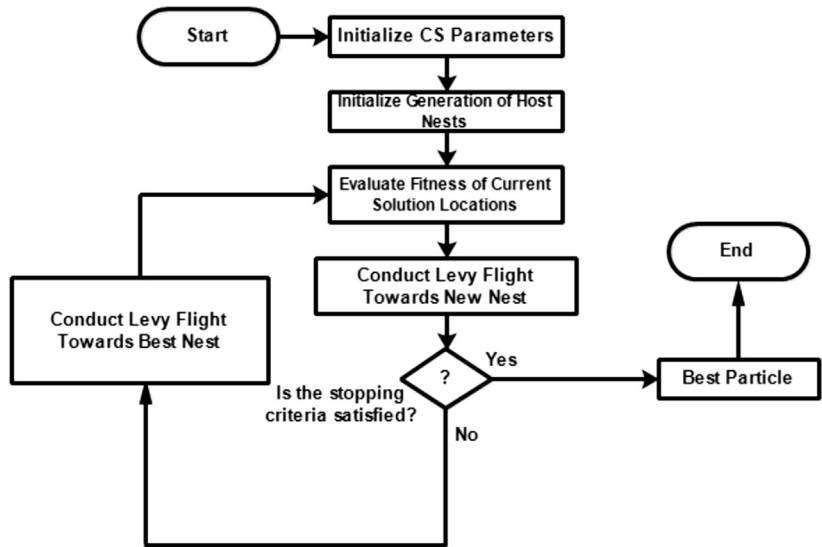


图 12. 布谷鸟搜索算法的应用流程图。

(Hajihassani et al., 2018)。粒子群优化是一种流行的优化技术，由于它的多功能性，在医学 (Eshtay et al., 2018)、金融 (Shie et al., 2012)、经济学 (Lee and Park, 2006)、安全与军事 (Cho et al., 2010; Cheng et al., 2008)、生物学、系统辨识等领域有着广泛的应用 (Khare and Rangnekar, 2013)。

许多学者使用粒子群优化来解决结构、水文和岩土土木工程领域中的难题 (Hajihassani et al., 2018)。最近，粒子群优化已被用于解决运筹学中的问题 (Yu et al., 2016a; Xia and Li, 2017)。一些应用也与机械工程密切相关 (Chen et al., 2013)，并且粒子群优化算法已被用于解决运筹学中的问题 (Che et al., 2012)。学术研究人员也使用粒子群优化来解决电子与通信工程领域的优化问题 (Ganguly et al., 2013)。Liu et al. (2013) 基于粒子群优化算法创建了一个用于云计算的作业调度模型，以减少能耗并提高利润。Zhang et al. (2015) 详细概述了粒子群优化技术的实现。群体机器人编队运动是另一个成功应用粒子群优化的领域 (Dada and Ramlan, 2015; Dada and Ramlana, 2013)。

PSO已被广泛应用于严重依赖PID控制器的自动控制系统。在Štimac等人 (2014)的工作中，采用PSO来调节主动磁轴承中PID控制器的设置，该轴承悬浮旋转轴并通过调节径向和轴向的电磁力使其保持悬浮状态。Chang和Chen (2014)为MIMO活动设计了一个PID控制系统。在MIMO控制系统中，使用改进的粒子群优化算法来调节PID控制增益。Rajesh和Ananda (2015)采用粒子群优化方法来优化PID控制器的设置，以便调整无人机中摄像头的位置。研究结果与传统PID控制器进行了比较。Medewar和Munje (2015)使用PSO方法来校准永磁直流电机各种暂态条件下的PID控制器。实验结果表明，所提出的方法提高了抗噪性和稳态检测精度。PSO还被用于调节倒立摆控制的PID控制变量 (Joseph和Dada, 2018)，以及直流电机速度控制 (Bassi等人, 2011年)。

4.10. 布谷鸟 优化 算法

Yang和Deb (2009)于2009年提出了布谷鸟搜索 (CS) 元启发式进化优化算法。布谷鸟是一种启发CS概念的鸟类。布谷鸟是吸引人的鸟类，不仅因为它们美妙的歌声，更因为它们主动的繁殖方式。布谷鸟在成长过程中会将蛋产在其他宿主鸟类或物种的巢中。这种寄生现象被称为强制巢寄生。关键在于布谷鸟能够产下可以复制或模仿宿主巢蛋的蛋，使得宿主巢中的母鸟无法区分布谷鸟蛋和自己的孩子。布谷鸟的蛋将在宿主蛋之前孵化，从而确认布谷鸟对巢的控制，并因此延长其生存期。

布谷鸟优化算法(COA)的概念如图12所示。CS的伪代码、方程和应用细节可参见Yang和Deb (2009年)；Bhandari等人 (2014年)；Shair等人 (2015年)；Mamizadeh等人 (2018年)；Rajabioun (2011年)。

4.10.1. 布谷鸟 搜索 算法 用于 PID 整定

最近，CS几乎已应用于函数优化 (Fister等人, 2014年)、工程优化 (Yang和Deb, 2010年)、图像处理 (Brajevic和Tuba, 2014年)、调度 (Burnwal和Deb, 2013年)、规划 (Akbarzadeh和Shadkam, 2015年)、特征选择 (Abd El Aziz和Hassanien, 2018年)、预测 (Wu等人, 2019年)以及实际应用 (Expósito-Izquierdo和Expósito-Márquez, 2017年)等所有领域和范畴。工程优化似乎只是众多应用之一。CS及其衍生算法已几乎应用于研究、技术和制造的每个领域。

CS技术已被用于许多研究中整定PID控制器参数。Kumar等人 (2014)提出了针对倒立摆监测的PID参数整定、范德波尔振荡器的实验验证，以及作为非线性系统示例的倾覆问题稳定性研究，采用布谷鸟优化方法。Bingul和Karahana(2018)提出了一种新的AVR系统中PID参数整定方法，利用CS算法和新的频域性能基准。在他们的论文中，提供了一种利用CS优化算法设计PID控制器的新方法，以改善LED驱动电路中升降压变换器的性能(Verma等人, 2016; Nimisha和Senthilkumar, 2018)。CS优化方法最近被用于优化各种不同工业设备的PID参数。光伏系统的最大功率点跟踪(MPPT)(Mosaad等人, 2019b)、感应电机调速 (Peram等人, 2018)，以及优化

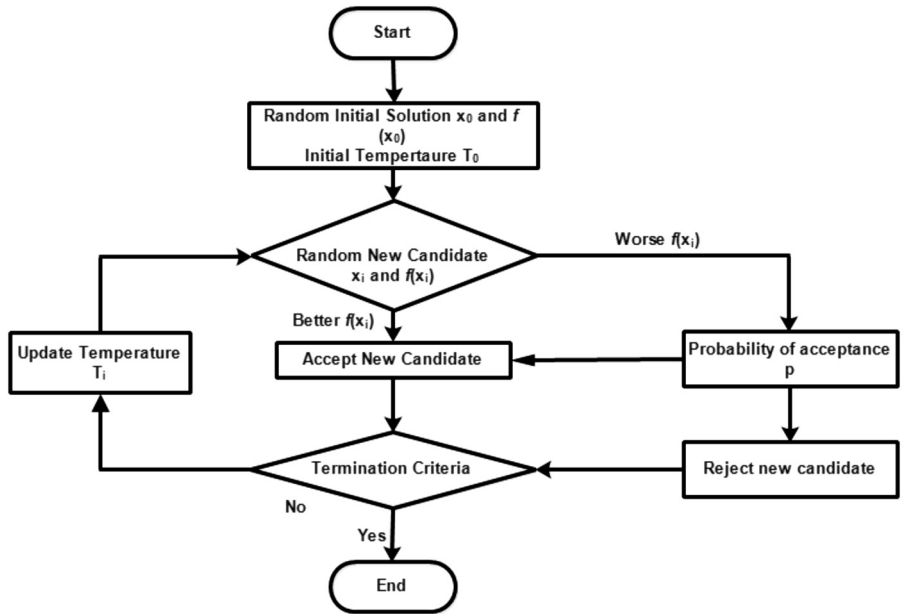


Fig. 13. Flowchart for Simulated annealing algorithm.

complicated stability radius via single objective constrained optimization are instances of such plants (Solihin and Akmeliawati, 2018).

4.11. Simulated annealing

In 1953, Metropolis et al. (1953) was the first to propose a Simulated Annealing (SA) approach. It resembles the rearranging of particles in a body to a crystalline state, which is accompanied by a drop in temperature. The constituents of a body wander laterally within a region whose amplitude is defined by its temperature. As long as cooling is gradual enough, atoms or particles can position themselves in lower and lower energy levels, finally reaching the lowest energy state (crystalline state) (Baselga, 2011). SA is based on the Monte-Carlo iterative approach (Jia and Lichti, 2017).

The flowchart for SA is presented in Fig. 13. Further readings and descriptive algorithms on SA can be found in Fouskakis and Draper (2002); Suman and Kumar (2006); Eglese (1990).

4.11.1. Simulated annealing for PID tuning

SA is a method for proffering solution to difficult optimization problems with satisfactory outcomes. Over the years, South Africa has attracted a lot of attention and interest. Kirkpatrick et al. (1983) and Černý (1985) sparked the current interest. These papers showed how to use a model for simulating solid annealing provided by Metropolis et al. (1953) to tackle optimization issues where the objective function to be minimized relates to the energy of the solid's states.

SA has been used to solve problems in a variety of fields, including computer VLSI design (Wong et al., 2012), image processing (Tavares et al., 2019), molecular physics (Wu et al., 2016), chemistry (Hernandez et al., 2019), process optimization (Sibaliija, 2018), and job shop scheduling (Cruz-Chávez et al., 2017). There has also been development on theoretical results from a mathematical examination of the method, as well as several computing studies evaluate the effectiveness of SA with the effectiveness of other techniques for a variety of situations.

Since the SA algorithm selects control parameters in its own evolutionary process and does not need any rule of thumb, it has diffused rapidly into many application areas. Recently, some researchers have used SA algorithm in tuning of PID control parameters. An intelligent concept for PID controller parameters tuning is suggested using SA for multi-objective problem and its results was compared with basic PID which shows SA to be better than conventional or basic PID controller

in terms of tuning PID parameters (Bansal et al., 2014). Suman et al. (2010) introduces a novel multi-objective optimization approach based on SA dubbed Orthogonal Simulated Annealing (OSA).

Yachen and Yueming (2008) used SA algorithm for tuning different PID controllers in practical applications. SA was used to tune PID parameters for motion control system having constant disturbance. The experimented results were compared, showing that different controller was superior in different control requirement. Each controller has its own advantage and proper scope.

SA was recently employed to fine-tune the characteristics of a permanent magnet brushless dc motor (PMBLDCMPID)'s controller (Shatnawi and Bayoumi, 2019). The response parameters index, maximum overshoot, rise time, and settling time for each controller are tuned to reduce the controller's parameters. The SA optimization approach is used to tune the system. The technique determines the best PID/PI controller parameter for fixed-structure PID/PI controllers.

SA has been applied for tuning PID parameters of different processes like AVR system (Lahcene et al., 2017), attitude control system for a re-entry capsule (Vijay and Banu, 2016) and DC-DC power converters (Alqudah et al., 2014). Further readings on application of SA can be seen in Siddique and Adeli (2016); Suman and Kumar (2006); Sibaliija (2018); Varty (2017).

4.12. Grey wolf optimizer

Grey Wolf Optimization (GWO) is a recently proposed metaheuristic optimization algorithm (Mirjalili et al., 2014). Grey wolves' social structure and intelligent hunting tactics motivate GWO. Grey wolves are generally at the top of the hierarchy in their habitats. Grey wolves usually dwell in groups of 5–12 people. Grey wolves, in particular, have a well-defined social hierarchy. The grey wolf social hierarchy and duties are depicted in Fig. 14. The main distinction between GWO and other metaheuristic optimization algorithms is the social dominant hierarchy, whose primary goal is to develop the candidate solution during each iteration. As a result, GWO imitates grey wolf hunting behavior as regards locating and attacking prey (Goldbogen et al., 2013).

Muro et al. (2011) stated the major stages of grey wolves hunting actions as follows:

- i. Trailing, pursuing, and advancing the prey.
- ii. Following, encircling, and harassing the prey until it stops moving.
- iii. Attacking in the direction of prey.

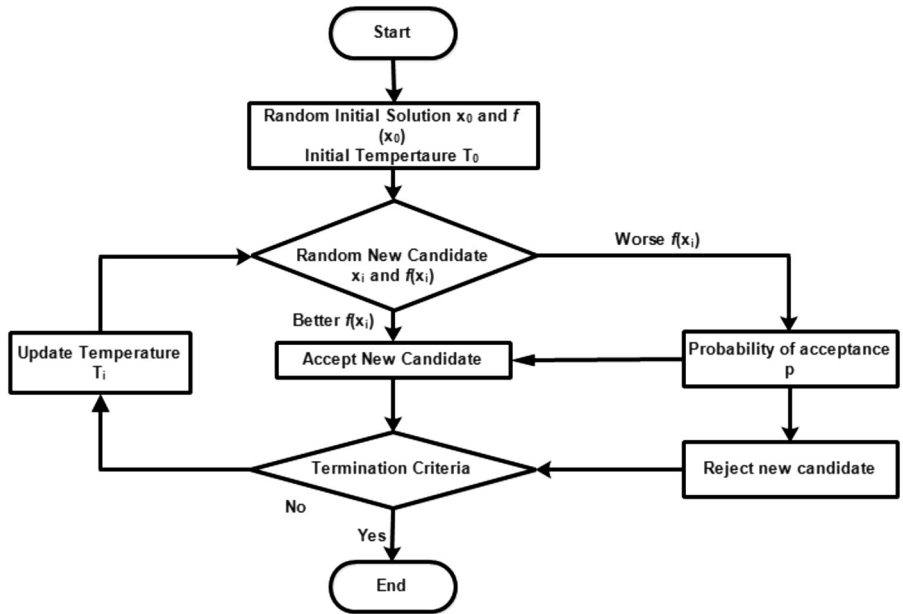


图 13.模拟退火算法流程图。

复杂 稳定性 半径 通过 单 目标 约束 优化 是 实例 的 这些 设备 (Solihin 和 Akmeliawati, 2018)

4.11. 模拟退火

1953年, Metropolis等人 (1953) 首次提出了模拟退火 (SA) 方法。它类似于物体中粒子重新排列至晶态的过程, 伴随温度下降。物体内的成分在一个幅度由温度决定的区域内横向移动。只要冷却足够缓慢, 原子或粒子就能逐渐定位到越来越低的能量水平, 最终达到最低能量状态 (晶态) (Baselga, 2011)。SA基于蒙特卡洛迭代法 (Jia and Lichti, 2017)。

SA的流程图如图13所示。关于SA的更多阅读材料和描述性算法可参考 Fouskakis和Draper (2002); Suman和Kumar (2006); Eglese (1990)。

4.11.1. 模拟退火 用于 PID 整定

SA是一种为困难优化问题提供满意解决方案的方法。多年来, 南非吸引了大量关注和兴趣。Kirkpatrick等(1983)和Černý(1985)引发了当前的兴趣。这些论文展示了如何使用Metropolis等(1953)提供的模拟固体退火的模型来解决优化问题, 其中要最小化的目标函数与固体状态的能量相关。

SA已被用于解决多个领域的问题, 包括计算机VLSI设计(Wong等, 2012)、图像处理(Tavares等, 2019)、分子物理学(Wu等, 2016)、化学(Hernandez等, 2019)、过程优化(Sibaliija, 2018)和作业车间调度(Cruz-Chávez等, 2017)。此外, 对该方法的数学检验也取得了理论成果, 并且多项计算研究评估了SA在各种情况下的有效性与其他技术相比的优势。

由于SA算法在其自身的进化过程中选择控制参数, 无需任何经验法则, 因此它迅速扩散到许多应用领域。最近, 一些研究人员将SA算法用于PID控制参数的整定。提出了一种基于SA的多目标问题PID控制器参数整定的智能概念, 并将其结果与基本PID进行了比较, 结果表明SA优于传统或基本PID控制器。

在 PID 参数 整定 方面 (Bansal等人, 2014)。Suman等人 (2010) 提出了一种基于SA的 新颖 多目标 优化 方法, 称为 正交 模拟 退火 (OSA)。

Yachen 和 Yueming (2008) 使用 SA 算法在实际应用中整定不同的 PID 控制器。SA 被用来整定具有恒定干扰的运动控制系统的 PID 参数。实验结果进行了比较, 表明不同的控制器在不同的控制要求下表现更优。每个控制器都有其自身的优势和适用范围。

SA 最近被用于微调永磁无刷直流电机 (PMBLDCMPID) 控制器的特性 (Shatnawi 和 Bayoumi, 2019)。每个控制器的响应参数指标、最大超调量、上升时间和调节时间被整定以减小控制器的参数。SA优化方法用于整定系统。该技术确定了固定结构 PID/PI 控制器的最佳 PID/PI 控制器参数。

SA 已被应用于整定不同过程的 PID 参数, 如自动电压调节器系统 (Lahcene 等人, 2017)、再入舱的姿态控制系统 (Vijay 和 Banu, 2016) 和 DC-DC 电源转换器 (Alqudah 等人, 2014)。关于 SA 应用的进一步阅读可见 Siddique 和 Adeli (2016); Suman 和 Kumar (2006); Sibaliija (2018); Varty (2017)。

4.12. 灰狼 优化 器

灰狼优化(GWO)是近期提出的一种元启发式优化算法(Mirjalili 等人, 2014)。灰狼的社会结构和智能狩猎策略启发了GWO。灰狼通常在其栖息地的社会等级中处于顶端。灰狼通常以5-12只为一群居住。特别是, 灰狼具有明确的社会等级。灰狼的社会等级和职责如图14所示。GWO与其他元启发式优化算法的主要区别在于社会主导等级, 其主要目标是在每次迭代中发展候选解。因此, GWO模仿灰狼的狩猎行为, 涉及定位和攻击猎物(Goldbogen 等人, 2013)。

Muro 等人 (2011)将 灰狼 狩猎 行动 的 主要 阶段 描述 如下:

- i. 跟踪、追逐、和逼近猎物。ii. 跟随、包围、和骚扰猎物直到它停止移动。iii. 朝猎物方向攻击。



Fig. 14. Social hierarchy of grey wolves and their responsibilities.

4.12.1. GWO for PID tuning

The remarkable advantages of GWO has led to tremendous research applications from several critical research domains. These application domains can be categorized into: machine learning (Emary et al., 2016, 2015; Al-Tashi et al., 2019), engineering (Kohli and Arora, 2018; Yıldız and Yıldız, 2018), wireless sensor network (Kaushik et al., 2019; Rajakumar et al., 2017), environmental modeling (Tikhamarine et al., 2020; Sweidan et al., 2016), medical and bio informatics (Sharma et al., 2019, 2020) and image processing (Khairuzzaman and Chaudhury, 2017; Li et al., 2017a).

In control engineering, there has been an explosion in amount of publications that explore application of GWO algorithm for tuning PID controller parameters. The work in Li et al. (2017b) proposed GWO scheme to tune variables of a PI controller for a closed-loop condenser pressure control mechanism. Results of performance demonstrated the efficacy of Grey Wolf (GW) optimizer compared to GA and PSO. Also in Yadav et al. (2016), parameters of a basic PID controller was tuned using GWO algorithm for a magnetic levitation system. Likened to basic ZN tuning scheme, GW optimizer demonstrated superior results. The study in Das et al. (2015) investigated the application GW optimizer to tune the control variables of a PID controller for the regulation of speed of a second order DC motor system. Improved values of transient reaction stipulations were attained in comparison to PSO, ABC and ZN approaches. Regulation of control variables of PID controller for DC motor was similarly explored in Madadi and Motlagh (2014). Results of performance evaluation demonstrated superiority of GWO over PSO, by improving the dynamic performance of the system.

Some other applications of GWO for PID tuning include robotics (Şen and Kalyoncu, 2018), speed control of DC motor (Agarwal et al., 2018), inverted pendulum (Yadav et al., 2020), voltage regulation (Verma and Nagar, 2018) and wind turbine (Sule et al., 2020). A generic flowchart illustrating GWO algorithm is presented in Fig. 15.

4.13. Krill herd algorithm

The herding behavior of krill individuals inspired the Krill Herd (KH) algorithm. It is a new swarm intelligent algorithm that has outperformed conventional swarm intelligent algorithms. Gandomi and Alavi (2012) proposes a novel biologically motivated approach for tackling optimization issues called KH. The KH algorithm is based on the herding of krill swarms in response to biological and environmental events. Almost all of the necessary coefficients for using the KH method are determined from real-world experiments. Each krill's fitness function

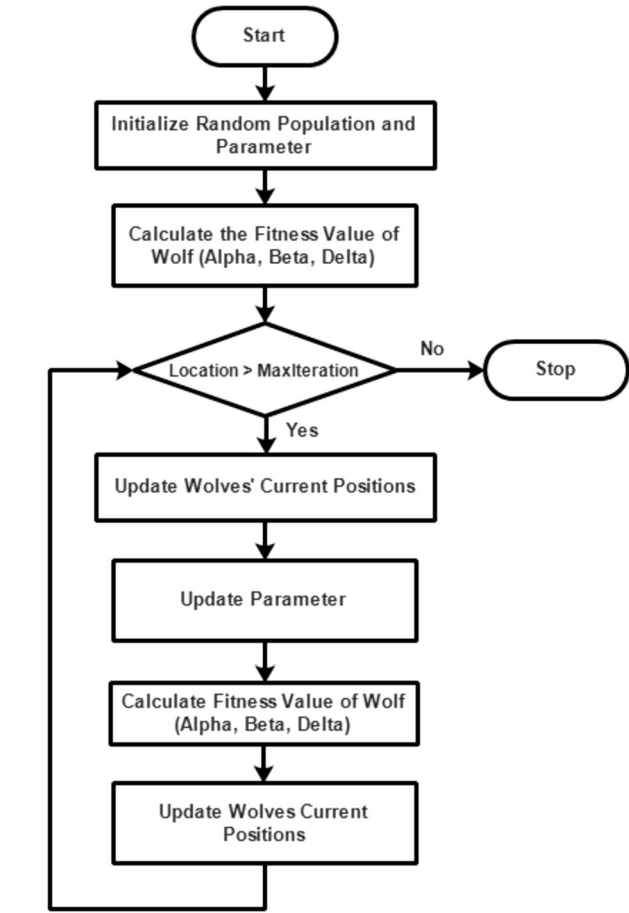


Fig. 15. Grey wolf optimization application flowchart.

is defined by its remoteness from food and maximal concentration of swarm.

KH's entire optimization technique may be broken down into three simple steps. These three acts are taken into account while updating each krill. The distance between the food position and the krill's location defines the objective function. Because of its remarkable application performance, KH has gotten a lot of attention from academics,

4.12.1. 灰狼优化用于PID整定

GWO 的显著优势促使其在多个关键研究领域得到了大量研究应用。这些应用领域可分为：机器学习 (Emary et al., 2016, 2015 2019 2018 ; Al-Tashi et al.,), 工程 (Kohli and Arora, ; Yıldız and Yıldız, 2018)、无线传感器网络 (Kaushik et al., 2019; Rajakumar et al., 2017)、环境建模 (Tikhamarine et al., 2020; Sweidan et al., 2016)、医学与生物信息学 (Sharma et al., 2019, 2020) 和图像处理 (Khairuzzaman and Chaudhury, 2017; Li et al., 2017a)。

在控制工程中，探索应用灰狼优化算法整定PID控制器参数的出版物数量激增。Li等 (2017b) 的工作提出了使用灰狼优化方案来整定闭环冷凝器压力控制机制的PI控制器变量。性能结果证明了灰狼优化器相比遗传算法和粒子群优化的有效性。同样，Yadav等 (2016) 使用灰狼优化算法对磁悬浮系统的基本PID控制器参数进行了整定。与基本的齐格勒-尼科尔斯整定方案相比，灰狼优化器表现出更优的结果。Das等 (2015) 的研究探讨了应用灰狼优化器整定PID控制器的控制变量，以调节二阶直流电机的速度。与粒子群优化、人工蜂群算法和齐格勒-尼科尔斯方法相比，获得了改进的暂态响应指标。Madadi和Motlagh (2014) 同样探索了直流电机PID控制器控制变量的调节。性能评估结果表明，灰狼优化通过改善系统的动态性能，优于粒子群优化。

GWO用于PID整定的其他应用包括机器人学 (Sen和Kalyoncu, 2018)、直流电机速度控制 (Agarwal et al., 2018)、倒立摆 (Yadav et al., 2020)、电压调节 (Verma和Nagar, 2018) 以及风力发电机 (Sule et al., 2020)。图15展示了说明GWO算法的通用流程图。

4.13. 磷虾群算法

磷虾个体的群集行为启发了磷虾群(KH)算法。这是一种新型群体智能算法，性能优于传统群体智能算法。Gandomi和Alavi(2012)提出了一种称为KH的生物启发性方法，用于解决优化问题。KH算法基于磷虾群响应生物和环境事件的群集行为。使用KH方法所需的几乎所有系数都来自真实实验。每个磷虾的适应度函数

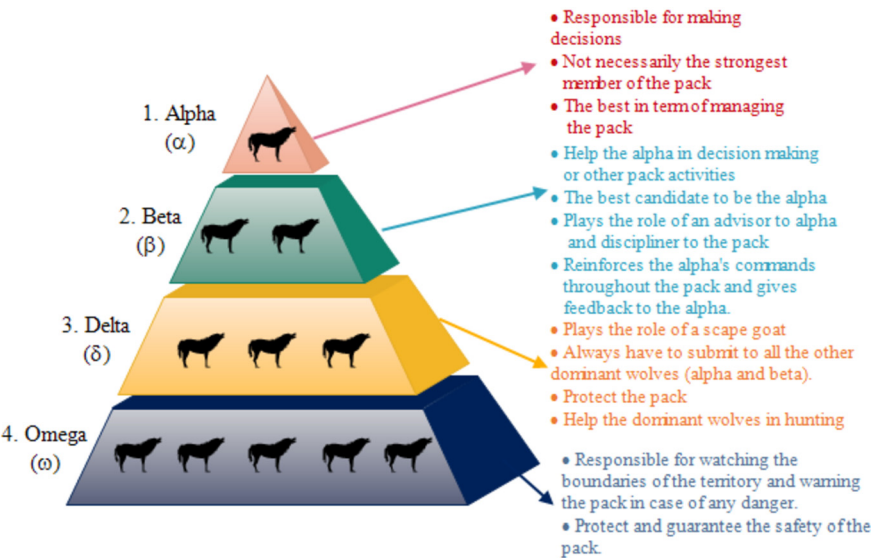


图 14.灰狼的社会等级及其职责。

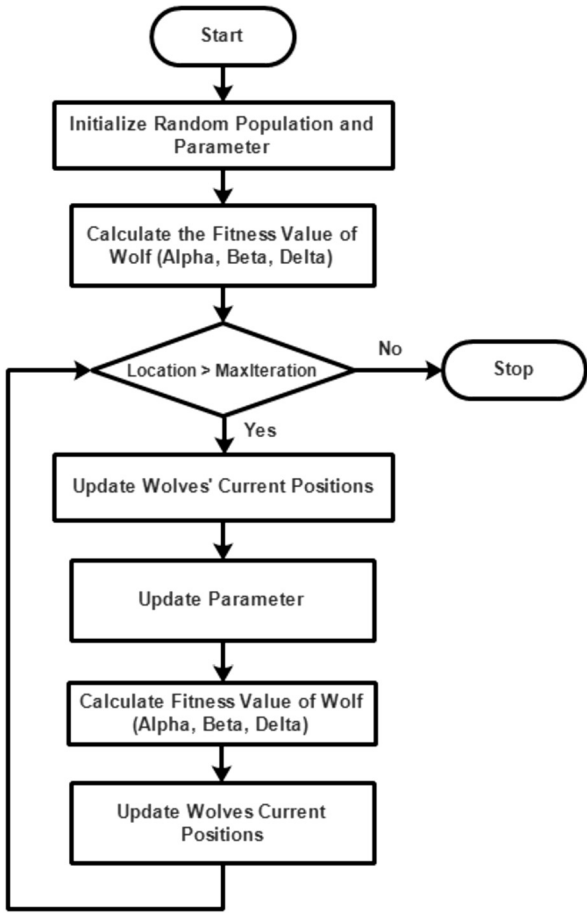


图15.灰狼优化应用流程图。

是根据其距食物的远近以及群体最大浓度来定义。

KH 的整个优化技术可以分解为三个简单的步骤。在更新每个磷虾时，会考虑这三个动作。食物位置与磷虾位置之间的距离定义了目标函数。由于其出色的应用性能，KH 受到了学术界、专家和科学家的广泛关注。

Table 2. Comprehensive list of some popular metaheuristic algorithms (1975–2014).

No.	Year of creation	Name of algorithm	Reference
1.	1975	Genetic Algorithm (GA).	Holland (1975)
2.	1977	Scatter Search (SS).	Cordón et al. (2006)
3.	1980	Genetic Programming.	Abraham et al. (2006)
4.	1983	Simulated Annealing (SA).	Kirkpatrick et al. (1983)
5.	1986	Tabu Search (TS).	Glover (1997)
6.	1986	Artificial Immune System (AIS).	Farmer et al. (1986)
7.	1989	Moscato presented Memetic Algorithm.	Qasem and Shamsuddin (2011)
8.	1992	Ant Colony Algorithm (ACO).	Dorigo et al. (1996)
9.	1993	Multi-Objective GA (MOGA).	Fonseca and Fleming (1993)
10.	1994	Reactive Search Optimization (RSO)	Battiti and Tecchiolli (1994)
11.	1995	Particle Swarm Optimization (PSO).	Eberhart and Kennedy (1995)
12.	1997	Differential Evolution (DE).	Storn and Price (1997)
13.	1997	Cross Entropy Method (CEM).	Rubinstein (1997)
14.	1999	Partial Optimization Metaheuristic Under Special Intensification Conditions (POPMUSIC)	Taillard and Voss (2002)
15.	2001	Harmony Search (HS).	Geem et al. (2001)
16.	2001	Bootstrap Algorithm (BA).	Hanseth and Aanestad (2003)
17.	2002	Evolution strategies	Beyer and Schwefel (2002)
18.	2004	Bees Optimization (BO).	Nakrani and Tovey (2004)
19.	2005	Glowworm Swarm Optimization (GSO).	Krishnanand and Ghose (2005)
20.	2005	Artificial Bee Colony Algorithm (ABC).	Karaboga (2005)
21.	2006	Invasive Weed Optimization (IWO)	Mehrabian and Lucas (2006)
22.	2006	Honey-bee Mating Optimization (HMO).	Haddad et al. (2006)
23.	2006	Cat swarm optimization	Chu et al. (2006)
24.	2007	Intelligent Water Drops (IWD).	Hosseini (2007)
25.	2007	Imperialist Competitive Algorithm (ICA).	Atashpaz-Gargari and Lucas (2007)
26.	2008	Water Drop Optimization (WDO)	Shah-Hosseini (2008)
27.	2007	Monkey Search (MS).	Mucherino and Seref (2007)
28.	2009	Firefly Algorithm (FA).	Yang (2009)
29.	2008	Plant Growth Optimization	Wang and Cheng (2008)
30.	2008	Biogeography - Based Optimization (BBO)	Simon (2008)
31.	2009	League Championship Algorithm (LCA).	Kashan (2009)
32.	2009	Gravitational Search Algorithm (GSA)	Rashedi et al. (2009)
33.	2009	Cuckoo Search (CS).	Yang and Deb (2009)
34.	2010	Bat Algorithm (BA).	Yang (2010)
35.	2010	Fireworks algorithm	Tan and Zhu (2010)
36.	2010	Chemical Reaction Algorithm	Ramaswamy and Sbalzarini (2010)
37.	2011	Galaxy-based Search Algorithm (GbSA).	Shah-Hosseini (2011)
38.	2011	Spiral Optimization (SO).	Tamura and Yasuda (2011)
39.	2011	Teaching-Learning-Based Optimization (TLBO) Algorithm.	Rao et al. (2011)
40.	2012	Krill Herd (KH) Algorithm.	Gandomi and Alavi (2012)
41.	2012	Flower Pollination Algorithm (FPA)	Yang (2012b)
42.	2012	Differential Search Algorithm (DSA).	Civicioglu (2012)
43.	2013	Firefly Algorithm (FA) with chaos	Gandomi et al. (2013)
44.	2014	Colliding Bodies Optimization (CBO)	Kaveh and Mahdavi (2014)
45.	2014	Animal Migration Optimization	Li et al. (2014c)
46.	2014	Centripetal Accelerated Particle Swarm Optimization (CAPSO)	Beheshti and Shamsuddin (2014)
47.	2014	Grey Wolf optimization (GWO)	Mirjalili et al. (2014)
48.	2014	Bumble Bees Mating Optimization	Marinakis and Marinaki (2014)
49.	2014	Crisscross Optimization Algorithm (COA)	Meng et al. (2014)

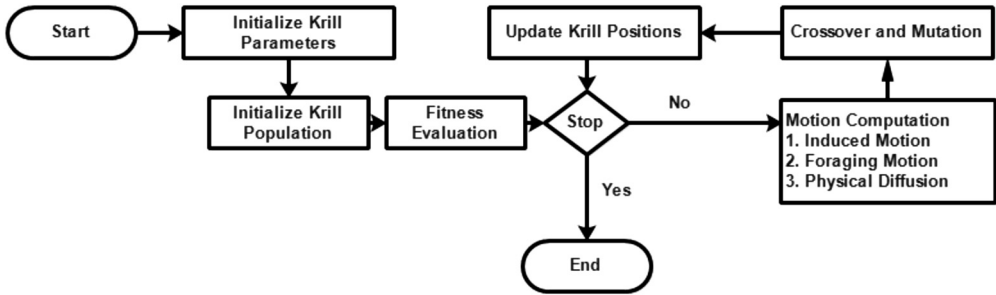


Fig. 16. Basic Krill Head application flow diagram.

experts, and scientists. Further details on KH algorithm can be seen in Gandomi and Alavi (2012). Fig. 16 presents basic KH application flow diagram. Other variants of KH algorithm can be seen in Li et al. (2014a); Yaghoobi and Mojallali (2016); Wang et al. (2019).

4.13.1. KH algorithm for PID tuning

KH algorithm has been employed to solve several engineering problems. These engineering applications of KH algorithm as grouped by Wang et al. (2019) include; continuous optimization, combinatorial

表2. 一些流行的元启发式算法（1975–2014）的综合列表。

No.	创建年份	名称 算法 名称	参考文献
1.	1975	遗传 算法 (GA)。	Holland (1975)
2.	1977	分散 搜索 (SS)。	Cordón等人 (2006年)
3.	1980	遗传编程。	Abraham等人 (2006年)
4.	1983	模拟 退火 (SA)。	Kirkpatrick等 (1983)
5.	1986	禁忌 搜索 (TS)。	Glover (1997)
6.	1986	人工 免疫 系统 (AIS)。	Farmer等 (1986)
7.	1989	Moscato 提出了 模因 算法。	Qasem和Shamsuddin (2011)
8.	1992	蚁群 算法 (ACO) 。	Dorigo等人 (1996)
9.	1993	多目标 遗传算法 (MOGA) 。	Fonseca和Fleming (1993)
10.	1994	反应式搜索优化 (RSO)	Battiti和Tecchiolli (1994)
11.	1995	粒子 群 优化 (粒子群优化)。	Eberhart和Kennedy (1995)
12.	1997	差分进化 (DE) 。	Storn和Price (1997)
13.	1997	交叉 熵 方法 (CEM)。	鲁宾斯坦 (1997)
14.	1999	部分 优化 元启发式 在 特殊 强化 条件 下 (POPMUSIC)	Taillard和Voss (2002)
15.	2001	和声 搜索 (HS) 。	Geem等人 (2001)
16.	2001	Bootstrap 算法 (BA)。	Hanseth和Aanestad (2003)
17.	2002	进化 策略	Beyer和Schwefel (2002)
18.	2004	蜜蜂 优化 (BO)。	Nakrani和Tovey (2004)
19.	2005	萤火虫 群 优化 (GSO)。	Krishnanand和Ghose (2005年)
20.	2005	人工 蜂 群 算法 (ABC)。	Karaboga (2005年)
21.	2006	入侵 杂草 优化 (IWO)	Mehrabian和Lucas (2006年)
22.	2006	Honey-bee Mating 优化 (HMO)。	Haddad等人(2006年)
23.	2006	猫 群 优化	Chu等人(2006年)
24.	2007	智能 水滴 算法 (IWD)。	Hosseini (2007)
25.	2007	帝国 竞争 算法 (ICA)。	Atashpaz-Gargari和Lucas (2007)
26.	2008	水滴 优化 算法 (WDO)	Shah-Hosseini (2008)
27.	2007	猴子 搜索 算法。	Mucherino和Seref (2007)
28.	2009	萤火虫 算法 (FA)。	Yang (2009年)
29.	2008	植物 生长 优化	Wang 和 Cheng (2008)
30.	2008	生物地理学 优化 (BBO)	Simon (2008)
31.	2009	联盟 锦标赛 算法 (LCA)。	卡山 (2009年)
32.	2009	引力 搜索 算法 (GSA)	Rashedi等人 (2009年)
33.	2009	布谷鸟 搜索 (CS)。	Yang和Deb (2009年)
34.	2010	蝙蝠算法 (BA)。	Yang (2010)
35.	2010	烟花 算法	Tan 和 Zhu (2010)
36.	2010	化学反应 算法	Ramaswamy 和 Sbalzarini (2010)
37.	2011	基于银河的搜索算法 (GbSA)。	Shah-Hosseini (2011)
38.	2011	螺旋优化 imiz 算法 (SO)。	Tamura 和 安田 (20 11)
39.	2011	基于教与学的优化 (TLBO) 算法。	Rao等人 (2011)
40.	2012	磷虾群 (KH) 算法。	Gandomi和Alavi (2012)
41.	2012	花朵 授粉 算法 (FPA)	Yang (2012b)
42.	2012	差分 搜索 算法 (DSA)。	西维乔格鲁 (2012)
43.	2013	带有混沌的萤火虫算法 (FA)	Gandomi等人 (2013)
44.	2014	碰撞 体 优化 (CBO)	Kaveh和Mahdavi (2014)
45.	2014	动物 迁徙 优化	Li等人 (2014c)
46.	2014	向心 加速 粒子 群 优化 (CAPSO)	Beheshti and Shamsuddin (2014)
47.	2014	灰狼 优化 算法 (GWO)	Mirjalili等人 (2014)
48.	2014	大黄蜂 交配 优化 算法	Marinakis 和 Marinaki (2014)
49.	2014	纵横 优化 算法 (COA)	Meng 等人 (2014)

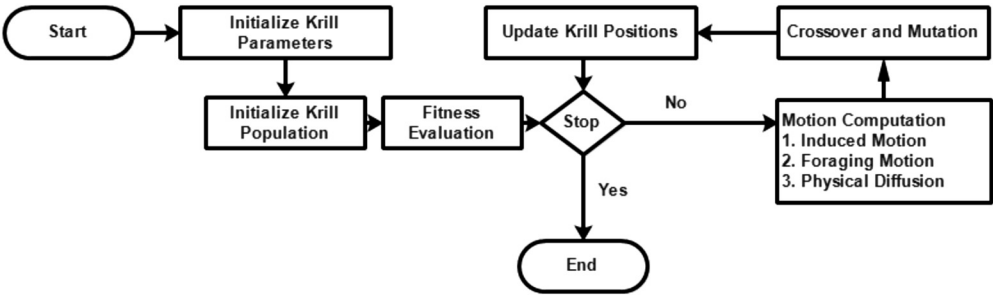


图 16.基本 Krill Head 应用流程图。

专家和科学家的广泛关注。关于 KH 算法的更多细节见 Gandomi 和 Alavi (2012)。图 16 展示了基本的 KH 应用流程图。KH 算法的其他变体见 Li 等 (2014a)、Yaghoobi 和 Mojallali (2016)、Wang 等 (2019)。

4.13.1. KH 算法 用于 PID 整定

KH算法已被用于解决多个工程问题。这些KH算法的工程应用由 Wang等人(2019)归类，包括：连续优化、组合

Table 3. Comprehensive list of some popular metaheuristic algorithms (2015–2020).

No.	Year of creation	Name of algorithm	Reference
50.	2015	Water Wave Optimization	Zheng (2015)
51.	2015	Ions Motion Algorithm	Javidy et al. (2015)
52.	2015	Social Spider Algorithm (SSA)	James and Li (2015)
53.	2015	Bull Optimization Algorithm	FINDIK (2015)
54.	2015	Elephant Herding Optimization	Wang et al. (2015)
55.	2016	Jaya Algorithm	Rao (2016)
56.	2016	Raven Roosting Optimization Algorithm	Brabazon et al. (2016)
57.	2016	FIFA World Cup Competitions Algorithm	Razmjooy et al. (2016)
58.	2016	Spider Monkey Optimization (SMO)	Al-Azza et al. (2016)
59.	2017	Smart Structured Algorithm (SSA)	Salmani and Eshghi (2017)
60.	2017	Opposition-Based Water Wave Optimization	Wu et al. (2017)
61.	2017	Primal-Dual Asynchronous Particle Swarm Optimization (pdAPSO)	Dada et al. (2017)
62.	2018	Primal-Dual Particle Swarm Optimization (pdPSO)	Dada and Ramlan (2018)
63.	2018	Farm Fertility.	Shayanfar and Gharehchopogh (2018)
64.	2018	Coyote Optimization Algorithm (COA).	Pierezan and Coelho (2018)
65.	2019	Wolf Pack Search and Local Search (WPS-LS).	Dong et al. (2019)
66.	2019	Chaotic Dolphin Swarm Algorithm (CDSA)	Qiao and Yang (2019)
67.	2019	Emperor Penguins Colony (EPC).	Harifi et al. (2019)
68.	2019	Pathfinder Algorithm (PFA)	Yapici and Cetinkaya (2019)
69.	2019	Squirrel Search Algorithm (SSA)	Jain et al. (2019)
70.	2019	Algorithm of the Innovative Gunner (AIG).	Pijarski and Kacejko (2019)
71.	2020	Butterfly Optimization Algorithm (BOA).	Arora and Singh (2019)
72.	2020	Levy Flight Distribution (LFD).	Houssein et al. (2020)
73.	2020	Shuffled Shepherd Optimization Algorithm (SSOA).	Kaveh and Zaerreza (2020)

optimization, constrained optimization, multi-objective optimization, dynamic and noisy environment applications. Supplementary specific application areas of KH include: electric and power systems (Bulbul et al., 2018), wireless and network systems (Kalaiselvi and Radhakrishnan, 2015; Karthick and Palanisamy, 2019), neural network training (Kowalski and Lukasik, 2016), robotics (Rao et al., 2018), clustering (Abualigah et al., 2017) and medicine (Kumari and Arumugam, 2015). Also, KH has been used for design of industrial controllers. An example of such can be seen in Younesi and Tohidi (2015).

The KH algorithm has been shown to be stable and robust in the tuning of PID settings. The KH method was used to propose optimal tuning of PID controller parameters for frequency stability in an isolated wind-diesel power system (Shayanfar et al., 2015). The simulation was run in four different scenarios: a step change in diesel side load, a step change in wind speed, a random change in diesel side load, and a random change in wind speed. The Bee Colony algorithm was used to compare the outcomes of the KH technique. When compared to BC-based PID controllers (BC-PID), the results showed that KH-based PID controllers (KH-PID) perform better. Optimal tuning of PID parameters for First Order Plus Dead Time system (FOPDT) and fourth order system in Alikhani et al. (2013). The results suggest that the KH algorithm is beneficial for controller tuning. The PID-KH algorithm was used to propose and illustrate optimal control of a state microgrid system (Regad et al., 2020). An objective function is used to optimize the gains of PID controllers. The simulation results were shown, and using the KH algorithm improves the system’s performance in comparison to the GA and PSO focusing on PID. The efficiency of the system has improved. Other PID-controlled factories that use KH are: Power System Stabilizer (Mukherjee et al., 2019), Power System Restoration (PSR) Indices (Chandrasekar et al., 2017), load frequency load control (Alam et al., 2016).

Furthermore, several other variants of KH algorithm have also been implemented for PID controller tuning. Some of such proposals can be seen in Yaghoobi and Mojallali (2016).

4.14. Whale optimization algorithm

Mirjalili and Lewis (2016) proposed the Whale Optimization Algorithm (WOA) in 2016. It is a stochastic metaheuristic algorithm. WOA was inspired by the hunting activity or tactic of humpback whales. WOA’s methodology treats each and every solution as a whale. In this

method, a whale tries to fill a new place in a search region that is thought to be the group’s best component. Whales use two search tactics to locate prey and attack. Prey encirclement and the development of bubble nets are two of these search techniques. In terms of optimization, as whales look for prey, they explore the search space, and exploitation happens during attack activity. Whales, like humans, have spindle cells that control emotions, judgment, and social behavior (Hof and Van der Gucht, 2007).

The implementation flow diagram for WOA is presented in Fig. 17. Whale optimization algorithm is formulated based on the following steps:

- i. Initialization of population number, maximum iteration number and design variables.
- ii. For every design variable, compute fitness function for each search agent and pick up best solution (optimum) and other search agent tries to update their position to the best one.
- iii. Update the position for each search agent.

Further readings and details of WOA can be seen in Mirjalili and Lewis (2016); Gharehchopogh and Gholizadeh (2019); Adhirai et al. (2018); Mirjalili et al. (2020); Mohammed et al. (2019).

4.14.1. Whale optimization algorithm for PID tuning

WOA is employed effectively to backup applications of several areas of study. An example of area of application includes engineering optimization problems. Articulating appropriate problems and defining suitable parameters and objective functions are the main concerns that can aid effectual problem solving while handling engineering applications. Appropriate usage of operators, variable settings, and population illustration are the major concerns regarding optimization algorithms.

WOA applications in engineering are classified into: Electrical engineering (Mirjalili et al., 2020), Civil Engineering (Kaveh and Rastegar Moghaddam, 2018), Classification, Clustering, Image Processing, Mechanical Engineering (Algabalawy et al., 2018), Robotics (Elhosseini et al., 2019), Computer Networks (Pham et al., 2020), Industrial Engineering (Khalilpourazari et al., 2019), Task Scheduling (Chen et al., 2020), and Control Engineering (Saha and Saikia, 2017), applications.

Though controllers may overcome some stability glitches nonetheless they do not resolve the problems of nonlinear loads, variable operating points and time delays. Optimization methods are used to over-

表3. 一些流行元启发式算法（2015–2020）的 综合 列表。

No.	创建年份	算法 名称 算法	参考文献
50.	2015	水 波 优化	Zheng (2015)
51.	2015	离子 运动 算法	Javidy等人 (2015)
52.	2015	社会 蜘蛛 算法 (SSA)	James和Li (2015)
53.	2015	公牛 优化 算法	Findik (2015)
54.	2015	大象 群 优化	Wang等人 (2015)
55.	2016	Jaya 算法	Rao (2016)
56.	2016	渡鸦 栖息 优化 算法	Brabazon等人 (2016)
57.	2016	国际足联 世界杯 比赛 算法	Razmjooy等人 (2016)
58.	2016	蜘蛛猴 优化 (SMO)	Al-Azza等人 (2016)
59.	2017	智能 结构化 算法 (SSA)	Salmani 和 Eshghi (2017)
60.	2017	基于对立的学习 水 波 优化	Wu 等人 (2017)
61.	2017	原始-对偶 异步 粒子 群 优化 (pdAPSO)	Dada 等人 (2017)
62.	2018	原始-对偶 粒子 群 优化 (pdPSO)	Dada 和 Ramlan (2018)
63.	2018	农场 生育力。	Shayanfar 和 Gharehchopogh (2018)
64.	2018	郊狼 优化 算法 (COA)。	Pierezan 与 Coelho (2018)
65.	2019	狼 群 搜索 与 局部 搜索 (WPS-LS) 。	Dong等人(2019)
66.	2019	混沌 海豚 群 算法 (CDSA)	乔和Yang(2019)
67.	2019	帝王 企鹅 群落 (进化规划群落, EPC) 。	Harifi等人(2019)
68.	2019	寻路者 算法 (PFA)	Yapici and Cetinkaya (2019)
69.	2019	松鼠 搜索 算法 (SSA)	Jain 等 (2019)
70.	2019	创新炮手 算法 (AIG)。	Pijarski 和 Kacejko (2019)
71.	2020	蝴蝶 优化 算法 (BOA)。	Arora和Singh(2019)
72.	2020	Levy 飞行 分布 (LFD)。	Houssein等人(2020)
73.	2020	混洗 牧羊犬 优化 算法 (SSOA)。	Kaveh和Zaerreza(2020)

优化、约束优化、多目标优化、动态和噪声环境应用。KH的其他具体应用领域包括：电力系统（Bulbul等人，2018）、无线和网络系统（Kalaiselvi和Radhakrishnan，2015；Karthick和Palanisamy，2019）、神经网络训练（Kowalski和Łukasik，2016）、机器人学（Rao等人，2018）、聚类（Abualigah等人，2017）和医学（Kumari和Arumugam，2015）。此外，KH已被用于工业控制器的设计。Younesi和Tohidi（2015）中可以看到这样的例子。

KH算法已被证明在PID参数整定中具有稳定性和鲁棒性。KH方法被用于提出针对孤立风柴电力系统频率稳定性的PID控制器参数最优整定（Shayanfar等人，2015）。仿真在四种不同场景下进行：柴油侧负载阶跃变化、风速阶跃变化、柴油侧负载随机变化和风速随机变化。使用蜂群算法来比较KH技术的结果。与基于BC的PID控制器（BC-PID）相比，结果表明基于KH的PID控制器（KH-PID）性能更优。在Alikhani等人（2013）中，对一阶加死区系统（FOPDT）和四阶系统进行了PID参数最优整定。结果表明KH算法有益于控制器整定。PID-KH算法被用于提出并说明状态微电网系统的最优控制（Regad等人，2020）。使用目标函数优化PID控制器的增益。展示了仿真结果，与遗传算法和粒子群优化算法在PID上的表现相比，使用KH算法提高了系统性能。系统效率得到提升。其他使用KH的PID控制工厂包括：电力系统稳定器（Mukherjee等人，2019）、电力系统恢复（PSR）指标（Chandrasekar等人，2017）、负载频率负载控制（Alam等人，2016）。

此外，KH算法的其他几种变体也已用于PID控制器参数整定。其中一些提议可见于Yaghoobi和Mojallali（2016）。

4.14. 鲸鱼 优化 算法

Mirjalili和Lewis（2016）在2016年提出了鲸鱼优化算法（WOA）。它是一种随机元启发式算法。WOA的灵感来自座头鲸的捕食活动或策略。WOA的方法论将每个解视为一头鲸鱼。在这种

方法中，鲸鱼试图在搜索区域中找到一个被认为是群体最佳组成部分的新位置。鲸鱼使用两种搜索策略来定位猎物并进行攻击。猎物包围和气泡网的开发是这些搜索技术中的两种。在优化方面，当鲸鱼寻找猎物时，它们探索搜索空间，而开发发生在攻击活动中。与人类一样，鲸鱼拥有控制情绪、判断和社会行为的纺锤体细胞（Hof and Van der Gucht, 2007）。

WOA的 实现 流程图 如图 17 所示 。 鲸鱼 优化 算法 基于 以下步骤 进行 公式化 ：

- i. 初始化群体数量、最大迭代次数和设计变量。ii. 对于每个设计变量，计算每个搜索代理的适应度函数，选取最佳解（最优解），其他搜索代理尝试更新其位置到最佳位置。iii. 更新每个搜索代理的位置。

WOA的进一步阅读和详细信息可见于Mirjalili和Lewis（2016）；Gharehchopogh和Gholizadeh（2019）；Adhirai等人（2018）；Mirjalili等人（2020）；Mohammed等人（2019）。

4.14.1. 用于 PID 整定 的 鲸鱼 优化算法

WOA被有效应用于多个研究领域的备份应用。应用领域的一个例子包括工程优化问题。清晰阐述适当的问题并定义合适的参数和目标函数，是处理工程应用时有助于有效解决问题的关键。操作符的恰当使用、变量设置和群体表示是优化算法的主要关注点。

WOA在工程中的应用分类为：电气工程（Mirjalili et al., 2020）、土木工程（Kaveh and Rastegar Moghaddam, 2018）、分类、聚类、图像处理、机械工程（Algabalawy et al., 2018）、机器人学（Elhosseini et al., 2019）、计算机网络（Pham et al., 2020）、工业工程（Khalilpourazari et al., 2019）、任务调度（Chen et al., 2020）以及控制工程（Saha and Saikia, 2017）应用。

尽管 控制器 可能 克服 一些 稳定性 问题，然而 它们 并 没有 解决 非线性 负载、可变 操作 点 和 时 延 的问题。优化 方法 被 用来 通过 整定 控制器 参数 来 克服

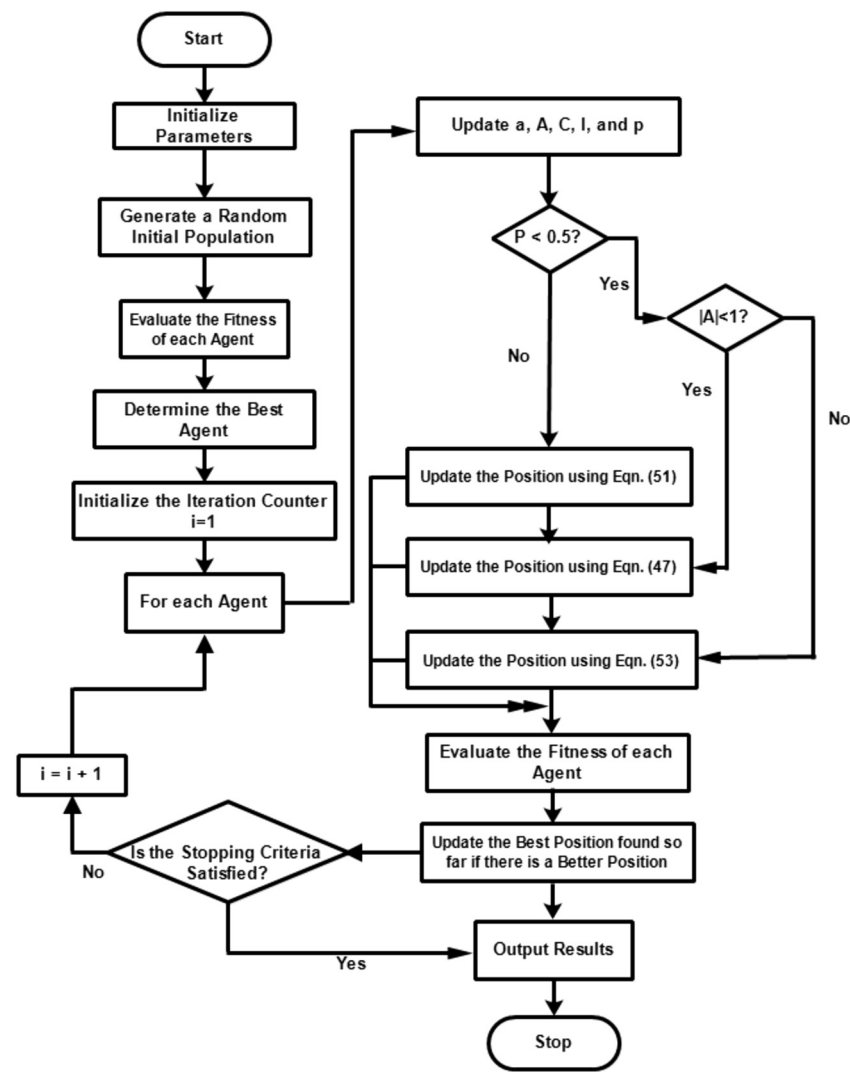


Fig. 17. Whale optimization implementation flow diagram.

come these problems through tuning the controller parameters. One of such techniques is WOA. Tuning of PID controller parameters using WOA for Load Frequency Control (LFC) of two-area interconnected non-reheat thermal power systems in order to maintain the system frequency at a prescheduled value in the presence of disturbances was presented in Bhatt et al. (2019). In Debnath et al. (2018), the parameters of a non-integer type PID controller was successfully tunes using Whale Optimization technique for the adjustment of frequency and interline power oscillations in multi-generation type unified system. Whale optimizer algorithm to regulate PID parameters for tracking control of 2-DOF robot manipulator was presented in Loucif et al. (2020). The results obtained witness the efficacy of WOA-PID in terms settling time, errors, and convergence time as well as its robustness for tuning parameters PID controller for robot tracking control with or without disturbances. Other plants controlled by PID and tuned with WOA include; AVR system (Mosaad et al., 2019a), DC-DC buck converter design (Hekimoğlu et al., 2018) and high-performance drilling machine (Kumar and Kumar, 2018).

5. Open research problems

Notwithstanding the accomplishments of modern metaheuristic algorithms, a number of important concerns remain unresolved. Heuristic algorithms' operating principles are well understood, although comprehending why they operate is only partially understood. Unfortunately, it

is difficult to mathematically explain why these algorithms are so successful, despite some interesting previous efforts (Yang, 2011). These are, in fact, unsolved open research problems. To the best of our knowledge, except for mathematical study of SA and PSO, the convergence of any other methods has not been proven using mathematical models. As a result, any mathematical examination will provide important insight into algorithms. This will be useful for advancing more effective variants of these algorithms or demonstrating inventive approaches of constructing new algorithms.

Furthermore, it is only poorly understood why distinct components of heuristics and metaheuristics interact in an intelligent and stable manner to produce effective algorithms that aggregate under specified conditions. Why, for example, does a well-balanced mix of randomization and determinism produce a far more effective algorithm (than a pure deterministic or random algorithm)? How do you tell whether you have reached a state of stability? How can you demonstrate that memory usage can significantly improve an algorithm's search efficiency? What are the circumstances? It can be demonstrated from the well-known No-Free-Lunch assertions of 1997 (Wolpert et al., 1997) that they have been validated for single-objective optimization, but not for multiobjective optimization. What are the implications for algorithm design if they are proven to be true (or not) for multiobjective optimization?

Similarly, there is no universally accepted metric for comparing the performance of diverse algorithms; nonetheless, the assumed absolute

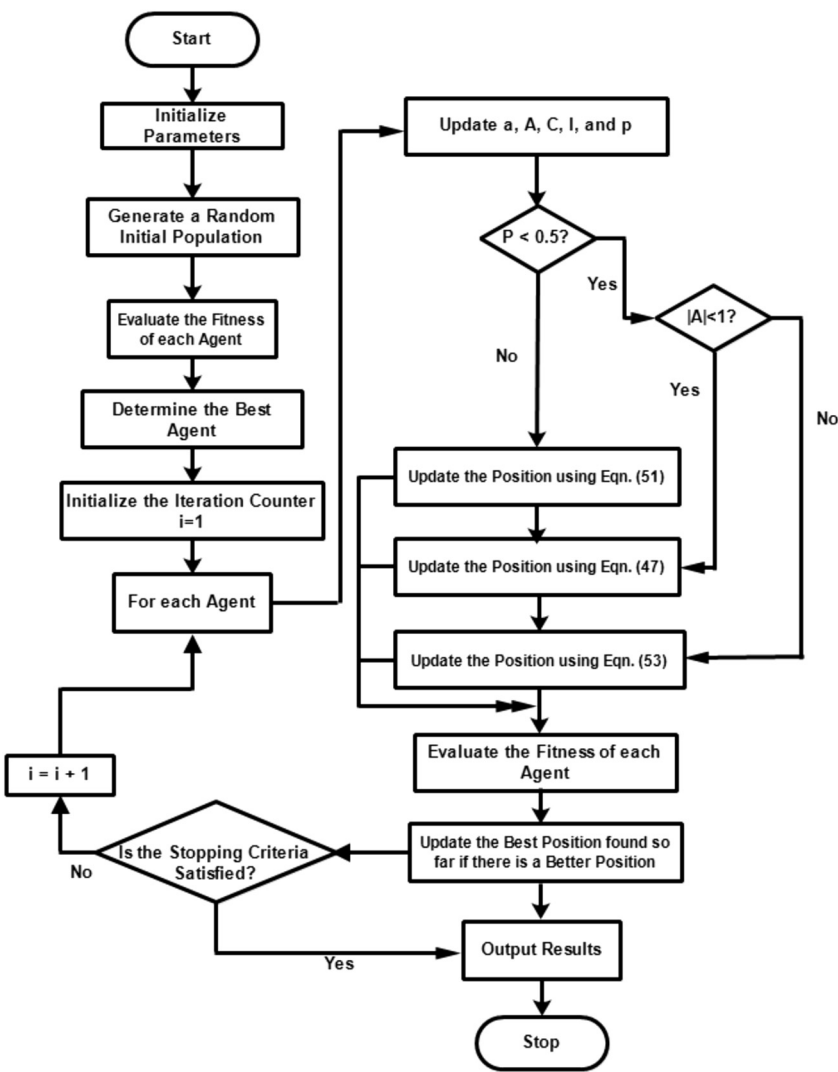


图 17. 鲸鱼优化实现流程图。

这些问题。其中一种技术是WOA。Bhatt等人(2019)提出了使用WOA对两区域互联无再热火力发电系统的负荷频率控制(LFC)进行PID控制器参数整定，以在存在扰动情况下维持系统频率在预定值。在Debnath等人(2018)中，成功使用鲸鱼优化技术对非整数型PID控制器的参数进行整定，用于调节多代类型统一系统中的频率和联络线功率振荡。Loucif等人(2020)提出了使用鲸鱼优化算法调节PID参数以实现二自由度机器人操纵器的跟踪控制。所得结果证明了WOA-PID在调节时间、误差和收敛时间方面的有效性，以及其在有或没有扰动的情况下整定机器人跟踪控制PID控制器参数的鲁棒性。其他由PID控制并使用WOA整定的对象包括：AVR系统(Mosaad等人，2019a)、DC-DC降压变换器设计(Hekimoğlu等人，2018)和高性能钻机(Kumar和Kumar，2018)。

5. 开放研究问题

尽管现代元启发式算法取得了成就，但一些重要问题仍然未解。启发式算法的运行原理已被充分理解，尽管我们只是部分理解了它们为何有效。不幸的是，这

尽管此前有一些有趣的尝试 (Yang, 2011)，但从数学上解释这些算法为何如此成功仍然困难重重。这些实际上是尚未解决的开放性研究问题。据我们所知，除了对模拟退火和粒子群优化算法的数学研究外，其他任何方法的收敛性尚未通过数学模型得到证明。因此，任何数学检验都将为算法提供重要见解。这将有助于推进这些算法的更有效变体，或展示构建新算法的创新方法。

此外，目前人们对启发式算法和元启发式算法中不同组件如何以智能且稳定的方式相互作用，以产生在特定条件下聚合的有效算法这一点仍然知之甚少。例如，为什么随机化和确定性的良好平衡混合能产生比纯确定性或随机算法更有效的算法？如何判断是否已达到稳定状态？如何证明记忆的使用可以显著提高算法的搜索效率？其条件是什么？从1997年著名的无免费午餐论断 (Wolpert等人，1997) 可以看出，这些论断已在单目标优化中得到验证，但在多目标优化中尚未得到验证。如果这些论断在多目标优化中被证实为真（或假），对算法设计有何影响？

同样，也没有一种普遍接受的度量标准用于比较不同算法的性能；然而，假设的绝对

objective value and the number of functional evaluations are two commonly used measures. The attempt to provide a universal framework was made by Shilane et al. (2008). A formal theoretical investigation, on the other hand, has yet to be formed (Chiarandini et al., 2007). This could be a fantastic place for students or scholars to conduct further investigation.

Natural World has a nearly infinite number of problem-solving strategies. Individuals are undoubtedly urged to design more powerful and efficient new generation algorithms through diligent observation. In nature, intelligence is a result of biological evolution. In the future, some intelligent algorithms (or systems) may originate, evolve, and modify appropriately to provide efficient and intelligent solutions to NP-hard optimization problems. Furthermore, a current trend is to use simple metaheuristic algorithms to solve difficult optimization problems. Possibly the need for more complicated metaheuristic algorithms that can genuinely mirror the exact operational technique of some natural or biological systems motivates the development of self-regulating, self-evolving, and fully intelligent metaheuristics in the long term.

Although the essential components of most algorithms are well understood, how they interact to achieve efficiency remains a secret, motivating further intensive research. Convergence analysis of a small number of algorithms, such as PSO, demonstrates some knowledge; nevertheless, common mathematical analysis of metaheuristic algorithms remains unsolved and is an important research area. The choice and precise use of a suitable algorithm is required for the solution of an optimization issue. The nature of the optimization problem is largely determined by the algorithm chosen. There is no defined policy as to how and what to choose for large-scale nonlinear global optimization issues. Indeed, it is not definite if an effective algorithm exists, which is especially true for NP-hard problems, which represent the majority of real-world problems.

In summary some of these open research problems include:

- i. Some metaheuristic algorithms, such as GWO, can be improved as their fundamental deficiency, notwithstanding their capacity to handle vast numbers of variables and evading local solutions when dealing with complex problems.
- ii. A hybrid metaheuristic algorithm can be created by combining memetic algorithms to solve several of their individual weaknesses.
- iii. To handle dynamic optimization challenges, certain recently proposed metaheuristic algorithms have to be changed.
- iv. A neutral trade-off between local amplification and global heterogeneity is required (Yang, 2011; Blum and Roli, 2003).
- v. When solving optimization problems, the parameters of metaheuristic algorithms have to be fine-tuned (Dokeroglu et al., 2019).
- vi. Metaheuristic algorithms with no parameters have to be developed.
- vii. Statistical analysis approaches for evaluating the performance of metaheuristic algorithms have to be developed.
- viii. To improve performance and resilience in solving emerging problems and cases, transfer learning is incorporated into metaheuristic algorithms (Talbi, 2020).
- ix. Speeding up execution time (process time) of metaheuristic algorithms. Some metaheuristic algorithms like population-based are slow solvers (Alba et al., 2013).
- x. Theoretical and mathematical models are needed for metaheuristic components like exploration versus exploitation, convergence, searching ability and local optimum versus global optimum (Hussain et al., 2019).
- xi. Development of a self-adapting, self-tuning, and self-evolving scalable metaheuristic algorithm. This is to allow algorithms to deal with the complicated and highly unbalanced landscapes of large optimization problems with a large number of decision variables (Hussain et al., 2019).
- xii. How can awareness about the traits and attributes of particular optimization problem classes and cases be acquired, articulated, and evaluated? (Sala and Müller, 2020).

- xiii. How to exploit knowledge or assumptions on particular properties of problem sub-sets or classes, in the selection or development of effective optimization strategies? (Sala and Müller, 2020).

6. Conclusion

This research provides an in-depth look at metaheuristic algorithms and their application to PID controller tuning. An overview of the current state-of-the-art metaheuristic algorithms employed for parameter tuning of PID controllers is provided. The efforts made by different scholars and researchers to solving tuning problems of PID controller parameters using metaheuristic algorithms were discussed. The evolution of metaheuristic algorithms over the years for optimization problems solving was examined. The basic architecture of PID controller and procedures involved tuning its parameters were looked into. The study surveyed some issues related to PID controller tuning and performance metrics utilized to measure effectiveness of tuning approaches for PID control. The challenges of metaheuristic optimization algorithms in efficiently handling issues of tuning was presented. A highlight of some open research problems associated with metaheuristic algorithms was offered. In general, the figure and volume of literature we reviewed revealed that significant progress have been made and will still be made in this area. Having presented open problems on metaheuristic algorithms, further research to improve the effectiveness of metaheuristic algorithms need to be done. This will make the development of metaheuristic algorithms remain an active research area for academicians and industrial practitioners researching metaheuristic optimization approaches for effective optimization and controller tuning. The goal and aspiration of this research is that researchers and students will use this study as a spring board or background for qualitative research in this field.

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References

- Aabid, M., Elakkary, A., Sefiani, N., 2017. PID parameters optimization using ant-colony algorithm for human heart control. In: 2017 23rd International Conference on Automation and Computing. ICAC. IEEE, pp. 1–6.
- Abachizadeh, M., Yazdi, M.R.H., Yousefi-Koma, A., 2010. Optimal tuning of PID controllers using artificial bee colony algorithm. In: 2010 IEEE/ASME International Conference on Advanced Intelligent Mechatronics. IEEE, pp. 379–384.
- Abbasi, E., Naghavi, N., 2017. Offline auto-tuning of a PID controller using extended classifier system (XCS) algorithm. J. Adv. Comput. Eng. Technol. 3, 41–44.
- Abd El Aziz, M., Hassanien, A.E., 2018. Modified cuckoo search algorithm with rough sets for feature selection. Neural Comput. Appl. 29, 925–934.

目标值和函数评估次数是两种常用的度量标准。Shilane 等人 (2008) 尝试提供了一个通用框架。另一方面，形式化理论探究尚未形成 (Chiarandini 等人, 2007)。这可能是学生或学者进行进一步研究的绝佳领域。

自然界拥有几乎无限的问题解决策略。毫无疑问，通过勤奋的观察，人们被激励设计出更强大、更高效的新一代算法。在自然界中，智能是生物进化的结果。未来，一些智能算法（或系统）可能会起源、进化并适当修改，从而为NP难优化问题提供高效且智能的解决方案。此外，当前的趋势是使用简单的元启发式算法来解决困难的优化问题。可能需要更复杂的元启发式算法，这些算法能够真实地反映某些自然或生物系统的精确操作技术，从而在长期内推动自我调节、自我进化以及全智能元启发式的发展。

尽管大多数算法的基本组成部分已得到很好的理解，但它们如何相互作用以实现效率仍然是一个秘密，这促使了进一步深入的研究。对少量算法（如PSO）的收敛性分析展示了部分知识；然而，对元启发式算法的通用数学分析仍未解决，且是一个重要的研究领域。解决优化问题需要选择并精确使用合适的算法。优化问题的性质在很大程度上取决于所选算法。对于大规模非线性全局优化问题，没有明确的策略规定如何选择以及选择什么。事实上，是否存在一个有效的算法并不确定，这对于代表大多数实际问题的NP难问题尤其如此。

总之，一些这些开放性研究问题包括：

- i. 一些元启发式算法，如灰狼优化，尽管能够处理大量变量并在处理复杂问题时避免局部解，但因其根本缺陷仍可改进。
- ii. 通过结合模因算法可以创建混合元启发算法，以解决它们各自的弱点。
- iii. 为处理动态优化挑战，某些最近提出的元启发式算法必须进行更改。
- iv. 需要在局部放大和全局异质性之间进行中性权衡 (Yang, 2011; Blum 和 Roli, 2003)。
- v. 在解决优化问题时，元启发式算法的参数必须进行微调 (Dokeroglu 等人, 2019)。
- vi. 必须开发无参数的元启发式算法。
- vii. 必须开发用于评估元启发式算法性能的统计分析方法。
- viii. 为了提高解决新兴问题和案例的性能和韧性，将迁移学习融入元启发式算法 (Talbi, 2020)。
- ix. 加速元启发式算法的执行时间（处理时间）。一些元启发式算法（如基于群体的）是慢速求解器 (Alba 等人, 2013)。
- x. 元启发式组件（如探索与利用、收敛、搜索能力、局部最优与全局最优）需要理论和数学模型 (Hussain 等人, 2019)。
- xi. 开发自适应、自整定和自进化的可扩展元启发式算法。这是为了让算法能够处理具有大量决策变量的大型优化问题的复杂且高度不平衡的景观 (Hussain 等人, 2019)。
- xii. 如何获取、阐述和评估特定优化问题类别和案例的特性和属性？ (Sala 和 Müller, 2020)。

- xiii. 如何利用关于问题子集或类别特定属性的知识或假设，来选择或开发有效的优化策略？ (Sala 和 Müller, 2020)。

6. 结论

本研究深入探讨了元启发式算法及其在PID控制器参数整定中的应用。概述了当前用于PID控制器参数整定的最先进的元启发式算法。讨论了不同学者和研究人员在利用元启发式算法解决PID控制器参数整定问题方面所做的努力。审视了元启发式算法多年来在解决优化问题方面的演变。研究了PID控制器的基本架构以及整定其参数所涉及的过程。调查了与PID控制器参数整定相关的一些问题，以及用于衡量PID控制整定方法有效性的性能指标。提出了元启发式优化算法在有效处理整定问题方面所面临的挑战。重点介绍了一些与元启发式算法相关的开放性研究问题。总体而言，我们审查的文献数量和规模表明，该领域已经取得了显著进展，并且还将继续取得进展。在提出元启发式算法的开放性问题上，需要进一步研究以提高元启发式算法的有效性。这将使元启发式算法的开发成为学者和工业从业者研究用于有效优化和控制器整定的元启发式优化方法的活跃研究领域。本研究的目的是希望研究人员和学生能够以此研究作为在该领域进行定性研究的跳板或背景知识。

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参考文献

- Aabid, M., Elakkary, A., Sefiani, N., 2017年. 使用蚁群算法优化PID参数用于人类心脏控制. 见: 2017年第23届自动化与计算国际会议. ICAC. IEEE, 第1-6页. Abachizadeh, M., Yazdi, M.R.H., Yousefi-Koma, A., 2010年. 使用人工蜂群算法最优整定PID控制器. 见: 2010年IEEE/ASME先进智能机电一体化国际会议. IEEE, 第379-384页. Abbasi, E., Naghavi, N., 2017年. 使用扩展分类器系统(XCS)算法的PID控制器离线自整定. J. Adv. Comput. Eng. Technol. 3, 41-44. Abd El Aziz, M., Hassanien, A.E., 2018年. 基于粗糙集的改进布谷鸟搜索算法用于特征选择. Neural Comput. Appl. 29, 925-934.

Abdel-Basset, M., Abdel-Fatah, L., Sangaiah, A.K., 2018. Metaheuristic algorithms: a comprehensive review. In: Computational Intelligence for Multimedia Big Data on the Cloud with Engineering Applications. Elsevier, pp. 185–231.

Abraham, A., Nedjah, N., de Macedo Mourelle, L., 2006. Evolutionary computation: from genetic algorithms to genetic programming. In: Genetic Systems Programming. Springer, pp. 1–20.

Abu-Mouti, F.S., El-Hawary, M.E., 2012. Overview of artificial bee colony (ABC) algorithm and its applications. In: 2012 IEEE International Systems Conference SysCon 2012. IEEE, pp. 1–6.

Abualigah, L.M., Khader, A.T., Hanandeh, E.S., Gandomi, A.H., 2017. A novel hybridization strategy for krill herd algorithm applied to clustering techniques. Appl. Soft Comput. 60, 423–435.

Adhirai, S., Mahapatra, R., Singh, P., 2018. The whale optimization algorithm and its implementation in Matlab. Int. J. Comput. Inf. Eng. 12, 815–822.

Adubi, S.A., Misra, S., 2014. A comparative study on the ant colony optimization algorithms. In: 2014 11th International Conference on Electronics, Computer and Computation. ICECCO. IEEE, pp. 1–4.

Agarwal, J., Parmar, G., Gupta, R., Sikander, A., 2018. Analysis of grey wolf optimizer based fractional order PID controller in speed control of DC motor. Microsyst. Technol. 24, 4997–5006.

Agarwal, S., Yadav, D., Verma, A., 2017. Speed control of PMSM drive using bacterial foraging optimization. In: 2017 4th IEEE Uttar Pradesh Section International Conference on Electrical, Computer and Electronics. UPCon. IEEE, pp. 84–90.

Agrawal, V., Sharma, H., Bansal, J.C., 2012. Bacterial foraging optimization: a survey. In: Proceedings of the International Conference on Soft Computing for Problem Solving. SocProS 2011, December 20–22, 2011. Springer, pp. 227–242.

Akbarzadeh, A., Shadkam, E., 2015. The study of cuckoo optimization algorithm for production planning problem. arXiv preprint. arXiv:1508.01310.

Al-Azza, A.A., Al-Jodah, A.A., Harackiewicz, F.J., 2016. Spider monkey optimization (SMO): a novel optimization technique in electromagnetics. In: 2016 IEEE Radio and Wireless Symposium. RWS. IEEE, pp. 238–240.

Al-Betar, M.A., Awadallah, M.A., Faris, H., Yang, X.-S., Khader, A.T., Alomari, O.A., 2018. Bat-inspired algorithms with natural selection mechanisms for global optimization. Neurocomputing 273, 448–465.

Al-Tashi, Q., Kadir, S.J.A., Rais, H.M., Mirjalili, S., Alhussian, H., 2019. Binary optimization using hybrid grey wolf optimization for feature selection. IEEE Access 7, 39496–39508.

Alam, M.S., Singh, A., Guha, D., 2016. Optimal solutions of load frequency control problem using oppositional krill herd algorithm. In: 2016 IEEE First International Conference on Control, Measurement and Instrumentation. CMI. IEEE, pp. 6–10.

Alba, E., Luque, G., Nesmachnow, S., 2013. Parallel metaheuristics: recent advances and new trends. Int. Trans. Oper. Res. 20, 1–48.

Algabalawy, M.A., Abdelaziz, A.Y., Mekhamer, S.F., Aleem, S.H.A., 2018. Considerations on optimal design of hybrid power generation systems using whale and sine cosine optimization algorithms. J. Electr. Syst. Inf. Technol. 5, 312–325.

Ali, A., Majhi, S., 2006. Design of optimum PID controller by bacterial foraging strategy. In: 2006 IEEE International Conference on Industrial Technology. IEEE, pp. 601–605.

Alikhani, A., Suratgar, A.A., Nouri, K., Nouredanesh, M., Salimi, S., 2013. Optimal PID tuning based on krill herd optimization algorithm. In: The 3rd International Conference on Control, Instrumentation, and Automation. IEEE, pp. 11–15.

de Almeida, B.S.G., Leite, V.C., 2019. Particle swarm optimization: a powerful technique for solving engineering problems. In: Ser, J.D., Villar, E., Osaba, E. (Eds.), Swarm Intelligence. IntechOpen, Rijeka. Chapter 3.

Alqudah, A., Malkawi, A., Alwadie, A., et al., 2014. Adaptive control of DC-DC converter using simulated annealing optimization method. J. Signal Inf. Process. 5, 198.

Alsmadi, M.K., 2018. Query-sensitive similarity measure for content-based image retrieval using meta-heuristic algorithm. J. King Saud Univ., Comput. Inf. Sci. 30, 373–381.

Amaral, J.M., Tanscheit, R., Pacheco, M.A., 2018. Tuning PID controllers through genetic algorithms. Complex Syst. 2, 3.

Ang, K.H., Chong, G., Li, Y., 2005. PID control system analysis, design, and technology. IEEE Trans. Control Syst. Technol. 13, 559–576.

Annisa, J., Darus, I.Z.M., Tokhi, M.O., Mohamaddan, S., 2018. Implementation of PID based controller tuned by evolutionary algorithm for double link flexible robotic manipulator. In: 2018 International Conference on Computational Approach in Smart Systems Design and Applications. ICASSDA. IEEE, pp. 1–5.

Anthony, K., Ho, P., 2020. Fundamental of PID control. <https://pdhonline.com/courses/e331/e331content.pdf>.

Arora, S., Singh, S., 2019. Butterfly optimization algorithm: a novel approach for global optimization. Soft Comput. 23, 715–734.

Åström, K.J., Hägglund, T., 1984. Automatic tuning of simple regulators with specifications on phase and amplitude margins. Automatica 20, 645–651.

Åström, K.J., Hägglund, T., 1995. PID Controllers: Theory, Design, and Tuning, vol. 2. Instrument Society of America Research, Triangle Park, NC.

Åström, K.J., Hägglund, T., 2004. Revisiting the Ziegler–Nichols step response method for PID control. J. Process Control 14, 635–650.

Atashpaz-Gargari, E., Lucas, C., 2007. Imperialist competitive algorithm: an algorithm for optimization inspired by imperialistic competition. In: 2007 IEEE Congress on Evolutionary Computation, pp. 4661–4667.

Azad, A.S., Vasant, P., Watada, J., Al Sokkalingam, R., 2020. Meta-heuristic approaches for the optimization of hydropower energy: a review. In: Handbook of Research on Smart Technology Models for Business and Industry, pp. 351–375.

Azar, A.T., Ammar, H.H., Ibrahim, Z.F., Ibrahim, H.A., Mohamed, N.A., Taha, M.A., 2019. Implementation of PID controller with pso tuning for autonomous vehicle. In: International Conference on Advanced Intelligent Systems and Informatics. Springer, pp. 288–299.

Azman, A.A., Rahiman, M.H.F., Mohammad, N.N., Marzaki, M.H., Taib, M.N., Ali, M.F., 2017. Modeling and comparative study of PID Ziegler Nichols (ZN) and Cohen-Coon (CC) tuning method for multi-tube aluminum sulphate water filter (MTAS). In: 2017 IEEE 2nd International Conference on Automatic Control and Intelligent Systems. I2CACIS. IEEE, pp. 25–30.

Bagis, A., Senberber, H., 2017. ABC algorithm based PID controller design for higher order oscillatory systems. Elektron. Elektrotech. 23, 3–9.

Baker, G., 2011. PID tuning of plants with time delay using root locus.

Bansal, H.O., Sharma, R., Shreeraman, P., 2012. PID controller tuning techniques: a review. J. Control Eng. Technol. 2, 168–176.

Bansal, R., Jain, M., Bhushan, B., 2014. Designing of multi-objective simulated annealing algorithm tuned PID controller for a temperature control system. In: 2014 6th IEEE Power India International Conference. PIICON. IEEE, pp. 1–6.

Baselga, S., 2011. Second order design of geodetic networks by the simulated annealing method. J. Surv. Eng. 137, 167–173.

Bassi, S.J., Mishra, M.K., Omizegba, E.E., 2011. Automatic tuning of proportional-integral-derivative (PID) controller using particle swarm optimization (PSO) algorithm. Int. J. Artif. Intell. Appl. 2, 25.

Battiti, R., Tecchiolli, G., 1994. The reactive tabu search. ORSA J. Comput. 6, 126–140.

Bayoumi, E.H.E., Salem, F., 2012. PID controller for series-parallel resonant converters using bacterial foraging optimization. Electromotion Sci. J. 19, 64–78.

Beheshti, Z., Shamsuddin, S.M.H., 2013. A review of population-based meta-heuristic algorithms. Int. J. Adv. Soft Comput. Appl. 5, 1–35.

Beheshti, Z., Shamsuddin, S.M.H., 2014. CAPSO: centripetal accelerated particle swarm optimization. Inf. Sci. 258, 54–79.

Benbouabdallah, K., Zhu, Q.-d., 2013. Bacterial foraging oriented by particle swarm optimization of a Lyapunov-based controller for mobile robot target tracking. In: 2013 Ninth International Conference on Natural Computation. ICNC. IEEE, pp. 506–511.

Bennett, S., 1993. A History of Control Engineering, 1930–1955. IET, p. 47.

Bernstein, D.S., 2002. What makes some control problems hard? IEEE Control Syst. Mag. 22, 8–19.

Beyer, H.-G., Schwefel, H.-P., 2002. Evolution strategies – a comprehensive introduction. Nat. Comput. 1, 3–52.

Bhandari, A.K., Singh, V.K., Kumar, A., Singh, G.K., 2014. Cuckoo search algorithm and wind driven optimization based study of satellite image segmentation for multilevel thresholding using Kapur's entropy. Expert Syst. Appl. 41, 3538–3560.

Bharat, S., Ganguly, A., Chatterjee, R., Basak, B., Sheet, D.K., Ganguly, A., 2019. A review on tuning methods for PID controller. Asian J. Conver. Technol.

Bhatt, R., Parmar, G., Gupta, R., 2019. Performance analysis of WOA optimised PID controllers for LFC of interconnected thermal power systems. Int. J. Comput. Syst. Eng. 5, 179–192.

Bhawana, S., Neelu, J., 2017. Tuning techniques of PID controller: a review. Int. J. Emerg. Technol. 8 (1), 481–485. Special Issue NCETST-2017.

Bingul, Z., Karahan, O., 2018. A novel performance criterion approach to optimum design of PID controller using cuckoo search algorithm for avr system. J. Franklin Inst. 355, 5534–5559.

Biswas, K., Vasant, P.M., Vintaned, J.A.G., Watada, J., 2020. A review of metaheuristic algorithms for optimizing 3D well-path designs. Arch. Comput. Methods Eng.

Biswas, S., Kundu, S., Bose, D., Das, S., 2012. Cooperative co-evolutionary teaching-learning based algorithm with a modified exploration strategy for large scale global optimization. In: International Conference on Swarm, Evolutionary, and Memetic Computing. Springer, pp. 467–475.

Blum, C., Roli, A., 2003. Metaheuristics in combinatorial optimization: overview and conceptual comparison. ACM Comput. Surv. 35, 268–308.

Boussaïd, I., Lepagnot, J., Siarry, P., 2013. A survey on optimization metaheuristics. Inf. Sci. 237, 82–117.

Boyd, S., Baratt, C., Norman, S., 1990. Linear controller design: limits of performance via convex optimization. Proc. IEEE 78, 529–574.

Brabazon, A., Cui, W., O'Neill, M., 2016. The raven roosting optimisation algorithm. Soft Comput. 20, 525–545.

Brajevic, I., Tuba, M., 2014. Cuckoo search and firefly algorithm applied to multilevel image thresholding. In: Cuckoo Search and Firefly Algorithm. Springer, pp. 115–139.

Brogliato, B., Lozano, R., 1994. Adaptive control of first-order nonlinear systems with reduced knowledge of the plant parameters. IEEE Trans. Autom. Control 39, 1764–1768.

Bulbul, S.M.A., Pradhan, M., Roy, P.K., Pal, T., 2018. Opposition-based krill herd algorithm applied to economic load dispatch problem. Ain Shams Eng. J. 9, 423–440.

Burnwal, S., Deb, S., 2013. Scheduling optimization of flexible manufacturing system using cuckoo search-based approach. Int. J. Adv. Manuf. Technol. 64, 951–959.

Černý, V., 1985. Thermodynamical approach to the traveling salesman problem: an efficient simulation algorithm. J. Optim. Theory Appl. 45, 41–51.

Chaib, L., Choucha, A., Arif, S., 2017. Optimal design and tuning of novel fractional order PID power system stabilizer using a new metaheuristic bat algorithm. Ain Shams Eng. J. 8, 113–125.

Chandrasekar, K., Paramasivam, B., Chidambaram, I.A., 2017. Evaluation of power system restoration indices using krill herd algorithm based optimized pi+ controller for a restructured power system with facts devices. J. Eng. Appl. Sci. 12, 4973–4989.

Abdel-Basset, M., Abdel-Fatah, L., Sangaiah, A.K., 2018. 元启发式算法：全面综述。见：云上多媒体大数据工程应用的计算智能。Elsevier, 第185-231页。Abraham, A., Nedjah, N., de Macedo Mourelle, L., 2006. 进化计算：从遗传算法到遗传编程。见：遗传系统编程。Springer, 第1-20页。Abu-Mouti, F.S., El-Hawary, M.E., 2012. 人工蜂群算法(ABC)及其应用概述。见：2012年IEEE国际系统会议SysCon 2012. IEEE, 第1-6页。Abualigah, L.M., Khader, A.T., Hanandeh, E.S., Gandomi, A.H., 2017. 一种用于聚类技术的磷虾群算法新型混合策略。应用软计算 60, 423-435。Adhirai, S., Mahapatra, R., Singh, P., 2018. 鲸鱼优化算法及其在Matlab中的实现。国际计算机与信息工程杂志 12, 815-822。Adubi, S.A., Misra, S., 2014. 蚁群优化算法比较研究。见：2014年第11届国际电子、计算机与计算会议。ICECCO. IEEE, 第1-4页。Agarwal, J., Parmar, G., Gupta, R., Sikander, A., 2018. 基于灰狼优化器的分数阶PID控制器在直流电机速度控制中的分析。微系统技术 24, 4997-5006。Agarwal, S., Yadav, D., Verma, A., 2017. 使用细菌觅食优化的永磁同步电机驱动速度控制。见：2017年IEEE北方部分会第四届国际电气、计算机与电子会议。UPCON. IEEE, 第84-90页。Agrawal, V., Sharma, H., Bansal, J.C., 2012. 细菌觅食优化：综述。见：软计算解决问题国际会议论文集 SocProS 2011, 2011年12月20-22日。Springer, 第227-242页。Akbarzadeh, A., Shadkam, E., 2015. 布谷鸟优化算法在生产规划问题中的研究。arXiv预印本。arXiv:1508.01310。Al-Azza, A.A., Al-Jodah, A.A., Harackiewicz, F.J., 2016. 蜘蛛猴优化：电磁学中的一种新型优化技术。见：2016年IEEE无线电与无线研讨会。RWS. IEEE, 第238-240页。Al-Betar, M.A., Awadallah, M.A., Faris, H., Yang, X.-S., Khader, A.T., Alomari, O.A., 2018. 基于自然选择机制的蝙蝠启发算法用于全局优化。Neurocomputing 273, 448-465。Al-Tashi, Q., Kadir, S.J.A., Rais, H.M., Mirjalili, S., Alhussian, H., 2019. 使用混合灰狼优化的二值优化用于特征选择。IEEE Access 7, 39496-39508。Alam, M.S., Singh, A., Guha, D., 2016. 使用对立的磷虾群算法求解负荷频率控制问题的最优解。见：2016年IEEE第一届控制、测量与仪器国际会议。CMI. IEEE, 第6-10页。Alba, E., Luque, G., Nesmachnow, S., 2013. 并行元启发式：最新进展与新趋势。国际运筹学交易 20, 1-48。Algabalawy, M.A., Abdelaziz, A.Y., Mekhamer, S.F., Aleem, S.H.A., 2018. 使用鲸鱼和正弦余弦优化算法设计混合发电系统最优方案的考虑。电气系统与信息技术杂志 5, 312-325。Ali, A., Majhi, S., 2006. 基于细菌觅食策略的最优PID控制器设计。见：2006年IEEE工业技术国际会议。IEEE, 第601-605页。Alikhani, A., Suratgar, A.A., Nouri, K., Nouredanesh, M., Salimi, S., 2013. 基于磷虾群优化算法的最优PID整定。见：第三届控制、仪器与自动化国际会议。IEEE, 第11-15页。de Almeida, B.S.G., Leite, V.C., 2019. 粒子群优化：解决工程问题的强大技术。见：Ser, J.D., Villar, E., Osaba, E. (编著), 群体智能。IntechOpen, 里耶卡。第3章。Alqudah, A., Malkawi, A., Alwadie, A., 等, 2014. 使用模拟退火优化方法的DC-DC变换器自适应控制。信号与信息处理杂志 5, 198。Alsmadi, M.K., 2018. 基于元启发式算法的查询敏感相似度量用于基于内容的图像检索。沙特国王大学学报：计算机与信息科学 30, 373-381。Amaral, J.M., Tanscheit, R., Pacheco, M.A., 2018. 通过遗传算法整定PID控制器。复杂系统 2, 3。Ang, K.H., Chong, G., Li, Y., 2005. PID控制系统分析、设计与技术。IEEE控制系统技术汇刊 13, 559-576。Annisa, J., Darus, I.Z.M., Tokhi, M.O., Mohamaddan, S., 2018. 使用进化算法整定的基于PID的控制器在双连杆柔性机械臂中的实现。见：2018年智能系统设计与应用计算方法国际会议。ICASSDA. IEEE, 第1-5页。Anthony, K., Ho, P., 2020. PID控制基础。https://pdhonline.com/courses/e331/e331content.pdf。Arora, S., Singh, S., 2019. 蝴蝶优化算法：一种全局优化的新方法。软计算 23, 715-734。Åström, K.J., Hägglund, T., 1984. 带相位裕度和幅值裕度规范的简单调节器自动整定。Automatica 20, 645-651。Åström, K.J., Hägglund, T., 1995. PID控制器：理论、设计与整定, 第2卷。美国仪器学会研究, 北卡罗来纳州三角研究园。Åström, K.J., Hägglund, T., 2004. 重新审视齐格勒-尼科尔阶梯响应法用于PID控制。过程控制杂志 14, 635-650。Atashpaz-Gargari, E., Lucas, C., 2007. 帝国主义竞争算法：一种受帝国主义竞争启发的优化算法。见：2007年IEEE进化计算大会, 第4661-4667页。Azad, A.S., Vasant, P., Watada, J., Al Sokkalingam, R., 2020. 元启发式方法用于水电能源优化：综述。见：商业与工业智能技术模型研究手册, 第351-375页。

Azar, A.T., Ammar, H.H., Ibrahim, Z.F., Ibrahim, H.A., Mohamed, N.A., Taha, M.A., 2019. 基于PSO整定的自主车辆PID控制器实现。见：国际先进智能系统与信息学术会议。Springer, 第288-299页。Azman, A.A., Rahiman, M.H.F., Mohammad, N.N., Marzaki, M.H., Taib, M.N., Ali, M.F., 2017. 多管硫酸铝水过滤器 (MTAS) 的PID齐格勒-尼科尔斯特 (ZN) 和科恩-库恩 (CC) 整定方法建模与比较研究。见：2017年IEEE第二届自动控制与智能系统国际会议 (I2CACIS)。IEEE, 第25–30页。Bagis, A., Senberber, H., 2017. 基于ABC算法的高阶振荡系统PID控制器设计。Elektron. Elektrotech. 23, 3–9。Baker, G., 2011. 利用根轨迹的时滞对象PID整定。Bansal, H.O., Sharma, R., Shreeraman, P., 2012. PID控制器整定技术综述。J. Control Eng. Technol. 2, 168–176。Bansal, R., Jain, M., Bhushan, B., 2014. 用于温度控制系统的多目标模拟退火算法整定PID控制器设计。见：2014年第六届IEEE电力印度国际会议 (PIICON)。IEEE, 第1–6页。Baselga, S., 2011. 模拟退火法的大地测量网络二阶设计。J. Surv. Eng. 137, 167–173。Bassi, S.J., Mishra, M.K., Omizegba, E.E., 2011. 使用粒子群优化 (PSO) 算法的比例-积分-微分 (PID) 控制器自动整定。Int. J. Artif. Intell. Appl. 2, 25。Battiti, R., Tecchiolli, G., 1994. 反应式禁忌搜索。ORSA J. Comput. 6, 126–140。Bayoumi, E.H.E., Salem, F., 2012. 使用细菌觅食优化的串并联谐振变换器PID控制器。Electromotion Sci. J. 19, 64–78。Beheshti, Z., Shamsuddin, S.M.H., 2013. 基于群体的元启发式算法综述。Int. J. Adv. Soft Comput. Appl. 5, 1–35。Beheshti, Z., Shamsuddin, S.M.H., 2014. CAPSO: 向心加速粒子群优化。Inf. Sci. 258, 54–79。Benbouabdallah, K., Zhu, Q.-d., 2013. 细菌觅食用于移动机器人目标跟踪的Lyapunov控制器。见：2013年第九届自然计算国际会议 (ICNC)。IEEE, 第506–511页。Bennett, S., 1993. 控制工程史：1930-1955。IET, 第47页。Bernstein, D.S., 2002. 是什么让某些控制问题变得困难？IEEE Control Syst. Mag. 22, 8–19。Beyer, H.-G., Schwefel, H.-P., 2002. 进化策略——全面介绍。Nat. Comput. 1, 3–52。Bhandari, A.K., Singh, V.K., Kumar, A., Singh, G.K., 2014. 基于Kapur熵的卫星图像多级阈值分割的布谷鸟搜索算法和凤驱动优化研究。Expert Syst. Appl. 41, 3538–3560。Bharat, S., Ganguly, A., Chatterjee, R., Basak, B., Sheet, D.K., Ganguly, A., 2019. PID控制器整定方法综述。Asian J. Conver. Technol. Bhatt, R., Parmar, G., Gupta, R., 2019. 用于互联电力系统LFC的WOA优化PID控制器性能分析。Int. J. Comput. Syst. Eng. 5, 179–192。Bhawana, S., Neelu, J., 2017. PID控制器整定技术：综述。Int. J. Emerg. Technol. 8 (1), 481–485。特刊NCETST-2017。Bingul, Z., Karahan, O., 2018. 一种用于AVR系统布谷鸟搜索算法最优PID控制器设计的新性能准则方法。J. Franklin Inst. 355, 5534–5559。Biswas, K., Vasant, P.M., Vintaned, J.A.G., Watada, J., 2020. 优化三维并眼轨迹设计的元启发式算法综述。Arch. Comput. Methods Eng. Biswas, S., Kundu, S., Bose, D., Das, S., 2012. 具有改进探索策略的大规模全局优化合作协同进化教与学算法。见：国际群体、进化与模因计算会议。Springer, 第467–475页。Blum, C., Roli, A., 2003. 组合优化中的元启发式方法：概述与概念比较。ACM Comput. Surv. 35, 268–308。Boussaïd, I., Lepagnot, J., Siarry, P., 2013. 优化元启发式方法综述。Inf. Sci. 237, 82–117。Boyd, S., Baratt, C., Norman, S., 1990. 线性控制器设计：通过凸优化的性能极限。Proc. IEEE 78, 529–574。Brabazon, A., Cui, W., O'Neill, M., 2016. 布谷鸟搜索和萤火虫算法应用于多级图像阈值分割。见：布谷鸟搜索与萤火虫算法。Springer, 第115–139页。Brogliato, B., Lozano, R., 1994. 对象参数知识较少的一阶非线性系统自适应控制。IEEE Trans. Autom. Control 39, 1764–1768。Bulbul, S.M.A., Pradhan, M., Roy, P.K., Pal, T., 2018. 应用于经济负荷分配问题的反向磷虾群算法。Ain Shams Eng. J. 9, 423–440。Burnwal, S., Deb, S., 2013. 基于布谷鸟搜索的柔性制造系统调度优化。Int. J. Adv. Manuf. Technol. 64, 951–959。Černý, V., 1985. 旅行商问题的热力学方法：一种有效的模拟算法。J. Optim. Theory Appl. 45, 41–51。Chaib, L., Choucha, A., Arif, S., 2017. 使用新型元启发式蝙蝠算法的最优分数阶PID电力系统稳定器设计与整定。Ain Shams Eng. J. 8, 113–125。Chandrasekar, K., Paramasivam, B., Chidambaram, I.A., 2017. 使用基于磷虾群算法的优化PI+ 控制器评估重构电力系统带FACTS设备的恢复指标。J. Eng. Appl. Sci. 12, 4973–4989。

Chang, W.-D., Chen, C.-Y., 2014. PID controller design for MIMO processes using improved particle swarm optimization. *Circuits Syst. Signal Process.* 33, 1473–1490.

Chang, W.-D., Hwang, R.-C., Hsieh, J.-G., 2003. A multivariable on-line adaptive PID controller using auto-tuning neurons. *Eng. Appl. Artif. Intell.* 16, 57–63.

Chatterjee, S., Mukherjee, V., 2016. PID controller for automatic voltage regulator using teaching–learning based optimization technique. *Int. J. Electr. Power Energy Syst.* 77, 418–429.

Che, Z., Chiang, T.-A., Che, Z.-G., 2012. Using analytic network process and turbo particle swarm optimization algorithm for non-balanced supply chain planning considering supplier relationship management. *Trans. Inst. Meas. Control* 34, 720–735.

Chen, J., Tang, Y., Ge, R., An, Q., Guo, X., 2013. Reliability design optimization of composite structures based on PSO together with FEA. *Chin. J. Aeronaut.* 26, 343–349.

Chen, X., Cheng, L., Liu, C., Liu, Q., Liu, J., Mao, Y., Murphy, J., 2020. A WOA-based optimization approach for task scheduling in cloud computing systems. *IEEE Syst. J.*

Chen, X., Mei, C., Xu, B., Yu, K., Huang, X., 2018a. Quadratic interpolation based teaching-learning-based optimization for chemical dynamic system optimization. *Knowl.-Based Syst.* 145, 250–263.

Chen, X., Xu, B., Yu, K., Du, W., 2018b. Teaching-learning-based optimization with learning enthusiasm mechanism and its application in chemical engineering. *J. Appl. Math.* 2018.

Cheng, J., Li, Y., Ren, W., 2008. Application of enhanced particle swarm optimization algorithm in anti-aircraft fire distribution. *Ordnance Ind. Autom.* 4.

Chiarandini, M., Paquete, L., Preuss, M., Ridge, E., 2007. Experiments on metaheuristics: Methodological overview and open issues. *Technical Report Citeseer*.

Chiha, I., Liouane, N., Borne, P., 2012. Tuning PID controller using multiobjective ant colony optimization. *Appl. Comput. Intell. Soft Comput.* 2012.

Cho, Y., Smith, J.S., Smith, A.E., 2010. Optimizing tactical military manets with a specialized pso. In: *IEEE Congress on Evolutionary Computation*. IEEE, pp. 1–6.

Chu, S.-C., Tsai, P.-W., Pan, J.-S., 2006. Cat swarm optimization. In: *Pacific Rim International Conference on Artificial Intelligence*. Springer, pp. 854–858.

Civicioglu, P., 2012. Transforming geocentric Cartesian coordinates to geodetic coordinates by using differential search algorithm. *Comput. Geosci.* 46, 229–247.

Cohen, G., 1953. Theoretical consideration of retarded control. *Trans. Am. Soc. Mech. Eng.* 75, 827–834.

Cooper, M.A., Smeresky, B., 2020. An overview of evolutionary algorithms toward space-craft attitude control. In: *Advances in Spacecraft Attitude Control*. IntechOpen.

Cordón, O., Damas, S., Santamaría, J., 2006. A fast and accurate approach for 3d image registration using the scatter search evolutionary algorithm. *Pattern Recognit. Lett.* 27, 1191–1200.

Cruz-Bernal, A., 2013. Meta-heuristic optimization techniques and its applications in robotics. In: *Recent Advances on Meta-Heuristics and Their Application to Real Scenarios*, p. 53.

Cruz-Chávez, M.A., Martínez-Rangel, M.G., Cruz-Rosales, M.H., 2017. Accelerated simulated annealing algorithm applied to the flexible job shop scheduling problem. *Int. Trans. Oper. Res.* 24, 1119–1137.

Dada, E.G., Joseph, S.B., Okunade, O.A., 2017. Primal-dual asynchronous particle swarm optimisation (pdAPSO) hybrid metaheuristic algorithm for solving global optimisation problems. *Am. J. Eng. Res.* 6, 60–61.

Dada, E.G., Ramlan, E.I., 2015. A hybrid primal-dual-PSO (pdipmPSO) algorithm for swarm robotics flocking strategy. In: *2015 Second International Conference on Computing Technology and Information Management, ICCTIM, IEEE*, pp. 93–98.

Dada, E.G., Ramlan, E.I., 2018. PDPSO: the fusion of primal-dual interior point method and particle swarm optimization algorithm. *Malays. J. Comput. Sci.* 31, 17–34.

Dada, I.G., Ramlana, I., 2013. A novel control algorithm for multi-robot pattern formation. *Int. J. Adv. Res. Comput. Sci. Tech.* 4, 10–17.

Das, K.R., Das, D., Das, J., 2015. Optimal tuning of PID controller using GWO algorithm for speed control in DC motor. In: *2015 International Conference on Soft Computing Techniques and Implementations, ICSCIT, IEEE*, pp. 108–112.

Das, S., Biswas, A., Dasgupta, S., Abraham, A., 2009. Bacterial foraging optimization algorithm: theoretical foundations, analysis, and applications. In: *Foundations of Computational Intelligence Volume 3*. Springer, pp. 23–55.

Datta, A., Nandakumar, S., 2017. A survey on bio inspired meta heuristic based clustering protocols for wireless sensor networks. *IOP Conf. Ser., Mater. Sci. Eng.* 263, 052026.

Debnath, M.K., Patel, N.C., Choudhury, S., 2018. Load frequency control based on non-integer type PID controller tuned by WOA. In: *2018 2nd IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems, ICPEICES, IEEE*, pp. 114–119.

Dey, N., Ashour, A.S., 2018. Meta-heuristic algorithms in medical image segmentation: a review. In: *Advancements in Applied Metaheuristic Computing*. IGI Global, pp. 185–203.

Ding, L., Wu, H., Yao, Y., 2015. Chaotic artificial bee colony algorithm for system identification of a small-scale unmanned helicopter. *Int. J. Aerosp. Eng.* 2015.

Divya, N., Nirmalkumar, A., 2018. A survey on tuning of PID controller for industrial process using soft computing techniques. *Int. J. Pure Appl. Math.* 118, 663–667.

Dixit, M.L., Pragati Joshi, Ajay, 2016. A survey: PID optimization for automatic voltage regulator. *Int. J. Eng. Sci. Res. Tech.* 5 (6), 404–406.

Dokeroglu, T., Sevinc, E., Kucukylmaz, T., Cosar, A., 2019. A survey on new generation metaheuristic algorithms. *Comput. Ind. Eng.* 137, 106040.

Dong, R., Wang, S., Wang, G., Wang, X., 2019. Hybrid optimization algorithm based on wolf pack search and local search for solving traveling salesman problem. *J. Shanghai Jiaotong Univ.* 24, 41–47.

Donyagard Vahed, N., Ghobaei-Arani, M., Sourì, A., 2019. Multiobjective virtual machine placement mechanisms using nature-inspired metaheuristic algorithms in cloud environments: a comprehensive review. *Int. J. Commun. Syst.* 32, e4068.

Dorigo, M., 1992. Optimization, learning and natural algorithms. PhD thesis. Politecnico di Milano, Italy.

Dorigo, M., Birattari, M., Stutzle, T., 2006. Ant colony optimization. *IEEE Comput. Intell. Mag.* 1, 28–39.

Dorigo, M., Maniezzo, V., Colomi, A., 1996. Ant system: optimization by a colony of cooperating agents. *IEEE Trans. Syst. Man Cybern., Part B, Cybern.* 26, 29–41.

DrEO, J., Aumasson, J.-P., Tfaili, W., Siarry, P., 2007. Adaptive learning search, a new tool to help comprehending metaheuristics. *Int. J. Artif. Intell. Tools* 16, 483–505.

Eberhart, R., Kennedy, J., 1995. A new optimizer using particle swarm theory. In: *MHS'95. Proceedings of the Sixth International Symposium on Micro Machine and Human Science*. IEEE, pp. 39–43.

Eglese, R.W., 1990. Simulated annealing: a tool for operational research. *Eur. J. Oper. Res.* 46, 271–281.

El-Deen, A.T., Mahmoud, A.A.H., El-Sawi, A.R., 2015. Optimal PID tuning for DC motor speed controller based on genetic algorithm. *Int. Rev. Autom. Control* 8, 80–85.

El-Telbany, M.E., 2013. Tuning PID controller for DC motor: an artificial bees optimization approach. *Int. J. Comput. Appl.* 77.

Elhosseini, M.A., Haikal, A.Y., Badawy, M., Khashan, N., 2019. Biped robot stability based on an A-C parametric whale optimization algorithm. *J. Comput. Sci.* 31, 17–32.

Elshaer, R., Awad, H., 2020. A taxonomic review of metaheuristic algorithms for solving the vehicle routing problem and its variants. *Comput. Ind. Eng.* 140, 106242.

Elsrogy, W.M., Fkirin, M.A., Hassan, M.A.M., 2013. Speed control of DC motor using PID controller based on artificial intelligence techniques. In: *2013 International Conference on Control, Decision and Information Technologies, CoDIT, IEEE*, pp. 196–201.

Emary, E., Zawbaa, H.M., Grosan, C., Hassenian, A.E., 2015. Feature subset selection approach by gray-wolf optimization. In: *Afro-European Conference for Industrial Advancement*. Springer, pp. 1–13.

Emary, E., Zawbaa, H.M., Hassanien, A.E., 2016. Binary grey wolf optimization approaches for feature selection. *Neurocomputing* 172, 371–381.

Emmanuel, A.C., Inyama, H., 2015. A survey of controller design methods for a robot manipulator in harsh environments. *Eur. J. Eng. Technol.* 3, 64–73.

Eshtay, M., Faris, H., Obeid, N., 2018. Improving extreme learning machine by competitive swarm optimization and its application for medical diagnosis problems. *Expert Syst. Appl.* 104, 134–152.

Expósito-Izquierdo, C., Expósito-Márquez, A., 2017. A survey of the cuckoo search and its applications in real-world optimization problems. In: *Handbook of Research on Soft Computing and Nature-Inspired Algorithms*. IGI Global, pp. 541–555.

Farmer, J.D., Packard, N.H., Perelson, A.S., 1986. The immune system, adaptation, and machine learning. *Physica D* 22, 187–204.

Findik, O., 2015. Bull optimization algorithm based on genetic operators for continuous optimization problems. *Turk. J. Electr. Eng. Comput. Sci.* 23, 2225–2239.

Fister, D., Fister Jr., I., Fister, I., Šafarić, R., 2016. Parameter tuning of PID controller with reactive nature-inspired algorithms. *Robot. Auton. Syst.* 84, 64–75.

Fister, I., Yang, X.-S., Fister, D., 2014. Cuckoo search: a brief literature review. In: *Cuckoo Search and Firefly Algorithm*. Springer, pp. 49–62.

Fonseca, C.M., Fleming, P.J., 1993. Genetic algorithms for multiobjective optimization: formulation/discussion and generalization. In: *Icga*, pp. 416–423. Citeseer volume 93.

Fouskakis, D., Draper, D., 2002. Stochastic optimization: a review. *Int. Stat. Rev.* 70, 315–349.

Fraga-Gonzalez, L.F., Fuentes-Aguilar, R.Q., Garcia-Gonzalez, A., Sanchez-Ante, G., 2017. Adaptive simulated annealing for tuning PID controllers. *AI Commun.* 30, 347–362.

Freeman, R.A., Krstić, M., Kokotović, P., 1998. Robustness of adaptive nonlinear control to bounded uncertainties. *Automatica* 34, 1227–1230.

Fu, Z., Pun, C.-M., Gao, H., Lu, H., 2020. Endmember extraction of hyperspectral remote sensing images based on an improved discrete artificial bee colony algorithm and genetic algorithm. *Mob. Netw. Appl.* 25, 1033–1041.

Gandomi, A.H., Alavi, A.H., 2012. Krill herd: a new bio-inspired optimization algorithm. *Commun. Nonlinear Sci. Numer. Simul.* 17, 4831–4845.

Gandomi, A.H., Yang, X.-S., 2014. Chaotic bat algorithm. *J. Comput. Sci.* 5, 224–232.

Gandomi, A.H., Yang, X.-S., Talatahari, S., Alavi, A.H., 2013. Firefly algorithm with chaos. *Commun. Nonlinear Sci. Numer. Simul.* 18, 89–98.

Ganguly, S., Sahoo, N., Das, D., 2013. Multi-objective particle swarm optimization based on fuzzy-Pareto-dominance for possibilistic planning of electrical distribution systems incorporating distributed generation. *Fuzzy Sets Syst.* 213, 47–73.

Garpinger, O., Hägglund, T., Åström, K.J., 2014. Performance and robustness trade-offs in PID control. *J. Process Control* 24, 568–577.

Geem, Z.W., 2011. Solution quality improvement in chiller loading optimization. *Appl. Therm. Eng.* 31, 1848–1851.

Geem, Z.W., Kim, J.H., Loganathan, G.V., 2001. A new heuristic optimization algorithm: harmony search. *Simulation* 76, 60–68.

Gharehchopogh, F.S., Gholizadeh, H., 2019. A comprehensive survey: whale optimization algorithm and its applications. *Swarm Evol. Comput.* 48, 1–24.

Gholap, V., Naik Dessai, C., Bagyaveereswaran, V., 2017. PID controller tuning using metaheuristic optimization algorithms for benchmark problems. *Mater. Sci. Eng. Conf. Ser.* 263, 052021.

Ghosal, S., Darbar, R., Neogi, B., Das, A., Tibarewala, D.N., 2012. Application of swarm intelligence computation techniques in PID controller tuning: a review. In: *Proceedings*

Chang, W.-D., Chen, C.-Y., 2014. 使用改进粒子群优化的MIMO过程PID控制器设计. *Circuits Syst. Signal Process.* 33, 1473–1490. Chang, W.-D., Hwang, R.-C., Hsieh, J.-G., 2003. 使用自调谐神经网络的多变量在线自适应PID控制器. *Eng. Appl. Artif. Intell.* 16, 57–63. Chatterjee, S., Mukherjee, V., 2016. 使用教与学优化技术的自动电压调节器PID控制器. *Int. J. Electr. Power Energy Syst.* 77, 418–429. Che, Z., Chiang, T.-A., Che, Z.-G., 2012. 使用分析网络过程和涡轮粒子群优化算法考虑供应商关系管理的非平衡供应链规划. *Trans. Inst. Meas. Control* 34, 720–735. Chen, J., Tang, Y., Ge, R., An, Q., Guo, X., 2013. 基于PSO结合有限元分析的复合材料结构可靠性设计优化. *Chin. J. Aeronaut.* 26, 343–349. Chen, X., Cheng, L., Liu, C., Liu, Q., Liu, J., Mao, Y., Murphy, J., 2020. 云计算系统中任务调度的基于WOA的优化方法. *IEEE Syst. J.* Chen, X., Mei, C., Xu, B., Yu, K., Huang, X., 2018a. 基于二次插值的教与学优化用于化学动态系统优化. *Knowl.-Based Syst.* 145, 250–263. Chen, X., Xu, B., Yu, K., Du, W., 2018b. 具有学习热情机制的教与学优化及其在化学工程中的应用. *J. Appl. Math.* 2018. Cheng, J., Li, Y., Ren, W., 2008. 增强粒子群优化算法在防上火分配中的应用. *兵工自动化* 4. Chiarandini, M., Paquete, L., Preuss, M., Ridge, E., 2007. 元启发式实验: 方法论概述与开放问题. *Technical Report Citeseer*. Chiha, I., Liouane, N., Borne, P., 2012. 使用多目标蚁群优化整定PID控制器. *Appl. Comput. Intell. Soft Comput.* 2012. Cho, Y., Smith, J.S., Smith, A.E., 2010. 使用专用粒子群优化优化战术军事移动自组网. 见: *IEEE进化计算大会*. IEEE, 第1–6页. Chu, S.-C., Tsai, P.-W., Pan, J.-S., 2006. 猫群优化. *IntechOpen*. Cordón, O., Damas, S., Santamaría, J., 2006. 使用分散搜索进化算法的快速准确三维图像配准方法. *Pattern Recognit. Lett.* 27, 1191–1200. Cruz-Bernal, A., 2013. 元启发优化技术及其在机器人学中的应用. 见: *元启发式及其在真实场景中的应用最新进展*, 第53页. Cruz-Chávez, M.A., Martínez-Rangel, M.G., Cruz-Rosales, M.H., 2017. 应用于柔性作业车间调度问题的加速模拟退火算法. *Int. Trans. Oper. Res.* 24, 1119–1137. Dada, E.G., Joseph, S.B., Okunade, O.A., 2017. 用于解决全局优化问题的原始-对偶异步粒子群优化 (pdAPSO) 混合元启发算法. *Am. J. Eng. Res.* 6, 60–61. Dada, E.G., Ramlan, E.I., 2015. 用于群机器人聚集策略的混合原始-对偶-PSO (pdipmPSO) 算法. 见: *2015年第二届计算技术与信息管理国际会议 (ICCTIM)*. IEEE, 第93–98页. Dada, E.G., Ramlan, E.I., 2018. PDPSO: 原始-对偶内点法与粒子群优化算法的融合. *Malays. J. Comput. Sci.* 31, 17–34. Dada, I.G., Ramlana, I., 2013. 一种多机器人模式形成的新型控制算法. *Int. J. Adv. Res. Comput. Sci. Tech.* 4, 10–17. Das, K.R., Das, D., Das, J., 2015. 使用灰狼优化算法优化整定直流电机速度控制PID控制器. 见: *2015年软计算技术与实现国际会议 (ICSCIT)*. IEEE, 第108–112页. Das, S., Biswas, A., Dasgupta, S., Abraham, A., 2009. 细菌觅食优化算法: 理论基础、分析与应用. 见: *计算智能基础第3卷*. Springer, 第23–55页. Datta, A., Nandakumar, S., 2017. 无线传感器网络中基于生物启发元启发式的聚类协议综述. *IOP Conf. Ser., Mater. Sci. Eng.* 263, 052026. Debnath, M.K., Patel, N.C., Choudhury, S., 2018. 基于WOA整定的非整数型PID控制器负荷频率控制. 见: *2018年第二届IEEE电力电子、智能控制与能源系统国际会议 (ICPEICES)*. IEEE, 第114–119页. Dey, N., Ashour, A.S., 2018. 医学图像分割中的元启发式算法综述. 见: *应用元启发式计算进展*. IGI Global, 第185–203页. Ding, L., Wu, H., Yao, Y., 2015. 用于小型无人直升机系统辨识的混沌人工蜂群算法. *Int. J. Aerosp. Eng.* 2015. Divya, N., Nirmalkumar, A., 2018. 基于软计算技术的工业过程PID控制器整定综述. *Int. J. Pure Appl. Math.* 118, 663–667. Dixit, M.L., Pragati Joshi, Ajay, 2016. 自动电压调节器的PID优化综述. *Int. J. Eng. Sci. Res. Tech.* 5 (6), 404–406. Dokeroglu, T., Sevinc, E., Kucukylmaz, T., Cosar, A., 2019. 新一代元启发式算法综述. *Comput. Ind. Eng.* 137, 106040. Dong, R., Wang, S., Wang, G., Wang, X., 2019. 基于狼群搜索和局部搜索的混合优化算法求解旅行商问题. *J. Shanghai Jiaotong Univ.* 24, 41–47.

Donyagard Vahed, N., Ghobaei-Arani, M., Sourì, A., 2019. 云环境中使用自然启发式元启发算法的多目标虚拟机放置机制: 综述. *Int. J. Commun. Syst.* 32, e4068.

Dorigo, M., 1992. 优化、学习与自然算法. 博士论文. 米兰理工大学, 意大利. Dorigo, M., Birattari, M., Stutzle, T., 2006. 蚁群优化. *IEEE Comput. Intell. Mag.* 1, 28–39.

Dorigo, M., Maniezzo, V., Colomi, A., 1996. 蚁群系统: 通过合作代理群体进行优化. *IEEE Trans. Syst. Man Cybern., Part B, Cybern.* 26, 29–41.

DrEO, J., Aumasson, J.-P., Tfaili, W., Siarry, P., 2007. 自适应学习搜索: 一种帮助理解元启发式的新工具. *Int. J. Artif. Intell. Tools* 16, 483–505.

Eberhart, R., Kennedy, J., 1995. 使用粒子群理论的新优化器. 在: *MHS'95. 第六届国际微型机械与人类科学研讨会论文集*. IEEE, 第39–43页.

Eglese, R.W., 1990. 模拟退火: 运筹学工具. *Eur. J. Oper. Res.* 46, 271–281.

El-Deen, A.T., Mahmoud, A.A.H., El-Sawi, A.R., 2015. 基于遗传算法的直流电机速度控制器最优PID整定. *Int. Rev. Autom. Control* 8, 80–85.

El-Telbany, M.E., 2013. Tuning PID controller for DC motor: an artificial bees optimization approach. *Int. J. Comput. Appl.* 77.

Elhosseini, M.A., Haikal, A.Y., Badawy, M., Khashan, N., 2019. 基于A-C参数鲸鱼优化算法的双足机器人稳定性. *J. Comput. Sci.* 31, 17–32.

Elshaer, R., Awad, H., 2020. 求解车辆路径问题及其变体的元启发式算法分类综述. *Comput. Ind. Eng.* 140, 106242.

Elsrogy, W.M., Fkirin, M.A., Hassan, M.A.M., 2013. 基于人工智能技术的PID控制器直流电机速度控制. 在: *2013年国际控制、决策与信息技术会议, CoDIT, IEEE*, 第196–201页.

Emary, E., Zawbaa, H.M., Grosan, C., Hassenian, A.E., 2015. 基于灰狼优化的特征子集选择方法. 在: *非洲-欧洲工业进步会议, Spring*, 第1–13页.

Emary, E., Zawbaa, H.M., Hassanien, A.E., 2015. 用于特征选择的二元灰狼优化方法. *Neurocomputing* 172, 371–381.

Emmanuel, A.C., Inyama, H., 2015. 恶劣环境下机器人机械臂控制器设计方法综述. *Eur. J. Eng. Technol.* 3, 64–73.

Eshtay, M., Faris, H., Obeid, N., 2018. 通过竞争性群体优化改进极限学习机及其在医学诊断问题中的应用. *Expert Syst. Appl.* 104, 134–152.

Expósito-Izquierdo, C., Expósito-Márquez, A., 2017. 布谷鸟搜索及其在现实世界优化问题中的应用综述. 在: *软计算与自然启发算法研究手册*. IGI Global, 第541–555页.

Farmer, J.D., Packard, N.H., Perelson, A.S., 1986. 免疫系统、适应与机器学习. *Physica D* 22, 187–204.

Findik, O., 2015. 基于遗传算子的公牛优化算法. *Neurocomputing* 172, 371–381.

Fister, D., Fister Jr., I., Fister, I., Šafarić, R., 2016. 参数调优的PID控制器具有反应性自然启发式算法. *Robot. Auton. Syst.* 84, 64–75.

Fister, I., Yang, X.-S., Fister, D., 2014. Cuckoo search: a brief literature review. In: *Cuckoo Search and Firefly Algorithm*. Springer, pp. 49–62.

Fonseca, C.M., Fleming, P.J., 1993. Genetic algorithms for multiobjective optimization: formulation/discussion and generalization. In: *Icga*, pp. 416–423. Citeseer volume 93.

Fouskakis, D., Draper, D., 2002. Stochastic optimization: a review. *Int. Stat. Rev.* 70, 315–349.

Fraga-Gonzalez, L.F., Fuentes-Aguilar, R.Q., Garcia-Gonzalez, A., Sanchez-Ante, G., 2017. Adaptive simulated annealing for tuning PID controllers. *AI Commun.* 30, 347–362.

Freeman, R.A., Krstić, M., Kokotović, P., 1998. Robustness of adaptive nonlinear control to bounded uncertainties. *Automatica* 34, 1227–1230.

Fu, Z., Pun, C.-M., Gao, H., Lu, H., 2020. Endmember extraction of hyperspectral remote sensing images based on an improved discrete artificial bee colony algorithm and genetic algorithm. *Mob. Netw. Appl.* 25, 1033–1041.

Gandomi, A.H., Alavi, A.H., 2012. Krill herd: a new bio-inspired optimization algorithm. *Commun. Nonlinear Sci. Numer. Simul.* 17, 4831–4845.

Gandomi, A.H., Yang, X.-S., 2014. Chaotic bat algorithm. *J. Comput. Sci.* 5, 224–232.

Gandomi, A.H., Yang, X.-S., Talatahari, S., Alavi, A.H., 2013. Firefly algorithm with chaos. *Commun. Nonlinear Sci. Numer. Simul.* 18, 89–98.

Ganguly, S., Sahoo, N., Das, D., 2013. Multi-objective particle swarm optimization based on fuzzy-Pareto-dominance for possibilistic planning of electrical distribution systems incorporating distributed generation. *Fuzzy Sets Syst.* 213, 47–73.

Garpinger, O., Hägglund, T., Åström, K.J., 2014. Performance and robustness trade-offs in PID control. *J. Process Control* 24, 568–577.

Geem, Z.W., 2011. Solution quality improvement in chiller loading optimization. *Appl. Therm. Eng.* 31, 1848–1851.

Geem, Z.W., Kim, J.H., Loganathan, G.V., 2001. A new heuristic optimization algorithm: harmony search. *Simulation* 76, 60–68.

Gharehchopogh, F.S., Gholizadeh, H., 2019. A comprehensive survey: whale optimization algorithm and its applications. *Swarm Evol. Comput.* 48, 1–24.

Gholap, V., Naik Dessai, C., Bagyaveereswaran, V., 2017. PID controller tuning using metaheuristic optimization algorithms for benchmark problems. *Mater. Sci. Eng. Conf. Ser.* 263, 052021.

Ghosal, S., Darbar, R., Neogi, B., Das, A., Tibarewala, D.N., 2012. Application of swarm intelligence computation techniques in PID controller tuning: a review. In: *Proceedings*

of the International Conference on Information Systems Design and Intelligent Applications 2012. INDIA 2012, Held in Visakhapatnam, India, January 2012. Springer, pp. 195–208.

Glover, F., 1997. Tabu search and adaptive memory programming advances, applications and challenges. In: *Interfaces in Computer Science and Operations Research*. Springer, pp. 1–75.

Gogna, A., Tayal, A., 2013. Metaheuristics: review and application. *J. Exp. Theor. Artif. Intell.* 25, 503–526.

Goldbogen, J.A., Friedlaender, A.S., Calambokidis, J., Mckenna, M.F., Simon, M., Nowacek, D.P., 2013. Integrative approaches to the study of baleen whale diving behavior, feeding performance, and foraging ecology. *BioScience* 63, 90–100.

Goldenberg, D.E., 1989. Genetic algorithms in search, optimization and machine learning. Grimm, G., Messina, M.J., Tuna, S.E., Teel, A.R., 2007. Nominally robust model predictive control with state constraints. *IEEE Trans. Autom. Control* 52, 1856–1870.

Haber, R.E., Haber-Haber, R., Del Toro, R.M., Alique, J.R., 2007. Using simulated annealing for optimal tuning of a PID controller for time-delay systems. An application to a high-performance drilling process. In: *International Work-Conference on Artificial Neural Networks*. Springer, pp. 1155–1162.

Haddad, O.B., Afshar, A., Mariño, M.A., 2006. Honey-bees mating optimization (HBMO) algorithm: a new heuristic approach for water resources optimization. *Water Resour. Manag.* 20, 661–680.

Hägglund, T., Åström, K.J., 1991. Industrial adaptive controllers based on frequency response techniques. *Automatica* 27, 599–609.

Hahn, J., 2006. Handbook of PI and PID controller tuning rules-[book review]. *IEEE Control Syst. Mag.* 26, 92–93.

Hajihassani, M., Armaghani, D.J., Kalatehjari, R., 2018. Applications of particle swarm optimization in geotechnical engineering: a comprehensive review. *Geotech. Geolog. Eng.* 36, 705–722.

Hambali, N., Masngut, A., Ishak, A.A., Janin, Z., 2014. Process controllability for flow control system using Ziegler-Nichols (ZN), Cohen-Coon (CC) and Chien-Hrones-Reswick (CHR) tuning methods. In: 2014 IEEE International Conference on Smart Instrumentation, Measurement and Applications. ICSIMA. IEEE, pp. 1–6.

Hang, C.C., Åström, K.J., Ho, W.K., 1991. Refinements of the Ziegler–Nichols tuning formula. In: *IEE Proceedings D (Control Theory and Applications)*, vol. 138. IET, pp. 111–118.

Hanseth, O., Aanestad, M., 2003. Design as bootstrapping. On the evolution of ICT networks in health care. *Methods Inf. Med.* 42, 385–391.

Harifi, S., Khalilian, M., Mohammadzadeh, J., Ebrahimnejad, S., 2019. Emperor penguins colony: a new metaheuristic algorithm for optimization. *Evol. Intell.* 12, 211–226.

Harris, T.J., 1989. Assessment of control loop performance. *Can. J. Chem. Eng.* 67, 856–861.

Hekimoğlu, B., Ekinci, S., Kaya, S., 2018. Optimal PID controller design of DC-DC buck converter using whale optimization algorithm. In: 2018 International Conference on Artificial Intelligence and Data Processing. IDAP. IEEE, pp. 1–6.

Herlambang, T., Rahmalia, D., Yulianto, T., 2019. Particle swarm optimization (PSO) and ant colony optimization (ACO) for optimizing PID parameters on autonomous underwater vehicle (AUV) control system. In: *Journal of Physics: Conference Series*. In: IOP Publishing, vol. 1211. 012039.

Hernandez, J., Salto, C., Minetti, G., Carnero, M., Sanchez, M.C., 2019. Hybrid simulated annealing for optimal cost instrumentation in chemical plants. *Chem. Eng. Trans.* 74, 709–714.

Ho, W.K., Gan, O., Tay, E.B., Ang, E., 1996. Performance and gain and phase margins of well-known PID tuning formulas. *IEEE Trans. Control Syst. Technol.* 4, 473–477.

Hof, P.R., Van der Gucht, E., 2007. Structure of the cerebral cortex of the humpback whale, megaptera novaeangliae (cetacea, mysticeti, balaenopteridae). The anatomical record: advances in integrative anatomy and evolutionary biology. *Adv. Integr. Anat. Evol. Biol.* 290, 1–31.

Holland, J.H., 1975. *Adaptation in Natural and Artificial Systems* Ann Arbor, vol. 1. The University of Michigan Press, p. 975.

Hosseini, H.S., 2007. Problem solving by intelligent water drops. In: 2007 IEEE Congress on Evolutionary Computation. IEEE, pp. 3226–3231.

Houssein, E.H., Saad, M.R., Hashim, F.A., Shaban, H., Hassaballah, M., 2020. Lévy flight distribution: a new metaheuristic algorithm for solving engineering optimization problems. *Eng. Appl. Artif. Intell.* 94, 103731.

Huang, B., Shah, S., Kwok, E., 1997. Good, bad or optimal? Performance assessment of multivariable processes. *Automatica* 33, 1175–1183.

Huang, B., Shah, S.L., 1999. Performance Assessment of Control Loops: Theory and Applications. Springer Science & Business Media.

Hussain, K., Salleh, M.N.M., Cheng, S., Shi, Y., 2019. Metaheuristic research: a comprehensive survey. *Artif. Intell. Rev.* 52, 2191–2233.

Hussain, K.M., Zepherin, R.A.R., Shantha, M., 2014. Comparison of PID controller tuning methods with genetic algorithm for FOPTD system. *Criterion* 16, 17.

Hussien, A.G., Amin, M., Abd El Aziz, M., 2020. A comprehensive review of moth-flame optimisation: variants, hybrids, and applications. *J. Exp. Theor. Artif. Intell.* 32, 705–725.

Huynh, B.-P., Su, S.-F., Kuo, Y.-L., 2020. Vision/position hybrid control for a hexa robot using bacterial foraging optimization in real-time pose adjustment. *Symmetry* 12, 564.

Jackson, W.G., Özcan, E., John, R.I., 2018. Move acceptance in local search metaheuristics for cross-domain search. *Expert Syst. Appl.* 109, 131–151.

Jain, M., Singh, V., Rani, A., 2019. A novel nature-inspired algorithm for optimization: squirrel search algorithm. *Swarm Evol. Comput.* 44, 148–175.

Jain, N., Gupta, R., Parmar, G., 2012. A review of soft computing techniques for PID controller tuning. *Int. J. Adv. Electr. Electron. Eng.* 1, 221–224.

James, J., Li, V.O., 2015. A social spider algorithm for global optimization. *Appl. Soft Comput.* 30, 614–627.

Jasim, M.H., 2018. Tuning of a PID controller by bacterial foraging algorithm for position control of DC servo motor. *Eng. Technol. J.* 36.

Javidy, B., Hatamlou, A., Mirjalili, S., 2015. Ions motion algorithm for solving optimization problems. *Appl. Soft Comput.* 32, 72–79.

Jayabarathi, T., Raghunathan, T., Gandomi, A., 2018. The bat algorithm, variants and some practical engineering applications: a review. In: *Nature-Inspired Algorithms and Applied Optimization*. Springer, pp. 313–330.

Jayanth, J., Koliwad, S., Kumar, T.A., 2015. Classification of remote sensed data using artificial bee colony algorithm. *Egypt. J. Remote Sens. Space Sci.* 18, 119–126.

Jhankal, N.K., Adhyaru, D., 2011. Bacterial foraging optimization algorithm: a derivative free technique. In: 2011 Nirma University International Conference on Engineering. IEEE, pp. 1–4.

Jia, F., Lichti, D., 2017. A comparison of simulated annealing, genetic algorithm and particle swarm optimization in optimal first-order design of indoor TLS networks. *ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci.* 4.

Jin, Q., Shi, Y., Liu, Q., Chu, M., Zhang, Y., 2018. Graphical robust PID tuning for disturbance rejection satisfying multiple objectives. *Chem. Eng. Commun.* 205, 1701–1711.

Johnson, M., Moradi, M., 2005. *PID Control*. Springer, London.

Joseph, S.B., Dada, E.G., 2018. Proportional-integral-derivative (PID) controller tuning for an inverted pendulum using particle swarm optimisation (PSO) algorithm. *FUDMA J. Sci.* (ISSN 2616-1370) 2, 72–78.

Joseph, S.B., Emmanuel, G.D., 2018. Automatic tuning of proportional integral derivative controller using genetic algorithm. *Pac. J. Sci. Technol.* 19, 51–57.

Kalaiselvi, D., Radhakrishnan, R., 2015. Multiconstrained qos routing using a differentially guided krill herd algorithm in mobile ad hoc networks. *Math. Probl. Eng.*

Kaliannan, J., Baskaran, A., Dey, N., Ashour, A.S., 2016. Ant colony optimization algorithm based PID controller for LFC of single area power system with non-linearity and boiler dynamics. *World J. Model. Simul.* 12, 3–14.

Kaloust, J., Qu, Z., 1997. Robust control design for nonlinear uncertain systems with an unknown time-varying control direction. *IEEE Trans. Autom. Control* 42, 393–399.

Kalra, M., Singh, S., 2015. A review of metaheuristic scheduling techniques in cloud computing. *Egypt. Inform. J.* 16, 275–295.

Karaboga, D., 2005. An idea based on honey bee swarm for numerical optimization. Technical Report Technical report-tr06. Erciyes University, Engineering Faculty.

Karaboga, D., 2010. Artificial bee colony algorithm. *Scholarpedia* 5, 6915.

Karthick, P., Palanisamy, C., 2019. Optimized cluster head selection using krill herd algorithm for wireless sensor network. *Automatika* 60, 340–348.

Kashan, A.H., 2009. League championship algorithm: a new algorithm for numerical function optimization. In: 2009 International Conference of Soft Computing and Pattern Recognition. IEEE, pp. 43–48.

Katal, N., Kumar, P., Narayan, S., 2014. Optimal PID controller for coupled-tank liquid-level control system using bat algorithm. In: 2014 International Conference on Power, Control and Embedded Systems. ICPCES. IEEE, pp. 1–4.

Kaushik, A., Indu, S., Gupta, D., 2019. A grey wolf optimization approach for improving the performance of wireless sensor networks. *Wirel. Pers. Commun.* 106, 1429–1449.

Kaveh, A., Mahdavi, V., 2014. Colliding bodies optimization: a novel meta-heuristic method. *Comput. Struct.* 139, 18–27.

Kaveh, A., Rastegar Moghaddam, M., 2018. A hybrid woa-cbo algorithm for construction site layout planning problem. *Sci. Iran.* 25, 1094–1104.

Kaveh, A., Zaerreza, A., 2020. Shuffled shepherd optimization method: a new meta-heuristic algorithm. In: *Engineering Computations*.

Kennedy, J., Eberhart, R., 1995. Particle swarm optimization. In: *Proceedings of ICNN'95-International Conference on Neural Networks*. In: IEEE, vol. 4, pp. 1942–1948.

Khairuzzaman, A.K.M., Chaudhury, S., 2017. Multilevel thresholding using grey wolf optimizer for image segmentation. *Expert Syst. Appl.* 86, 64–76.

Khalilpourazari, S., Pasandideh, S.H.R., Ghodrtnama, A., 2019. Robust possibilistic programming for multi-item eoq model with defective supply batches: whale optimization and water cycle algorithms. *Neural Comput. Appl.* 31, 6587–6614.

Khare, A., Rangnekar, S., 2013. A review of particle swarm optimization and its applications in solar photovoltaic system. *Appl. Soft Comput.* 13, 2997–3006.

Kim, D.H., Cho, J.H., 2005. Adaptive tuning of PID controller for multivariable system using bacterial foraging based optimization. In: *International Atlantic Web Intelligence Conference*. Springer, pp. 231–235.

Kim, D.H., Cho, J.H., 2006. A biologically inspired intelligent PID controller tuning for AVR systems. *Int. J. Control. Autom. Syst.* 4, 624–636.

Kim, D.H., Hong, W.P., Park, J.I., 2002. Auto-tuning of reference model based PID controller using immune algorithm. In: *Proceedings of the 2002 Congress on Evolutionary Computation. CEC'02 (Cat. No. 02TH8600)*. In: IEEE, vol. 1, pp. 483–488.

Kirkpatrick, S., Gelatt, C.D., Vecchi, M.P., 1983. Optimization by simulated annealing. *Science* 220, 671–680.

Kohli, M., Arora, S., 2018. Chaotic grey wolf optimization algorithm for constrained optimization problems. *J. Comput. Des. Eng.* 5, 458–472.

Korani, W.M., Dorrah, H.T., Emara, H.M., 2009. Bacterial foraging oriented by particle swarm optimization strategy for PID tuning. In: 2009 IEEE International Symposium on Computational Intelligence in Robotics and Automation. CIRA. IEEE, pp. 445–450.

Glover, F., 1997. 禁忌搜索与自适应记忆编程：进展、应用与挑战。在：计算机科学与运筹学中的接口。Springer, 第1–75页。Gogna, A., Tayal, A., 2013. 元启发式：综述与应用。J. Exp. Theor. Artif. Intell. 25, 503–526. Goldbogen, J.A., Friedlaender, A.S., Calambokidis, J., Mckenna, M.F., Simon, M., Nowacek, D.P., 2013. 须鲸潜水行为、摄食性能与觅食生态学研究的整合方法。BioScience 63, 90–100. Goldenberg, D.E., 1989. 搜索、优化与机器学习中的遗传算法。Grimm, G., Messina, M.J., Tuna, S.E., Teel, A.R., 2007. 带状态约束的名义鲁棒模型预测控制。IEEE Trans. Autom. Control 52, 1856–1870. Haber, R.E., Haber-Haber, R., Del Toro, R.M., Alique, J.R., 2007. 使用模拟退火对时滞系统PID控制器进行最优整定：在高性能钻孔过程中的应用。在：国际人工神经网络工作会议。Springer, 第1155–1162页。Haddad, O.B., Afshar, A., Mariño, M.A., 2006. 蜜蜂交配优化（HBMO）算法：一种用于水资源优化的新启发式方法。Water Resour. Manag. 20, 661–680. Hägglund, T., Åström, K.J., 1991. 基于频率响应技术的工业自适应控制器。Automatica 27, 599–609. Hahn, J., 2006. PI和PID控制器整定规则手册——[书评]。IEEE Control Syst. Mag. 26, 92–93. Hajihassani, M., Armaghani, D.J., Kalatehjari, R., 2018. 粒子群优化在岩土工程中的应用：综合综述。Geotech. Geolog. Eng. 36, 705–722. Hambali, N., Masngut, A., Ishak, A.A., Janin, Z., 2014. 使用齐格勒-尼科尔斯（ZN）、科恩-库恩（CC）和Chien-Hrones-Reswick（CHR）整定方法的流量控制系统过程可控性。在：2014年IEEE智能仪器、测量与应用国际会议。ICSIMA. IEEE, 第1–6页。Hang, C.C., Åström, K.J., Ho, W.K., 1991. 齐格勒-尼科尔斯整定公式的改进。在：IEE Proceedings D（控制理论与应用）。卷138. IET, 第111–118页。Hanseth, O., Aanestad, M., 2003. 设计作为自举：医疗保健中ICT网络的演化。Methods Inf. Med. 42, 385–391. Harifi, S., Khalilian, M., Mohammadzadeh, J., Ebrahimnejad, S., 2019. 帝企鹅群：一种新的元启发式优化算法。Evol. Intell. 12, 211–226. Harris, T.J., 1989. 控制回路性能评估。Can. J. Chem. Eng. 67, 856–861. Hekimoğlu, B., Ekinci, S., Kaya, S., 2018. 使用鲸鱼优化算法的DC-DC降压变换器最优PID控制器设计。在：2018年国际人工智能与数据处理会议。IDAP. IEEE, 第1–6页。Herlambang, T., Rahmalia, D., Yulianto, T., 2019. 粒子群优化（PSO）和蚁群优化（ACO）优化自主水下航行器（AUV）控制系统PID参数。在：IOP Publishing, 卷1211. 012039. Hernandez, J., Salto, C., Minetti, G., Carnero, M., Sanchez, M.C., 2019. 用于化工厂最优成本仪表配置的混合模拟退火。Chem. Eng. Trans. 74, 709–714. Ho, W.K., Gan, O., Tay, E.B., Ang, E., 1996. 已知PID整定公式的性能及增益和相位裕度。IEEE Trans. Control Syst. Technol. 4, 473–477. Hof, P.R., Van der Gucht, E., 2007. 座头鲸（megaptera novaeangliae）（鲸目，须鲸亚目，鳁鲸科）大脑皮层结构。解剖学记录：整合解剖学与进化生物学的进展。Adv. Integr. Anat. Evol. Biol. 290, 1–31. Holland, J.H., 1975. 自然与人工系统中的适应，安阿伯，卷1。密歇根大学出版社，第975页。Hosseini, H.S., 2007. 智能水滴求解问题。在：2007年IEEE进化计算大会。IEEE, 第3226–3231页。Houssein, E.H., Saad, M.R., Hashim, F.A., Shaban, H., Hassaballah, M., 2020. 莱维飞行分布：一种求解工程优化问题的新元启发式算法。Eng. Appl. Artif. Intell. 94, 103731. Huang, B., Shah, S., Kwok, E., 1997. 好、坏还是最优？多变量过程性能评估。Automatica 33, 1175–1183. Huang, B., Shah, S.L., 1999. 控制回路性能评估：理论与应用。Springer Science & Business Media. Hussain, K., Salleh, M.N.M., Cheng, S., Shi, Y., 2019. 元启发式研究：综合调查。Artif. Intell. Rev. 52, 2191–2233. Hussain, K.M., Zepherin, R.A.R., Shantha, M., 2014. 基于遗传算法的一阶纯滞后系统PID控制器整定方法比较。Criterion 16, 17. Hussien, A.G., Amin, M., Abd El Aziz, M., 2020. 基于细菌觅食优化的六足机器人视觉/位置混合控制实时位姿调整。Symmetry 12, 564. Jackson, W.G., Özcan, E., John, R.I., 2018. 局部搜索元启发式中跨域搜索的移动接受机制。Expert Syst. Appl. 109, 131–151.

Jain, M., Singh, V., Rani, A., 2019. 一种新型自然启发优化算法：松鼠搜索算法。Swarm Evol. Comput. 44, 148–175. Jain, N., Gupta, R., Parmar, G., 2012. 软计算技术在PID控制器整定中的应用综述。Int. J. Adv. Electr. Electron. Eng. 1, 221–224. James, J., Li, V.O., 2015. 一种用于全局优化的社会蜘蛛算法。Appl. Soft Comput. 30, 614–627. Jasim, M.H., 2018. 基于细菌觅食算法的直流伺服电机位置控制PID控制器整定。Eng. Technol. J. 36. Javidy, B., Hatamlou, A., Mirjalili, S., 2015. 用于求解优化问题的离子运动算法。Appl. Soft Comput. 32, 72–79. Jayabarathi, T., Raghunathan, T., Gandomi, A., 2018. 蝙蝠算法、变体及一些实际工程应用：综述。载于：自然启发算法与应用优化。Springer, 第313–330页。Jayanth, J., Koliwad, S., Kumar, T.A., 2015. 使用人工蜂群算法对遥感数据进行分类。Egypt. J. Remote Sens. Space Sci. 18, 119–126. Jhankal, N.K., Adhyaru, D., 2011. 细菌觅食优化算法：一种无导数技术。载于：2011年Nirma大学国际工程会议。IEEE, 第1–4页。Jia, F., Lichti, D., 2017. 模拟退火、遗传算法与粒子群优化在室内TLS网络最优一阶段设计中的比较。ISPRS Ann. Photogramm. Remote Sens. Spatial Inf. Sci. 4. Jin, Q., Shi, Y., Liu, Q., Chu, M., Zhang, Y., 2018. 满足多目标的抗扰动图形化鲁棒PID整定。Chem. Eng. Commun. 205, 1701–1711. Johnson, M., Moradi, M., 2005. PID控制。Springer, 伦敦。Joseph, S.B., Dada, E.G., 2018. 使用粒子群优化(PSO)算法对倒立摆进行比例-积分-微分(PID)控制器整定。FUD MA J. Sci. (ISSN 2616-1370) 2, 72–78. Joseph, S.B., Emmanuel, G.D., 2018. 使用遗传算法自动整定比例积分微分控制器。Pac. J. Sci. Technol. 19, 51–57. Kalaiselvi, D., Radhakrishnan, R., 2015. 移动自组织网络中基于差异引导磷虾群算法的多约束QoS路由。Math. Probl. Eng. Kaliannan, J., Baskaran, A., Dey, N., Ashour, A.S., 2016. 基于蚁群优化算法的PID控制器用于非线性及锅炉动力学的单区域电力系统LFC。World J. Model. Simul. 12, 3–14. Kaloust, J., Qu, Z., 1997. 具有未知时变控制方向的非线性不确定系统的鲁棒控制设计。ICPCEs. IEEE, 第1–4页。Kaushik, A., Indu, S., Gupta, D., 2019. 一种提高无线传感器网络性能的灰狼优化方法。Wirel. Pers. Commun. 106, 1429–1449. Kaveh, A., Mahdavi, V., 2014. 碰撞体优化：一种新型元启发式方法。Comput. Struct. 139, 18–27. Kaveh, A., Rastegar Moghaddam, M., 2018. 一种用于施工场地布局规划问题的混合WOA-CBO算法。Sci. Iran. 25, 1094–1104. Kaveh, A., Zaerreza, A., 2020. 洗牌牧羊人优化方法：一种新的元启发式算法。载于：工程计算。Kennedy, J., Eberhart, R., 1995. 粒子群优化。载于：ICNN'95国际神经网络会议论文集。载于：IEEE, 第4卷, 第1942–1948页。Khairuzzaman, A.K.M., Chaudhury, S., 2017. 使用灰狼优化器的多级阈值化用于图像分割。Expert Syst. Appl. 86, 64–76. Khalilpourazari, S., Pasandideh, S.H.R., Ghodrtnama, A., 2019. 具有缺陷供应批次的多物品EOQ模型的鲁棒可能性规划：鲸鱼优化算法与水循环算法。Neural Comput. Appl. 31, 6587–6614. Khare, A., Rangnekar, S., 2013. 粒子群优化及其在太阳能光伏系统中的应用综述。Appl. Soft Comput. 13, 2997–3006. Kim, D.H., Cho, J.H., 2005. 基于细菌觅食优化的多变量系统PID控制器自适应整定。载于：国际大西洋网络智能会议。Springer, 第231–235页。Kim, D.H., Cho, J.H., 2006. 一种用于AVR系统的生物启发智能PID控制器整定。Int. J. Control. Autom. Syst. 4, 624–636. Kim, D.H., Hong, W.P., Park, J.I., 2002. 使用免疫算法的参考模型PID控制器自整定。载于：2002年进化计算大会论文集。CEC'02 (Cat. No. 02TH8600)。载于：IEEE, 第1卷, 第483–488页。Kirkpatrick, S., Gelatt, C.D., Vecchi, M.P., 1983. 模拟退火优化。Science 220, 671–680. Kohli, M., Arora, S., 2018. 用于约束优化问题的混沌灰狼优化算法。J. Comput. Des. Eng. 5, 458–472. Korani, W.M., Dorrah, H.T., Emara, H.M., 2009. 粒子群优化策略导向的细菌觅食用于PID整定。载于：2009年IEEE计算智能在机器人自动化国际研讨会。CIRA. IEEE, 第445–450页。

Kotteeswaran, R., Sivakumar, L., 2013. Optimal partial-retuning of decentralised PI controller of coal gasifier using bat algorithm. In: International Conference on Swarm, Evolutionary, and Memetic Computing. Springer, pp. 750–761.

Kowalski, P.A., Łukasik, S., 2016. Training neural networks with krill herd algorithm. Neural Process. Lett. 44, 5–17.

Krishnakumar, K., Goldberg, D.E., 1992. Control system optimization using genetic algorithms. J. Guid. Control Dyn. 15, 735–740.

Krishnanand, K., Ghose, D., 2005. Detection of multiple source locations using a glow-worm metaphor with applications to collective robotics. In: Proceedings 2005 IEEE Swarm Intelligence Symposium, 2005. SIS 2005. IEEE, pp. 84–91.

Kumar, A., Bawa, S., 2018. Generalized ant colony optimizer: swarm-based meta-heuristic algorithm for cloud services execution. Computing, 1–24.

Kumar, A., Bawa, S., 2020. A comparative review of meta-heuristic approaches to optimize the SLA violation costs for dynamic execution of cloud services. Soft Comput. 24, 3909–3922.

Kumar, A., Kumar, D., Jarial, S., 2017. A review on artificial bee colony algorithms and their applications to data clustering. Cybern. Inf. Technol. 17, 3–28.

Kumar, A.A., Kumar, S.G., 2018. Application of whale optimization algorithm for tuning of a PID controller for a drilling machine. In: ICAARS 2018.

Kumar, N., Vidyarthi, D.P., 2016. A novel hybrid pso–ga meta-heuristic for scheduling of dag with communication on multiprocessor systems. Eng. Comput. 32, 35–47.

Kumar, P., Nema, S., Padhy, P., 2014. PID controller for nonlinear system using cuckoo optimization. In: 2014 International Conference on Control, Instrumentation, Communication and Computational Technologies. ICCICCT. IEEE, pp. 711–716.

Kumar, R., Singla, S., Chopra, V., 2015. Comparison among some well known control schemes with different tuning methods. J. Appl. Res. Technol. 13, 409–4015.

Kumari, S., Arumugam, M., 2015. Application of bio-inspired krill herd algorithm for breast cancer classification and diagnosis. Indian J. Sci. Technol. 8.

Kushwah, M., Patra, A., 2014. Tuning PID controller for speed control of DC motor using soft computing techniques-a review. Adv. Electron. Electr. Eng. 4, 141–148.

Lahcene, R., Abdeldjalil, S., Aissa, K., 2017. Optimal tuning of fractional order PID controller for AVR system using simulated annealing optimization algorithm. In: 2017 5th International Conference on Electrical Engineering-Boumerdes. ICEE-B. IEEE, pp. 1–6.

Latha, K., Rajinikanth, V., Surekha, P.M., 2013. PSO-based PID controller design for a class of stable and unstable systems. In: ISRN Artificial Intelligence.

Lazar, M., Heemels, W., 2009. Predictive control of hybrid systems: input-to-state stability result for sub-optimal solutions. Automatica 45, 180–185.

Lee, K.Y., Park, J.-B., 2006. Application of particle swarm optimization to economic dispatch problem: advantages and disadvantages. In: 2006 IEEE PES Power Systems Conference and Exposition. IEEE, pp. 188–192.

Lequin, O., Gevers, M., Triest, L., 1999. Optimizing the settling time with iterative feedback tuning. IFAC Proc. Vol. 32, 4659–4663.

Li, J., Tang, Y., Hua, C., Guan, X., 2014a. An improved krill herd algorithm: krill herd with linear decreasing step. Appl. Math. Comput. 234, 356–367.

Li, L., Sun, L., Guo, J., Qi, J., Xu, B., Li, S., 2017a. Modified discrete grey wolf optimizer algorithm for multilevel image thresholding. Comput. Intell. Neurosci. 2017.

Li, Q., Chen, H., Huang, H., Zhao, X., Cai, Z., Tong, C., Liu, W., Tian, X., 2017b. An enhanced grey wolf optimization based feature selection wrapped kernel extreme learning machine for medical diagnosis. Comput. Math. Methods Med. 2017.

Li, X., Zhang, J., Yin, M., 2014c. Animal migration optimization: an optimization algorithm inspired by animal migration behavior. Neural Comput. Appl. 24, 1867–1877.

Li, X.H., Yu, H.B., Yuan, M.Z., 2009. Design of an optimal PID controller based on Lyapunov approach. In: 2009 International Conference on Information Engineering and Computer Science. IEEE, pp. 1–5.

Liao, W., Hu, Y., Wang, H., 2014. Optimization of PID control for DC motor based on artificial bee colony algorithm. In: Proceedings of the 2014 International Conference on Advanced Mechatronic Systems. IEEE, pp. 23–27.

Lipták, B.G., 2002. Instrument Engineers' Handbook: Process Measurement and Analysis. CRC Press.

Liu, J., Luo, X.G., Zhang, X.M., Zhang, F., 2013. Job scheduling algorithm for cloud computing based on particle swarm optimization. In: Advanced Materials Research. In: Trans. Tech. Publ., vol. 662, pp. 957–960.

Loucif, F., Kechida, S., Sebbagh, A., 2020. Whale optimizer algorithm to tune PID controller for the trajectory tracking control of robot manipulator. J. Braz. Soc. Mech. Sci. Eng. 42, 1.

Lozano, R., Valera, A., Albertos, P., Arimoto, S., Nakayama, T., 1999. PD control of robot manipulators with joint flexibility, actuators dynamics and friction. Automatica 35, 1697–1700.

Luyben, W.L., 1986. Simple method for tuning SISO controllers in multivariable systems. Ind. Eng. Chem. Process Des. Dev. 25, 654–660.

Madadi, A., Motlagh, M.M., 2014. Optimal control of DC motor using grey wolf optimizer algorithm. Tech. J. Eng. Appl. Sci. 4, 373–379.

Mamizadeh, A., Genc, N., Rajabioun, R., 2018. Optimal tuning of PI controller for boost DC-DC converters based on cuckoo optimization algorithm. In: 2018 7th International Conference on Renewable Energy Research and Applications. ICRERA. IEEE, pp. 677–680.

Man, K.-F., Tang, K.-S., Kwong, S., 1996. Genetic algorithms: concepts and applications [in engineering design]. IEEE Trans. Ind. Electron. 43, 519–534.

Mange, J., Pace, S., Dunn, A., 2015. Optimization of PID controller parameters for automated ground vehicle control on dynamic terrain. In: Proceedings of the International Conference on Scientific Computing. CSC, p. 3.

Marinakis, Y., Marinaki, M., 2014. A bumble bees mating optimization algorithm for the open vehicle routing problem. Swarm Evol. Comput. 15, 80–94.

Martins, F.G., 2005. Tuning PID controllers using the ITAE criterion. Int. J. Eng. Educ. 21, 867.

McFarlane, D., Glover, K., 1992. A loop-shaping design procedure using h/sub infinity/synthesis. IEEE Trans. Autom. Control 37, 759–769.

Medewar, P., Munje, R., 2015. PSO based PID controller tuning for PMDC motor. In: 2015 International Conference on Energy Systems and Applications. IEEE, pp. 522–526.

Meena, D., Devanshu, A., 2017. Genetic algorithm tuned PID controller for process control. In: 2017 International Conference on Inventive Systems and Control. ICISC. IEEE, pp. 1–6.

Mehrabian, A.R., Lucas, C., 2006. A novel numerical optimization algorithm inspired from weed colonization. Ecol. Inform. 1, 355–366.

Meng, A.-b., Chen, Y.-c., Yin, H., Chen, S.-z., 2014. Crisscross optimization algorithm and its application. Knowl.-Based Syst. 67, 218–229.

Metropolis, N., Rosenbluth, A.W., Rosenbluth, M.N., Teller, A.H., Teller, E., 1953. Equation of state calculations by fast computing machines. J. Chem. Phys. 21, 1087–1092.

Mirjalili, S., Lewis, A., 2016. The whale optimization algorithm. Adv. Eng. Softw. 95, 51–67.

Mirjalili, S., Mirjalili, S.M., Lewis, A., 2014. Grey wolf optimizer. Adv. Eng. Softw. 69, 46–61.

Mirjalili, S., Mirjalili, S.M., Saremi, S., Mirjalili, S., 2020. Whale optimization algorithm: theory, literature review, and application in designing photonic crystal filters. In: Nature-Inspired Optimizers. Springer, pp. 219–238.

Mohammed, H.M., Umar, S.U., Rashid, T.A., 2019. A systematic and meta-analysis survey of whale optimization algorithm. Comput. Intell. Neurosci. 2019.

Mohammed, N.F., Song, E., Ma, X., Hayat, Q., 2014. Tuning of PID controller of synchronous generators using genetic algorithm. In: 2014 IEEE International Conference on Mechatronics and Automation. IEEE, pp. 1544–1548.

Moradi, M., 2003. New techniques for PID controller design. In: Proceedings of 2003 IEEE Conference on Control Applications. CCA 2003, vol. 2. IEEE, pp. 903–908.

Mosaad, A.M., Attia, M.A., Abdelaziz, A.Y., 2019a. Whale optimization algorithm to tune PID and PIDA controllers on AVR system. Ain Shams Eng. J. 10, 755–767.

Mosaad, M.I., abed el Raouf, M.O., Al-Ahmar, M.A., Banakher, F.A., 2019b. Maximum power point tracking of PV system based cuckoo search algorithm; review and comparison. Energy Proc. 162, 117–126.

Mucherino, A., Seref, O., 2007. Monkey search: a novel metaheuristic search for global optimization. In: AIP Conference Proceedings. In: AIP, vol. 953, pp. 162–173.

Mukherjee, P., Das, A., Bera, P.S., 2019. Design of PID power system stabilizer using oppositional krill herd algorithm for a single machine infinite bus system. In: International Conference on Innovation in Modern Science and Technology. Springer, pp. 827–844.

Muliadi, J., Kusumoputro, B., 2018. Neural network control system of UAV altitude dynamics and its comparison with the PID control system. J. Adv. Transp. 2018.

Muro, C., Escobedo, R., Spector, L., Coppinger, R., 2011. Wolf-pack (canis lupus) hunting strategies emerge from simple rules in computational simulations. Behav. Process. 88, 192–197.

Murrieta-Mendoza, A., Botez, R.M., Bunel, A., 2018. Four-dimensional aircraft en route optimization algorithm using the artificial bee colony. J. Aerosp. Inform. Syst. 15, 307–334.

Nagaraj, B., Subha, S., Rampriya, B., 2008. Tuning algorithms for PID controller using soft computing techniques. Int. J. Comput. Sci. Netw. Secur. 8, 278–281.

Nakrani, S., Tovey, C., 2004. On honey bees and dynamic server allocation in internet hosting centers. Adapt. Behav. 12, 223–240.

Natarajan, K., Gilbert, A.F., 1997. On direct PID controller tuning based on finite number of frequency response data. ISA Trans. 36, 139–149.

Nimisha, K., Senthilkumar, R., 2018. A survey on optimal tuning of PID controller for buck-boost converter using cuckoo-search algorithm. In: 2018 International Conference on Control, Power, Communication and Computing Technologies. ICCPCCT. IEEE, pp. 216–221.

Nisi, K., Nagaraj, B., Agalya, A., 2019. Tuning of a PID controller using evolutionary multi objective optimization methodologies and application to the pulp and paper industry. Int. J. Mach. Learn. Cybern. 10, 2015–2025.

Niu, B., Fan, Y., Tan, L., Rao, J., Li, L., 2010. A review of bacterial foraging optimization part II: applications and challenges. In: International Conference on Intelligent Computing. Springer, pp. 544–550.

Niu, B., Zhu, Y., He, X., Zeng, X., 2006. Optimum design of PID controllers using only a germ of intelligence. In: 2006 6th World Congress on Intelligent Control and Automation. In: IEEE, vol. 1, pp. 3584–3588.

Nojhan, 2007. Dreaming of metaheuristics. Available at <http://nojhan.free.fr/metah/index.php>.

Passino, K.M., 2002. Biomimicry of bacterial foraging for distributed optimization and control. IEEE Control Syst. Mag. 22, 52–67.

Patel, V.K., Savsani, V.J., 2016. A multi-objective improved teaching–learning based optimization algorithm (MO-ITLBO). Inf. Sci. 357, 182–200.

Patil, N., Lakhekar, G., 2017. Design of PID controller for cascade control process using genetic algorithm. In: 2017 International Conference on Intelligent Computing and Control Systems. ICICCS. IEEE, pp. 1089–1095.

Kotteeswaran, R., Sivakumar, L., 2013. 使用蝙蝠算法对煤气化炉分散PI控制器进行最优部分重调. 载于: 国际群体、进化与模拟计算会议. Springer, 第750–761页. Kowalski, P.A., Łukasik, S., 2016. 使用磷虾群算法训练神经网络. Neural Process. Lett. 44, 5–17. Krishnakumar, K., Goldberg, D.E., 1992. 使用遗传算法的控制系统优化. J. Guid. Control Dyn. 15, 735–740. Krishnanand, K., Ghose, D., 2005. 使用萤火虫隐喻检测多个源位置及其在集体机器人学中的应用. 载于: 2005年IEEE群体智能研讨会论文集. SIS 2005. IEEE, 第84–91页. Kumar, A., Bawa, S., 2018. 广义蚁群优化器: 用于云服务执行的基于群体的元启发式算法. Computing, 第1–24页. Kumar, A., Bawa, S., 2020. 元启发式方法优化云服务动态执行SLA违规成本的比较综述. Soft Comput. 24, 3909–3922. Kumar, A., Kumar, D., Jarial, S., 2017. 人工蜂群算法及其在数据聚类中的应用综述. Cybern. Inf. Technol. 17, 3–28. Kumar, A.A., Kumar, S.G., 2018. 鲸鱼优化算法在钻床PID控制器整定中的应用. 载于: ICAARS 2018. Kumar, N., Vidyarthi, D.P., 2016. 一种用于多处理器系统带通信DAG调度的新型混合PSO-GA元启发式. Eng. Comput. 32, 35–47. Kumar, P., Nema, S., Padhy, P., 2014. 使用布谷鸟优化算法的非线性系统PID控制器. 载于: 2014年控制、仪器、通信与计算技术国际会议. ICCICCT. IEEE, 第711–716页. Kumar, R., Singla, S., Chopra, V., 2015. 采用不同整定方法的一些著名控制方案的比较. J. Appl. Res. Technol. 13, 409–4015. Kumari, S., Arumugam, M., 2015. 生物启发磷虾群算法在乳腺癌分类与诊断中的应用. Indian J. Sci. Technol. 8. Kushwah, M., Patra, A., 2014. 使用软计算技术整定直流电机速度控制PID控制器综述. Adv. Electron. Electr. Eng. 4, 141–148. Lahcene, R., Abdeldjalil, S., Aissa, K., 2017. 使用模拟退火优化算法对AVR系统分数阶PID控制器进行最优整定. 载于: 2017年第五届国际电气工程-布默德斯会议. ICEE-B. IEEE, 第1–6页. Latha, K., Rajinikanth, V., Surekha, P.M., 2013. 基于PSO的稳定与不稳定系统PID控制器设计. 载于: ISRN Artificial Intelligence. Lazar, M., Heemels, W., 2009. 混合系统的预测控制: 次优化的输入到状态稳定性结果. Automatica 45, 180–185. Lee, K.Y., Park, J.-B., 2006. 粒子群优化在经济调度问题中的应用: 优缺点. 载于: 2006年IEEE PES电力系统会议与展览. IEEE, 第188–192页. Lequin, O., Gevers, M., Triest, L., 1999. 通过迭代反馈调优优化调节时间. IFAC Proc. Vol. 32, 4659–4663. Li, J., Tang, Y., Hua, C., Guan, X., 2014a. 一种改进的磷虾群算法: 线性递减步长磷虾群. Appl. Math. Comput. 234, 356–367. Li, L., Sun, L., Guo, J., Qi, J., Xu, B., Li, S., 2017a. 用于多级图像阈值化的改进离散灰狼优化算法. Comput. Intell. Neurosci. 2017. Li, Q., Chen, H., Huang, H., Zhao, X., Cai, Z., Tong, C., Liu, W., Tian, X., 2017b. 一种增强灰狼优化特征选择包裹核极学习机用于医学诊断. Comput. Math. Methods Med. 2017. Li, X., Zhang, J., Yin, M., 2014c. 动物迁移优化: 一种受动物迁移行为启发的优化算法. Neural Comput. Appl. 24, 1867–1877. Li, X.H., Yu, H.B., Yuan, M.Z., 2009. 基于Lyapunov方法的最优PID控制器设计. 载于: 2009年信息工程与计算机科学国际会议. IEEE, 第1–5页. Liao, W., Hu, Y., Wang, H., 2014. 基于人工蜂群算法的直流电机PID控制优化. 载于: 2014年先进机电系统国际会议论文集. IEEE, 第23–27页. Lipták, B.G., 2002. 仪器工程师手册: 过程测量与分析. CRC Press. Liu, J., Luo, X.G., Zhang, X.M., Zhang, F., 2013. 基于粒子群优化的云计算作业调度算法. 载于: Advanced Materials Research. 载于: Trans. Tech. Publ., 第662卷, 第957–960页. Loucif, F., Kechida, S., Sebbagh, A., 2020. 鲸鱼优化算法用于机器人操作臂轨迹跟踪控制PID控制器整定. J. Braz. Soc. Mech. Sci. Eng. 42, 1. Lozano, R., Valera, A., Albertos, P., Arimoto, S., Nakayama, T., 1999. 具有关节柔性、执行器动力学和摩擦的机器人操作臂PID控制. Automatica 35, 1697–1700. Luyben, W.L., 1986. 多变量系统中SISO控制器整定的简单方法. Ind. Eng. Chem. Process Des. Dev. 25, 654–660. Madadi, A., Motlagh, M.M., 2014. 使用灰狼优化算法的直流电机最优控制. Tech. J. Eng. Appl. Sci. 4, 373–379. Mamizadeh, A., Genc, N., Rajabioun, R., 2018. 基于布谷鸟优化算法的升压DC-DC变换器PI控制器最优整定. 载于: 2018年第七届国际可再生能源研究与应用会议. ICRERA. IEEE, 第677–680页. Man, K.-F., Tang, K.-S., Kwong, S., 1996. 遗传算法: 概念及其在工程设计中的应用. IEEE Trans. Ind. Electron. 43, 519–534.

Mange, J., Pace, S., Dunn, A., 2015. 动态地形上自动化地面车辆控制的PID控制器参数优化. 见: 国际科学计算会议论文集. CSC, 第3页. Marinakis, Y., Marinaki, M., 2014. 一种用于开放车辆路径问题的大黄蜂交配优化算法. Swarm Evol. Comput. 15, 80–94. Martins, F.G., 2005. 使用ITAE准则整定PID控制器. Int. J. Eng. Educ. 21, 867. McFarlane, D., Glover, K., 1992. 使用H ∞ 综合的回路成形设计程序. IEEE Trans. Autom. Control 37, 759–769. Medewar, P., Munje, R., 2015. 基于PSO的PMDC电机PID控制器整定. 见: 2015年国际能源系统与应用会议. IEEE, 第522–526页. Meena, D., Devanshu, A., 2017. 遗传算法整定PID控制器用于过程控制. 见: 2017年国际创新系统与控制会议. ICISC. IEEE, 第1–6页. Mehrabian, A.R., Lucas, C., 2006. 一种受杂草定殖启发的数值优化算法. Ecol. Inform. 1, 355–366. Meng, A.-b., Chen, Y.-c., Yin, H., Chen, S.-z., 2014. 纵横交叉优化算法及其应用. Knowl.-Based Syst. 67, 218–229. Metropolis, N., Rosenbluth, A.W., Rosenbluth, M.N., Teller, A.H., Teller, E., 1953. 快速计算的状态方程计算. J. Chem. Phys. 21, 1087–1092. Mirjalili, S., Lewis, A., 2016. 鲸鱼优化算法. Adv. Eng. Softw. 95, 51–67. Mirjalili, S., Mirjalili, S.M., Lewis, A., 2014. 灰狼优化器. Adv. Eng. Softw. 69, 46–61. Mirjalili, S., Mirjalili, S.M., Saremi, S., Mirjalili, S., 2020. 鲸鱼优化算法: 理论、文献综述及在光子晶体滤波器设计中的应用. 见: 自然启发优化器. Springer, 第219–238页. Mohammed, H.M., Umar, S.U., Rashid, T.A., 2019. 鲸鱼优化算法的系统性和荟萃分析调查. Comput. Intell. Neurosci. 2019. Mohammed, N.F., Song, E., Ma, X., Hayat, Q., 2014. 使用遗传算法整定同步发电机PID控制器. 见: 2014年IEEE国际机电一体化与自动化会议. IEEE, 第1544–1548页. Moradi, M., 2003. PID控制器设计新技术. 见: 2003年IEEE控制应用会议论文集. CCA 2003, 第2卷. IEEE, 第903–908页. Mosaad, A.M., Attia, M.A., Abdelaziz, A.Y., 2019a. 用于整定AVR系统PID和PIDA控制器的鲸鱼优化算法. Ain Shams Eng. J. 10, 755–767. Mosaad, M.I., abed el Raouf, M.O., Al-Ahmar, M.A., Banakher, F.A., 2019b. 基于布谷鸟搜索算法的光伏系统最大功率点跟踪; 综述与比较. Energy Proc. 162, 117–126. Mucherino, A., Seref, O., 2007. 猴子搜索: 一种用于全局优化的新型元启发式搜索. 见: AIP会议论文集. AIP, 第953卷, 第162–173页. Mukherjee, P., Das, A., Bera, P.S., 2019. 使用对立磷虾群算法对单机无穷大母线系统设计PID电力系统稳定器. 见: 国际现代科学与技术创新会议. Springer, 第827–844页. Muliadi, J., Kusumoputro, B., 2018. 无人机高度动力学神经网络控制系统及其与PID控制系统的比较. J. Adv. Transp. 2018. Muro, C., Escobedo, R., Spector, L., Coppinger, R., 2011. 狼群 (灰狼) 狩猎策略在计算模拟中从简单规则涌现. Behav. Process. 88, 192–197. Murrieta-Mendoza, A., Botez, R.M., Bunel, A., 2018. 使用人工蜂群算法的四维飞机航路优化算法. J. Aerosp. Inform. Syst. 15, 307–334. Nagaraj, B., Subha, S., Rampriya, B., 2008. 使用软计算技术的PID控制器整定算法. Int. J. Comput. Sci. Netw. Secur. 8, 278–281. Nakrani, S., Tovey, C., 2004. 关于蜜蜂与互联网托管中心的动态服务器分配. Adapt. Behav. 12, 223–240. Natarajan, K., Gilbert, A.F., 1997. 基于有限频率响应数据的直接PID控制器整定. ISA Trans. 36, 139–149. Nimisha, K., Senthilkumar, R., 2018. 使用布谷鸟搜索算法对升降压变换器进行PID控制器最优整定的综述. 见: 2018年国际控制、电力、通信与计算技术会议. ICCPCCT. IEEE, 第216–221页. Nisi, K., Nagaraj, B., Agalya, A., 2019. 使用进化多目标优化方法整定PID控制器及其在纸浆与造纸工业中的应用. Int. J. Mach. Learn. Cybern. 10, 2015–2025. Niu, B., Fan, Y., Tan, L., Rao, J., Li, L., 2010. 细菌觅食优化综述第二部分: 应用与挑战. 见: 智能计算国际会议. Springer, 第544–550页. Niu, B., Zhu, Y., He, X., Zeng, X., 2006. 仅用智能萌芽的PID控制器优化设计. 见: 2006年第六届世界智能控制与自动化大会. IEEE, 第1卷, 第3584–3588页. Nojhan, 2007. 元启发式梦想. 获取自 <http://nojhan.free.fr/metah/index.php>. Passino, K.M., 2002. 细菌觅食的仿生学用于分布式优化与控制. IEEE Control Syst. Mag. 22, 52–67. Patel, V.K., Savsani, V.J., 2016. 一种多目标改进教学与学优化算法 (MO-ITLBO). Inf. Sci. 357, 182–200. Patil, N., Lakhekar, G., 2017. 使用遗传算法设计串级控制过程的PID控制器. 见: 2017年国际智能计算与控制系统会议. ICICCS. IEEE, 第1089–1095页.

Peram, M., Mishra, S., Vemulapaty, M., Verma, B., Padhy, P.K., 2018. Optimal PI-PD and I-PD controller design using cuckoo search algorithm. In: 2018 5th International Conference on Signal Processing and Integrated Networks. SPIN. IEEE, pp. 643–646.

Pham, Q.-V., Mirjalili, S., Kumar, N., Alazab, M., Hwang, W.-J., 2020. Whale optimization algorithm with applications to resource allocation in wireless networks. IEEE Trans. Veh. Technol. 69, 4285–4297.

Pierezan, J., Coelho, L.D.S., 2018. Coyote optimization algorithm: a new metaheuristic for global optimization problems. In: 2018 IEEE Congress on Evolutionary Computation. CEC. IEEE, pp. 1–8.

Pijarski, P., Kacejko, P., 2019. A new metaheuristic optimization method: the algorithm of the innovative gunner (aig). Eng. Optim. 51, 2049–2068.

Pilla, R., Botcha, N., Gorripotu, T.S., Azar, A.T., 2019. Fuzzy PID controller for automatic generation control of interconnected power system tuned by glow-worm swarm optimization. In: International Conference on Application of Robotics in Industry Using Advanced Mechanisms. Springer, pp. 140–149.

Pirabakaran, K., Becerra, V., 2001. Automatic tuning of PID controllers using model reference adaptive control techniques. In: IECON'01. 27th Annual Conference of the IEEE Industrial Electronics Society (Cat. No. 37243), vol. 1. IEEE, pp. 736–740.

Prasad, L.B., Tyagi, B., Gupta, H.O., 2014. Optimal control of nonlinear inverted pendulum system using PID controller and LQR: performance analysis without and with disturbance input. Int. J. Autom. Comput. 11, 661–670.

Premkumar, K., Manikandan, B., 2016. Bat algorithm optimized fuzzy pd based speed controller for brushless direct current motor. Int. J. Eng. Sci. Technol. 19, 818–840.

Priyambada, S., Mohanty, P.K., Sahu, B.K., 2014. Automatic voltage regulator using TLBO algorithm optimized PID controller. In: 2014 9th International Conference on Industrial and Information Systems. ICIIS. IEEE, pp. 1–6.

Priyambodo, T.K., Putra, A.E., Dharmawan, A., 2015. Optimizing control based on ant colony logic for quadrotor stabilization. In: 2015 IEEE International Conference on Aerospace Electronics and Remote Sensing Technology. ICARES. IEEE, pp. 1–4.

Qasem, S.N., Shamsuddin, S.M., 2011. Memetic elitist Pareto differential evolution algorithm based radial basis function networks for classification problems. Appl. Soft Comput. 11, 5565–5581.

Qi, H., Altinakar, M.S., 2011. A conceptual framework of agricultural land use planning with BMP for integrated watershed management. J. Environ. Manag. 92, 149–155.

Qiao, W., Yang, Z., 2019. Modified dolphin swarm algorithm based on chaotic maps for solving high-dimensional function optimization problems. IEEE Access 7, 110472–110486.

Qin, S.J., 1998. Control performance monitoring a review and assessment. Comput. Chem. Eng. 23, 173–186.

Quwaider, M., Shatnawi, Y., 2020. Neural network model as internet of things congestion control using PID controller and immune-hill-climbing algorithm. Simul. Model. Pract. Theory 101, 102022.

Rahmani, M., Ghanbari, A., Ettefagh, M.M., 2016. Robust adaptive control of a bio-inspired robot manipulator using bat algorithm. Expert Syst. Appl. 56, 164–176.

Rahmani, M., Komijani, H., Ghanbari, A., Ettefagh, M.M., 2018. Optimal novel super-twisting PID sliding mode control of a MEMS gyroscope based on multi-objective bat algorithm. Microsyst. Technol. 24, 2835–2846.

Rajabioun, R., 2011. Cuckoo optimization algorithm. Appl. Soft Comput. 11, 5508–5518.

Rajakumar, R., Amudhavel, J., Dhavachelvan, P., Vengattaraman, T., 2017. Gwo-lpwsn: grey wolf optimization algorithm for node localization problem in wireless sensor networks. J. Comput. Netw. Commun. 2017.

Rajesh, R.J., Ananda, C.M., 2015. PSO tuned PID controller for controlling camera position in UAV using 2-axis gimbal. In: 2015 International Conference on Power and Advanced Control Engineering. ICPACE. IEEE, pp. 128–133.

Rajinikanth, V., Latha, K., 2012. I-PD controller tuning for unstable system using bacterial foraging algorithm: a study based on various error criterion. In: Applied Computational Intelligence and Soft Computing.

Rajinikanth, V., Satapathy, S.C., 2015. Design of controller for automatic voltage regulator using teaching learning based optimization. Proc. Technol. 21, 295–302.

Rakovic, S., Teel, A., Mayne, D., Astolfi, A., 2006. Simple robust control invariant tubes for some classes of nonlinear discrete time systems. In: Proceedings of the 45th IEEE Conference on Decision and Control. IEEE, pp. 6397–6402.

Ramachandran, R., Madasamy, B., Veerasamy, V., Saravanan, L., 2018. Load frequency control of a dynamic interconnected power system using generalised Hopfield neural network based self-adaptive PID controller. IET Gener. Transm. Distrib. 12, 5713–5722.

Ramaswamy, R., Sbalzarini, I.F., 2010. A partial-propensity variant of the composition-rejection stochastic simulation algorithm for chemical reaction networks. J. Chem. Phys. 132, 044102.

Ramos-Figueroa, O., Quiroz-Castellanos, M., Mezura-Montes, E., Schütze, O., 2020. Meta-heuristics to solve grouping problems: a review and a case study. Swarm Evol. Comput. 53, 100643.

Rani, B.S., Kumar, C.A., 2015. A comprehensive review on bacteria foraging optimization technique. In: Multi-Objective Swarm Intelligence. Springer, pp. 1–25.

Rao, D.C., Kabat, M.R., Das, P.K., Jena, P.K., 2018. Cooperative navigation planning of multiple mobile robots using improved krill herd. Arab. J. Sci. Eng. 43, 7869–7891.

Rao, R., 2016. Jaya: a simple and new optimization algorithm for solving constrained and unconstrained optimization problems. Int. J. Ind. Eng. Comput. 7, 19–34.

Rao, R., Patel, V., 2014. A multi-objective improved teaching-learning based optimization algorithm for unconstrained and constrained optimization problems. Int. J. Ind. Eng. Comput. 5, 1–22.

Rao, R.V., Kalyankar, V., 2013. Parameter optimization of modern machining processes using teaching–learning-based optimization algorithm. Eng. Appl. Artif. Intell. 26, 524–531.

Rao, R.V., Savsani, V.J., Vakharia, D., 2011. Teaching–learning-based optimization: a novel method for constrained mechanical design optimization problems. Comput. Aided Des. 43, 303–315.

Rao, R.V., Savsani, V.J., Vakharia, D., 2012. Teaching–learning-based optimization: an optimization method for continuous non-linear large scale problems. Inf. Sci. 183, 1–15.

Rashedi, E., Nezamabadi-Pour, H., Saryazdi, S., 2009. GSA: a gravitational search algorithm. Inf. Sci. 179, 2232–2248.

Razmjooy, N., Khalilpour, M., Ramezani, M., 2016. A new meta-heuristic optimization algorithm inspired by FIFA world cup competitions: theory and its application in PID designing for AVR system. J. Control Autom. Electr. Syst. 27, 419–440.

Regad, M., Helaimi, M., Taleb, R., Othman, A.M., Gabbar, H.A., 2020. Frequency control of microgrid with renewable generation using PID controller based krill herd. Indones. J. Electr. Eng. Inform. 8, 21–32.

Reis, V., Santos, M., Carmo, M., Ferreira, F., 2016. Control of level systems by Arduino via PC platform. In: 2016 20th International Conference on System Theory, Control and Computing. ICSTCC. IEEE, pp. 251–256.

Ribeiro, J., Santos, M., Carmo, M., Silva, M., 2017. Comparison of PID controller tuning methods: analytical/classical techniques versus optimization algorithms. In: 2017 18th International Carpathian Control Conference. ICC. IEEE, pp. 533–538.

Rodríguez-Molina, A., Mezura-Montes, E., Villarreal-Cervantes, M.G., Aldape-Pérez, M., 2020. Multi-objective meta-heuristic optimization in intelligent control: a survey on the controller tuning problem. Appl. Soft Comput., 106342.

Ruano, A.E., Ge, S.S., Guerra, T.M., Lewis, F.L., Principe, J.C., Colnarić, M., 2014. Computational intelligence in control. Annu. Rev. Control 38, 233–242.

Rubinstein, R.Y., 1997. Optimization of computer simulation models with rare events. Eur. J. Oper. Res. 99, 89–112.

Saad, M.S., Jamaluddin, H., Darus, I.Z., 2012. PID controller tuning using evolutionary algorithms. WSEAS Trans. Syst. Control 7, 139–149.

Saha, A., Saikia, L.C., 2017. Utilisation of ultra-capacitor in load frequency control under restructured STPP-thermal power systems using WOA optimised PIDN-FOPD controller. IET Gener. Transm. Distrib. 11, 3318–3331.

Sahu, B.K., Pati, S., Mohanty, P.K., Panda, S., 2015. Teaching–learning based optimization algorithm based fuzzy-PID controller for automatic generation control of multi-area power system. Appl. Soft Comput. 27, 240–249.

Sahu, R.K., Panda, S., Rout, U.K., Sahoo, D.K., 2016. Teaching learning based optimization algorithm for automatic generation control of power system using 2-DOF PID controller. Int. J. Electr. Power Energy Syst. 77, 287–301.

Sala, R., Müller, R., 2020. Benchmarking for metaheuristic black-box optimization: perspectives and open challenges. arXiv preprint. arXiv:2007.00541.

Salmani, M.H., Eshghi, K., 2017. A smart structural algorithm (SSA) based on infeasible region to solve mixed integer problems. Int. J. Appl. Metaheuristic Comput. 8, 24–44.

Sandoval, D., Soto, I., Adasme, P., 2015. Control of direct current motor using ant colony optimization. In: 2015 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies. CHILECON. IEEE, pp. 79–82.

dos Santos Coelho, L., 2009. Tuning of PID controller for an automatic regulator voltage system using chaotic optimization approach. Chaos Solitons Fractals 39, 1504–1514.

dos Santos Coelho, L., da Costa Silveira, C., 2006. Improved bacterial foraging strategy for controller optimization applied to robotic manipulator system. In: 2006 IEEE Conference on Computer Aided Control System Design, 2006 IEEE International Conference on Control Applications, 2006 IEEE International Symposium on Intelligent Control. IEEE, pp. 1276–1281.

dos Santos Coelho, L., Mariani, V.C., 2012. Firefly algorithm approach based on chaotic Tinkerbell map applied to multivariable PID controller tuning. Comput. Math. Appl. 64, 2371–2382.

Şen, M.A., Kalyoncu, M., 2018. Optimal tuning of PID controller using grey wolf optimizer algorithm for quadruped robot. Balkan J. Electr. Comput. Eng. 6, 29–35.

Sen, R., Pati, C., Dutta, S., Sen, R., 2015. Comparison between three tuning methods of PID control for high precision positioning stage. Mapan 30, 65–70.

Senberber, H., Bagis, A., 2017. Fractional PID controller design for fractional order systems using ABC algorithm. In: 2017 Electronics. IEEE, pp. 1–7.

Shah-Hosseini, H., 2008. Intelligent water drops algorithm: a new optimization method for solving the multiple knapsack problem. Int. J. Intell. Comput. Cybern. 1, 193–212.

Shah-Hosseini, H., 2011. Principal components analysis by the galaxy-based search algorithm: a novel metaheuristic for continuous optimisation. Int. J. Comput. Sci. Eng. 6, 132–140.

Shaheen, A.M., Spea, S.R., Farrag, S.M., Abido, M.A., 2018. A review of meta-heuristic algorithms for reactive power planning problem. Ain Shams Eng. J. 9, 215–231.

Shahrokh, M., Zomorodi, A., 2012. Comparison of PID controller tuning methods.

Shahrokh, M., Zomorodi, A., 2013. Comparison of PID Controller Tuning Methods. Department of Chemical & Petroleum Engineering Sharif University of Technology, pp. 1–2.

Shair, E.F., Khor, S., Abdullah, A., Jaafar, H., Saharuddin, N., Abidin, A.Z., 2015. Cuckoo search approach for cutting stock problem. Int. J. Inform. Electr. Eng. 5, 138.

Sharma, P., Gupta, A., Aggarwal, A., Gupta, D., Khanna, A., Hassanien, A.E., de Albuquerque, V.H.C., 2020. The health of things for classification of protein structure using improved grey wolf optimization. J. Supercomput. 76, 1226–1241.

Peram, M., Mishra, S., Vemulapaty, M., Verma, B., Padhy, P.K., 2018. 使用布谷鸟搜索算法设计最优PI-PD和I-PD控制器。见：2018年第五届信号处理与集成网络国际会议。SPIN. IEEE, 第643–646页。Pham, Q.-V., Mirjalili, S., Kumar, N., Alazab, M., Hwang, W.-J., 2020. 鲸鱼优化算法及其在无线网络资源分配中的应用。IEEE Trans. Veh. Technol. 69, 4285–4297。Pierezan, J., Coelho, L.D.S., 2018. 郊狼优化算法：一种用于全局优化问题的新型元启发式算法。见：2018年IEEE进化计算大会。CEC. IEEE, 第1–8页。Pijarski, P., Kacejko, P., 2019. 一种新的元启发式优化方法：创新炮手算法（AIG）。Eng. Optim. 51, 2049–2068。Pilla, R., Botcha, N., Gorripotu, T.S., Azar, A.T., 2019. 萤火虫群优化算法整定的互联电力系统自动发电控制模糊PID控制器。见：工业机器人应用先进机制国际会议。Springer, 第140–149页。Pirabakaran, K., Becerra, V., 2001. 使用模型参考自适应控制技术自动整定PID控制器。见：IECON'01. 第27届IEEE工业电子学会年会（目录号37243），第1卷。IEEE, 第736–740页。Prasad, L.B., Tyagi, B., Gupta, H.O., 2014. 使用PID控制器和LQR对非线性倒立摆系统进行最优控制：无扰动和带扰动输入的性能分析。Int. J. Autom. Comput. 11, 661–670。Premkumar, K., Manikandan, B., 2016. 蝙蝠算法优化的模糊PID速度控制器用于无刷直流电机。Int. J. Eng. Sci. Technol. 19, 818–840。Priyambada, S., Mohanty, P.K., Sahu, B.K., 2014. 使用TLBO算法优化PID控制器的自动电压调节器。见：2014年第九届国际工业与信息系统会议。ICIIS. IEEE, 第1–6页。Priyambodo, T.K., Putra, A.E., Dharmawan, A., 2015. 基于蚁群逻辑的四旋翼飞行器稳定优化控制。见：2015年IEEE航空航天电子与遥感技术国际会议。ICARES. IEEE, 第1–4页。Qasem, S.N., Shamsuddin, S.M., 2011. 基于记忆精英Pareto差分进化算法的径向基函数网络用于分类问题。Appl. Soft Comput. 11, 5565–5581。Qi, H., Altinakar, M.S., 2011. 结合BMP进行综合流域管理的农业土地利用规划概念框架。J. Environ. Manag. 92, 149–155。Qiao, W., Yang, Z., 2019. 基于混沌映射的改进海豚群算法求解高维函数优化问题。IEEE Access 7, 110472–110486。Qin, S.J., 1998. 控制性能监控：综述与评估。Comput. Chem. Eng. 23, 173–186。Quwaider, M., Shatnawi, Y., 2020. 结合PID控制器和免疫爬山算法的物联网拥塞控制神经网络模型。Simul. Model. Pract. Theory 101, 102022。Rahmani, M., Ghanbari, A., Ettefagh, M.M., 2016. 使用蝙蝠算法的仿生机器人操作臂鲁棒自适应控制。Expert Syst. Appl. 56, 164–176。Rahmani, M., Komijani, H., Ghanbari, A., Ettefagh, M.M., 2018. 基于多目标蝙蝠算法的MEMS陀螺仪最优新型超扭转PID滑模控制。Microsyst. Technol. 24, 2835–2846。Rajabioun, R., 2011. 布谷鸟优化算法。Appl. Soft Comput. 11, 5508–5518。Rajakumar, R., Amudhavel, J., Dhavachelvan, P., Vengattaraman, T., 2017. GWO-LPWSN: 用于无线传感器网络节点定位问题的灰狼优化算法。J. Comput. Netw. Commun. 2017。Rajesh, R.J., Ananda, C.M., 2015. PSO整定PID控制器用于控制两轴云台上的无人机相机位置。见：2015年电力与先进控制工程国际会议。ICPACE. IEEE, 第128–133页。Rajinikanth, V., Latha, K., 2012. 使用细菌觅食算法对不稳定系统进行I-PD控制器整定：基于各种误差准则的研究。见：应用计算智能与软计算。Rajinikanth, V., Satapathy, S.C., 2015. 使用教与学优化设计自动电压调节器控制器。Proc. Technol. 21, 295–302。Rakovic, S., Teel, A., Mayne, D., Astolfi, A., 2006. 一类非线性离散时间系统的简单鲁棒控制不变量。见：第45届IEEE决策与控制会议论文集。IEEE, 第6397–6402页。Ramachandran, R., Madasamy, B., Veerasamy, V., Saravanan, L., 2018. 基于广义Hopfield神经网络自适应PID控制器的动态互联电力系统负荷频率控制。IET Gener. Transm. Distrib. 12, 5713–5722。Ramaswamy, R., Sbalzarini, I.F., 2010. 化学反应网络组成-拒绝随机模拟算法的部分倾向变体。J. Chem. Phys. 132, 044102。Ramos-Figueroa, O., Quiroz-Castellanos, M., Mezura-Montes, E., Schütze, O., 2020. 求解分组问题的元启发式算法：综述与案例研究。Swarm Evol. Comput. 53, 100643。Rani, B.S., Kumar, C.A., 2015. 细菌觅食优化技术综述。见：多目标群体智能。Springer, 第1–25页。Rao, D.C., Kabat, M.R., Das, P.K., Jena, P.K., 2018. 使用改进磷虾群算法的多移动机器人协同导航规划。Arab. J. Sci. Eng. 43, 7869–7891。Rao, R., 2016. Jaya: 一种简单新颖的优化算法用于求解约束和无约束优化问题。Int. J. Ind. Eng. Comput. 7, 19–34。Rao, R., Patel, V., 2014. 一种用于无约束和约束优化问题的多目标改进教与学优化算法。Int. J. Ind. Eng. Comput. 5, 1–22。

Rao, R.V., Kalyankar, V., 2013. 使用教与学优化算法对现代加工过程进行参数优化。Eng. Appl. Artif. Intell. 26, 524–531。Rao, R.V., Savsani, V.J., Vakharia, D., 2011. 教与学优化：一种解决约束机械设计优化问题的新方法。Comput. Aided Des. 43, 303–315。Rao, R.V., Savsani, V.J., Vakharia, D., 2012. 教与学优化：一种用于连续非线性大规模问题的优化方法。Inf. Sci. 183, 1–15。Rashedi, E., Nezamabadi-Pour, H., Saryazdi, S., 2009. GSA: 一种引力搜索算法。Inf. Sci. 179, 2232–2248。Razmjooy, N., Khalilpour, M., Ramezani, M., 2016. 一种受FIFA世界杯比赛启发的元启发优化算法：理论及其在AVR系统PID设计中的应用。J. Control Autom. Electr. Syst. 27, 419–440。Regad, M., Helaimi, M., Taleb, R., Othman, A.M., Gabbar, H.A., 2020. 基于磷虾群算法和PID控制器的可再生能源微电网频率控制。Indones. J. Electr. Eng. Inform. 8, 21–32。Reis, V., Santos, M., Carmo, M., Ferreira, F., 2016. 通过PC平台基于Arduino的液位系统控制。2016年第20届系统理论、控制与计算国际会议。ICSTCC. IEEE, pp. 251–256。Ribeiro, J., Santos, M., Carmo, M., Silva, M., 2017. PID控制器整定方法比较：解析/经典技术与优化算法。2017年第18届国际喀尔巴阡控制会议。ICCC. IEEE, pp. 533–538。Rodríguez-Molina, A., Mezura-Montes, E., Villarreal-Cervantes, M.G., Aldape-Pérez, M., 2020. 智能控制中的多目标元启发优化：控制器调优问题综述。Appl. Soft Comput., 106342。Ruano, A.E., Ge, S.S., Guerra, T.M., Lewis, F.L., Principe, J.C., Colnarić, M., 2014. 控制中的计算智能。Annu. Rev. Control 38, 233–242。Rubinstein, R.Y., 1997. 稀有事件计算机仿真模型优化。Eur. J. Oper. Res. 99, 89–112。Saad, M.S., Jamaluddin, H., Darus, I.Z., 2012. 使用进化算法进行PID控制器调优。WSEAS Trans. Syst. Control 7, 139–149。Saha, A., Saikia, L.C., 2017. 在重组STPP火力发电系统负荷频率控制中利用超级电容器，采用WOA优化的PIDN-FOPD控制器。IET Gener. Transm. Distrib. 11, 3318–3331。Sahu, B.K., Pati, S., Mohanty, P.K., Panda, S., 2015. 基于教与学优化算法的模糊PID控制器用于多区域电力系统自动发电控制。Appl. Soft Comput. 27, 240–249。Sahu, R.K., Panda, S., Rout, U.K., Sahoo, D.K., 2016. 基于教与学优化算法的2-DOF PID控制器用于电力系统自动发电控制。Int. J. Electr. Power Energy Syst. 77, 287–301。Sala, R., Müller, R., 2020. 元启发黑箱优化基准测试：视角与公开挑战。arXiv预印本。arXiv:2007.00541。Salmani, M.H., Eshghi, K., 2017. 一种基于不可行域的智能结构算法(SSA)用于解决混合整数问题。Int. J. Appl. Metaheuristic Comput. 8, 24–44。Sandoval, D., Soto, I., Adasme, P., 2015. 基于蚁群优化的直流电机控制。2015年智利电气电子工程与信息技术会议。CHILECON. IEEE, pp. 79–82。dos Santos Coelho, L., 2009. 使用混沌优化方法对自动电压调节器系统进行PID控制器整定。Chaos Solitons Fractals 39, 1504–1514。dos Santos Coelho, L., da Costa Silveira, C., 2006. 改进的细菌觅食策略用于机器人操纵器系统控制器优化。2006年IEEE计算机辅助控制系统设计会议、2006年IEEE控制应用国际会议、2006年IEEE智能控制国际研讨会。IEEE, pp. 1276–1281。dos Santos Coelho, L., Mariani, V.C., 2012. 基于混沌Tinkerbell映射的萤火虫算法用于多变量PID控制器整定。Comput. Math. Appl. 64, 2371–2382。Sen, M.A., Kalyoncu, M., 2018. 基于灰狼优化算法的四足机器人PID控制器最优调优。Balkan J. Electr. Comput. Eng. 6, 29–35。Sen, R., Pati, C., Dutta, S., Sen, R., 2015. 高精度定位平台PID控制三种调优方法比较。Mapan 30, 65–70。Senberber, H., Bagis, A., 2017. 使用ABC算法设计分数阶系统的分数阶PID控制器。2017年电子学会议。IEEE, pp. 1–7。Shah-Hosseini, H., 2008. 智能水滴算法：一种解决多背包问题的新优化方法。Int. J. Intell. Comput. Cybern. 1, 193–212。Shah-Hosseini, H., 2011. 基于银河搜索算法的主成分分析：一种用于连续优化的新型元启发式算法。Int. J. Comput. Sci. Eng. 6, 132–140。Shaheen, A.M., Spea, S.R., Farrag, S.M., Abido, M.A., 2018. 元启发算法在无功规划问题中的综述。Ain Shams Eng. J. 9, 215–231。Shahrokh, M., Zomorodi, A., 2012. PID控制器整定方法比较。Shahrokh, M., Zomorodi, A., 2013. PID控制器整定方法比较。谢里夫科技大学化学与石油工程系, pp. 1–2。Shair, E.F., Khor, S., Abdullah, A., Jaafar, H., Saharuddin, N., Abidin, A.Z., 2015. 布谷鸟搜索方法在切割库存问题中的应用。Int. J. Inform. Electr. Eng. 5, 138。Sharma, P., Gupta, A., Aggarwal, A., Gupta, D., Khanna, A., Hassanien, A.E., de Albuquerque, V.H.C., 2020. 健康物联网中基于改进灰狼优化的蛋白质结构分类。J. Supercomput. 76, 1226–1241。

Sharma, P., Sundaram, S., Sharma, M., Sharma, A., Gupta, D., 2019. Diagnosis of Parkinson's disease using modified grey wolf optimization. *Cogn. Syst. Res.* 54, 100–115.

Shatnawi, M., Bayoumi, E., 2019. Brushless DC motor controller optimization using simulated annealing. In: 2019 International Conference on Electrical Drives & Power Electronics. EDPE. IEEE, pp. 292–297.

Shatnawi, Y., Quwaider, M., 2019. Congestion control in ATM networks using PID controller with immune algorithm. In: 2019 10th International Conference on Information and Communication Systems. ICICS. IEEE, pp. 19–24.

Shayanfar, H., Gharehchopogh, F.S., 2018. Farmland fertility: a new metaheuristic algorithm for solving continuous optimization problems. *Appl. Soft Comput.* 71, 728–746.

Shayanfar, H.A., Shayeghi, H., Younesi, A., 2015. Optimal PID controller design using krill herd algorithm for frequency stabilizing in an isolated wind-diesel system. In: Proceedings on the International Conference on Artificial Intelligence. ICAI. The Steering Committee of the World Congress in Computer Science, Computer, p. 516.

Shie, F.S., Chen, M.-Y., Liu, Y.-S., 2012. Prediction of corporate financial distress: an application of the America banking industry. *Neural Comput. Appl.* 21, 1687–1696.

Shilane, D., Martikainen, J., Dudoit, S., Ovaska, S.J., 2008. A general framework for statistical performance comparison of evolutionary computation algorithms. *Inf. Sci.* 178, 2870–2879.

Sibaliĵa, T., 2018. Application of simulated annealing in process optimization: a review. In: Simulated Annealing: Introduction, Applications and Theory, pp. 1–14.

Siddique, N., Adeli, H., 2016. Simulated annealing, its variants and engineering applications. *Int. J. Artif. Intell. Tools* 25, 1630001.

da Silva, B.R., da Silva Oliveira, L., de Alencar, D.B., das Neves Nascimento, E.R., Dinola, I.C.S., Nascimento, M.H.R., 2019. Analysis of robust PID control with pre-filter using the perfect itae performance criterion applied to the heavy-duty gas turbine fuel system. *Int. J. Adv. Eng. Res. Sci.* 6.

Simon, D., 2008. Biogeography-based optimization. *IEEE Trans. Evol. Comput.* 12, 702–713.

Şimşek, E.C., Köse, A., Şahin, M., Irmak, E., 2019. Optimization of PID parameters using ant colony algorithm for position control of DC motor. In: 2019 8th International Conference on Renewable Energy Research and Applications. ICRERA. IEEE, pp. 1047–1051.

Singh, K., Vasant, P., Elamvazuthi, I., Kannan, R., 2015. PID tuning of servo motor using bat algorithm. *Proc. Comput. Sci.* 60, 1798–1808.

Sivakumar, R., Deepa, P., Sankaran, D., 2016. A study on BFO algorithm based PID controller design for MIMO process using various cost functions. *Indian J. Sci. Technol.* 9, 1–6.

Skogestad, S., 2001. Probably the best simple PID tuning rules in the world. In: AICHE Annual Meeting. Reno, Nevada, vol. 77.

Solihin, M.I., Akmeliawati, R., 2018. Robust feedback controller design using cuckoo search optimization to maximize stability radius. In: International Conference on Robot Intelligence Technology and Applications. Springer, pp. 62–75.

Song, M.-P., Gu, G.-C., 2004. Research on particle swarm optimization: a review. In: Proceedings of 2004 International Conference on Machine Learning and Cybernetics (IEEE Cat. No. 04EX826), vol. 4. IEEE, pp. 2236–2241.

Soni, Y.K., Bhatt, R., 2013. BF-PSO optimized PID controller design using ISE, IAE, IATE and MSE error criteria. *Int. J. Adv. Res. Comput. Eng. Technol.* 2, 2333–2336.

Sorensen, E., 2007. An introduction to process control. In: Stefaan, S. (Ed.), Concepts of Chemical Engineering for Chemists. Royal Society of Chemistry, pp. 253–276.

Sreekanth, P., Hari, A., 2016. Genetic algorithm based self tuning regulator for ball and hoop system. In: 2016 Conference on Emerging Devices and Smart Systems. ICEDSS. IEEE, pp. 147–152.

Srinivas, P., Lakshmi, K.V., Kumar, V.N., 2014. A comparison of PID controller tuning methods for three tank level process. *Int. J. Adv. Res. Electr. Electron. Instrum. Eng.* 3, 6810–6820.

Štimac, G., Braut, S., Žigulić, R., 2014. Comparative analysis of PSO algorithms for PID controller tuning. *Chin. J. Mech. Eng.* 27, 928–936.

Stojanović, I., Brajević, I., Stanimirović, P.S., Kazakovtsev, L.A., Zdravev, Z., 2017. Application of heuristic and metaheuristic algorithms in solving constrained Weber problem with feasible region bounded by arcs. *Math. Probl. Eng.*

Storn, R., Price, K., 1997. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *J. Glob. Optim.* 11, 341–359.

Sule, A.H., Mokhtar, A.S., Jamian, J.J.B., Khidrani, A., Larik, R.M., 2020. Optimal tuning of proportional integral controller for fixed-speed wind turbine using grey wolf optimizer. *Int. J. Electr. Comput. Eng.* 10, 5251–5261.

Suman, B., Hoda, N., Jha, S., 2010. Orthogonal simulated annealing for multiobjective optimization. *Comput. Chem. Eng.* 34, 1618–1631.

Suman, B., Kumar, P., 2006. A survey of simulated annealing as a tool for single and multiobjective optimization. *J. Oper. Res. Soc.* 57, 1143–1160.

Sundareswaran, K., Nayak, P.S., 2012. Ant colony based feedback controller design for soft-starter fed induction motor drive. *Appl. Soft Comput.* 12, 1566–1573.

Sweidan, A.H., El-Bendary, N., Hassanien, A.E., Hegazy, O.M., Mohamed, A.-K., 2016. Grey wolf optimizer and case-based reasoning model for water quality assessment. In: The 1st International Conference on Advanced Intelligent System and Informatics. AISI2015, Beni Suef, Egypt, November 28–30, 2015. Springer, pp. 229–239.

Taillard, É.D., Voss, S., 2002. POPMUSIC – partial optimization metaheuristic under special intensification conditions. In: Essays and Surveys in Metaheuristics. Springer, pp. 613–629.

Talbi, E.-G., 2009. Metaheuristics: from Design to Implementation, vol. 74. John Wiley & Sons.

Talbi, E.-G., 2020. Machine learning into metaheuristics: a survey and taxonomy of data-driven metaheuristics.

Tamura, K., Yasuda, K., 2011. Primary study of spiral dynamics inspired optimization. *IEEJ Trans. Electr. Electron. Eng.* 6, S98–S100.

Tan, W., Marquez, H.J., Chen, T., 2004. Performance assessment of PID controllers. *Control Intell. Syst.* 32, 158–166.

Tan, W., Yuan, Y., Niu, Y., 1999. Tuning of PID controller for unstable process. In: Proceedings of the 1999 IEEE International Conference on Control Applications (Cat. No. 99CH36328), vol. 1. IEEE, pp. 121–124.

Tan, Y., Zhu, Y., 2010. Fireworks algorithm for optimization. In: International Conference in Swarm Intelligence. Springer, pp. 355–364.

Tang, W., Wu, Q., 2009. Biologically inspired optimization: a review. *Trans. Inst. Meas. Control* 31, 495–515.

Tavares, R.S., Sato, A.K., Martins, T.C., Lima, R.G., Tsuzuki, M.S.G., 2019. GPU acceleration of absolute EIT image reconstruction using simulated annealing. *Biomed. Signal Process. Control* 52, 445–455.

Tikhamarine, Y., Souag-Gamane, D., Ahmed, A.N., Kisi, O., El-Shafie, A., 2020. Improving artificial intelligence models accuracy for monthly streamflow forecasting using grey wolf optimization (GWO) algorithm. *J. Hydrol.* 582, 124435.

Valluru, S.K., Singh, M., 2017. Investigation of NARMA L-2 and artificial bee colony tuned PID controllers for bench scaled nonlinear dynamical system. *Int. J. Control Theory Appl.* 10, 363–374.

Varty, Z., 2017. Simulated annealing overview.

Verma, P., Patel, N., Nair, N.-K.C., Sikander, A., 2016. Design of PID controller using cuckoo search algorithm for buck-boost converter of led driver circuit. In: 2016 IEEE 2nd Annual Southern Power Electronics Conference. SPEC. IEEE, pp. 1–4.

Verma, S.K., Nagar, S.K., 2018. Design and optimization of fractional order PI λ D μ controller using grey wolf optimizer for automatic voltage regulator system. *Recent Adv. Electr. Electron. Eng.* 11, 217–226.

Vijay, D., Banu, U.S., 2016. PID controller tuned using simulated annealing for assured crew re-entry vehicle with PWPf thruster firing. In: 2016 Online International Conference on Green Engineering and Technologies. IC-GET. IEEE, pp. 1–6.

Villarreal-Cervantes, M.G., Alvarez-Gallegos, J., 2016. Off-line PID control tuning for a planar parallel robot using DE variants. *Expert Syst. Appl.* 64, 444–454.

Wang, C., Cheng, H.Z., 2008. Optimization of network configuration in large distribution systems using plant growth simulation algorithm. *IEEE Trans. Power Syst.* 23, 119–126.

Wang, G.-G., Deb, S., Coelho, L.d.S., 2015. Elephant herding optimization. In: 2015 3rd International Symposium on Computational and Business Intelligence. ISCBI. IEEE, pp. 1–5.

Wang, G.-G., Gandomi, A.H., Alavi, A.H., Gong, D., 2019. A comprehensive review of krill herd algorithm: variants, hybrids and applications. *Artif. Intell. Rev.* 51, 119–148.

Wang, J.-J., 2011. Simulation studies of inverted pendulum based on PID controllers. *Simul. Model. Pract. Theory* 19, 440–449.

Wang, Q., Spronck, P., Tracht, R., 2003. An overview of genetic algorithms applied to control engineering problems. In: Proceedings of the 2003 International Conference on Machine Learning and Cybernetics (IEEE Cat. No. 03EX693). In: IEEE, vol. 3, pp. 1651–1656.

Wang, Q.-G., Hang, C.-C., Bi, Q., 1999. A technique for frequency response identification from relay feedback. *IEEE Trans. Control Syst. Technol.* 7, 122–128.

Wolpert, D.H., Macready, W.G., et al., 1997. No free lunch theorems for optimization. *IEEE Trans. Evol. Comput.* 1, 67–82.

Wong, D., Leong, H.W., Liu, H., 2012. Simulated Annealing for VLSI Design, vol. 42. Springer Science & Business Media.

Wu, H., Xu, D., Yang, M., Zhang, X., 2016. Surface structure of hydroxyapatite from simulated annealing molecular dynamics simulations. *Langmuir* 32, 4643–4652.

Wu, X., Zhou, Y., Lu, Y., 2017. Elite opposition-based water wave optimization algorithm for global optimization. *Math. Probl. Eng.* 2017.

Wu, Z., Zhao, X., Ma, Y., Zhao, X., 2019. A hybrid model based on modified multi-objective cuckoo search algorithm for short-term load forecasting. *Appl. Energy* 237, 896–909.

Xia, S., Li, L., 2017. Application of greenhouse temperature prediction based on pso-rbf neural network. *Comput. Eng. Des.* 38, 744–748.

Xu, Y., Fan, P., Yuan, L., 2013. A simple and efficient artificial bee colony algorithm. *Math. Probl. Eng.*

Yachen, Z., Yueming, H., 2008. On PID controllers based on simulated annealing algorithm. In: 2008 27th Chinese Control Conference. IEEE, pp. 225–228.

Yadav, S., Nagar, S.K., Mishra, A., 2020. Tuning of parameters of PID controller using grey wolf optimizer. Available at SSRN 3575432.

Yadav, S., Verma, S.K., Nagar, S.K., 2016. Optimized PID controller for magnetic levitation system. *IFAC-PapersOnLine* 49, 778–782.

Yadav, S., Verma, S.K., Nagar, S.K., 2018. Performance enhancement of magnetic levitation system using teaching learning based optimization. *Alex. Eng. J.* 57, 2427–2433.

Yaghoobi, S., Mojjallali, H., 2016. Tuning of a PID controller using improved chaotic krill herd algorithm. *Optik* 127, 4803–4807.

Yang, B., Wang, J., Zhang, X., Yu, T., Yao, W., Shu, H., Zeng, F., Sun, L., 2020. Comprehensive overview of meta-heuristic algorithm applications on PV cell parameter identification. *Energy Convers. Manag.* 208, 112595.

Yang, X.-S., 2009. Firefly algorithms for multimodal optimization. In: International Symposium on Stochastic Algorithms. Springer, pp. 169–178.

Sharma, P., Sundaram, S., Sharma, M., Sharma, A., Gupta, D., 2019. 基于改进灰狼优化的帕金森病诊断. *Cogn. Syst. Res.* 54, 100–115. Shatnawi, M., Bayoumi, E., 2019. 基于模拟退火的无刷直流电机控制器优化. 2019年国际电力驱动与电力电子会议. EDPE. IEEE, pp. 292–297. Shatnawi, Y., Quwaider, M., 2019. 基于免疫算法和PID控制器的ATM网络拥塞控制. 2019年第10届国际信息与通信系统会议. ICICS. IEEE, pp. 19–24. Shayanfar, H., Gharehchopogh, F.S., 2018. 农田肥力: 一种解决连续优化问题的新元启发算法. *Appl. Soft Comput.* 71, 728–746. Shayanfar, H.A., Shayeghi, H., Younesi, A., 2015. 使用磷虾群算法设计最优PID控制器用于孤立风桨系统频率稳定. 国际人工智能会议论文集. ICAI. 世界计算机科学大会指导委员会, p. 516. Shie, F.S., Chen, M.-Y., Liu, Y.-S., 2012. 企业财务困境预测: 美国银行业的应用. *Neural Comput. Appl.* 21, 1687–1696. Shilane, D., Martikainen, J., Dudoit, S., Ovaska, S.J., 2008. 进化计算算法统计性能比较的通用框架. *Inf. Sci.* 178, 2870–2879. Sibaliĵa, T., 2018. 模拟退火在过程优化中的应用综述. 模拟退火: 介绍、应用与理论, pp. 1–14. Siddique, N., Adeli, H., 2016. 模拟退火及其变体与工程应用. *Int. J. Artif. Intell. Tools* 25, 1630001. da Silva, B.R., da Silva Oliveira, L., de Alencar, D.B., das Neves Nascimento, E.R., Dinola, I.C.S., Nascimento, M.H.R., 2019. 基于完美ITAE性能准则的鲁棒PID控制与预滤波器在重型燃气轮机燃料系统中的应用分析. *Int. J. Adv. Eng. Res. Sci.* 6. Simon, D., 2008. 基于生物地理学的优化. *IEEE Trans. Evol. Comput.* 12, 702–713. Simsek, E.C., Köse, A., Sahin, M., Irmak, E., 2019. 使用蚁群算法优化直流电机位置控制的PID参数. 2019年第8届国际可再生能源研究与应用会议. ICRERA. IEEE, pp. 1047–1051. Singh, K., Vasant, P., Elamvazuthi, I., Kannan, R., 2015. 基于蝙蝠算法的伺服电机PID整定. *Proc. Comput. Sci.* 60, 1798–1808. Sivakumar, R., Deepa, P., Sankaran, D., 2016. 基于BFO算法的PID控制器设计用于多输入多输出过程及各种代价函数研究. *Indian J. Sci. Technol.* 9, 1–6. Skogestad, S., 2001. 可能是世界上最好的简单PID整定规则. AICHE年会. 内华达州里诺, vol. 77. Solihin, M.I., Akmeliawati, R., 2018. 使用布谷鸟搜索优化最大化稳定半径的鲁棒反馈控制器设计. 国际机器人智能技术与应用会议. Springer, pp. 62–75. Song, M.-P., Gu, G.-C., 2004. 粒子群优化研究综述. 2004年国际机器学习与控制会议论文集(IEEE Cat. No. 04EX826), vol. 4. IEEE, pp. 2236–2241. Soni, Y.K., Bhatt, R., 2013. 基于BF-PSO优化的PID控制器设计采用ISE、IAE、IATE和MSE误差准则. *Int. J. Adv. Res. Comput. Eng. Technol.* 2, 2333–2336. Sorensen, E., 2007. 过程控制导论. Stefaan, S. (主编), 化学工程师的化学工程概念. 皇家化学学会, pp. 253–276. Sreekanth, P., Hari, A., 2016. 基于遗传算法的球环系统自整定调节器. 2016年新兴器件与智能系统会议. ICEDSS. IEEE, pp. 147–152. Srinivas, P., Lakshmi, K.V., Kumar, V.N., 2014. 三水箱液位过程PID控制器整定方法比较. *Int. J. Adv. Res. Electr. Electron. Instrum. Eng.* 3, 6810–6820. Štimac, G., Braut, S., Žigulić, R., 2014. PSO算法用于PID控制器整定的比较分析. *Chin. J. Mech. Eng.* 27, 928–936. Stojanović, I., Brajević, I., Stanimirović, P.S., Kazakovtsev, L.A., Zdravev, Z., 2017. 启发式和元启发式算法在弧形边界可行域约束韦伯问题中的应用. *Math. Probl. Eng. Storn, R., Price, K., 1997. 差分进化——一种简单高效的连续空间全局优化启发式算法. J. Glob. Optim.* 11, 341–359. Sule, A.H., Mokhtar, A.S., Jamian, J.J.B., Khidrani, A., Larik, R.M., 2020. 基于灰狼优化器的定速风力涡轮机比例积分控制器最优整定. *Int. J. Electr. Comput. Eng.* 10, 5251–5261. Suman, B., Hoda, N., Jha, S., 2010. 多目标优化的正交模拟退火. *Comput. Chem. Eng.* 34, 1618–1631. Suman, B., Kumar, P., 2006. 模拟退火作为单目标和多目标优化工具的综述. *J. Oper. Res. Soc.* 57, 1143–1160. Sundareswaran, K., Nayak, P.S., 2012. 基于蚁群的软启动供电感应电机驱动反馈控制器设计. *Appl. Soft Comput.* 12, 1566–1573. Sweidan, A.H., El-Bendary, N., Hassanien, A.E., Hegazy, O.M., Mohamed, A.-K., 2016. 灰狼优化器和案例推理模型用于水质评估. 第一届国际先进智能系统与信息学会议. AISI2015, 贝尼苏韦夫, 埃及, 2015年11月28–30日. Springer, pp. 229–239. Taillard, É.D., Voss, S., 2002. POPMUSIC – 特殊强化条件下的部分优化元启发式. 元启发式论文与综述. Springer, pp. 613–629. Talbi, E.-G., 2009. 元启发式: 从设计到实现, vol. 74. John Wiley & Sons.

Talbi, E.-G., 2020. 机器学习在元启发式算法中的应用: 数据驱动元启发式算法的综述与分类. Tamura, K., Yasuda, K., 2011. 螺旋动力学启发优化的初步研究. *IEEJ Trans. Electr. Electron. Eng.* 6, S98–S100. Tan, W., Marquez, H.J., Chen, T., 2004. PID控制器性能评估. *Control Intell. Syst.* 32, 158–166. Tan, W., Yuan, Y., Niu, Y., 1999. 不稳定过程的PID控制器整定. In: Pro- ceedings of the 1999 IEEE International Conference on Control Applications (Cat. No. 99CH36328), vol. 1. IEEE, pp. 121–124. Tan, Y., Zhu, Y., 2010. 烟花算法优化. In: International Conference in S warm Intelligence. Springer, pp. 355–364. Tang, W., Wu, Q., 2009. 生物启发优化综述. *Trans. Inst. Meas. Control* 31, 495–515. Tavares, R.S., Sato, A.K., Martins, T.C., Lima, R.G., Tsuzuki, M.S.G., 2019. 使用模拟退火的绝对EIT图像重建GPU加速. *Biomed. Signal Process. Control* 52, 445–455. Tikhamarine, Y., Souag-Gamane, D., Ahmed, A.N., Kisi, O., El-Shafie, A., 2016. 使用布谷鸟搜索算法设计LED驱动电路升降压变换器的PID控制器. In: 2016 IEEE 2nd Annual Southern Power Electronics Conference. SPEC. IEEE, pp. 1–4. Verma, S.K., Nagar, S.K., 2018. 使用灰狼优化器设计和优化自动电压调节器系统的分数阶PI λ D μ 控制器. *Recent Adv. Electr. Electron. Eng.* 11, 217–226. Vijay, D., Banu, U.S., 2016. 使用模拟退火整定PID控制器用于可靠人员再入飞行器与PWPf推力器点火. In: 2016 Online International Con- ference on Green Engineering and Technologies. IC-GET. IEEE, pp. 1–6. Villarreal-Cervantes, M.G., Alvarez-Gallegos, J., 2016. 基于差分进化变体的平面并联机器人离线PID控制整定. *Expert Syst. Appl.* 64, 444–454. Wang, C., Cheng, H.Z., 2008. 使用植物生长模拟算法优化大型配电系统网络配置. *IEEE Trans. Power Syst.* 23, 119–126. Wang, G.-G., Deb, S., Coelho, L.d.S., 2015. 象群优化. In: 2015 3rd International Symposium on Computational and Business Intelligence. ISCBI. IEEE, pp. 1–5. Wang, G.-G., Gandomi, A.H., Alavi, A.H., Gong, D., 2019. 磷虾群算法综述: 变体、混合与应用. *Artif. Intell. Rev.* 51, 119–148. Wang, J.-J., 2011. 基于PID控制器的倒立摆仿真研究. *Simul. Model. Pract. Theory* 19, 440–449. Wang, Q., Spronck, P., Tracht, R., 2003. 遗传算法在控制工程问题中的应用综述. In: Proceedings of the 2003 International Conference on Machine Learning and Cybernetics (IEEE Cat. No. 03EX693). In: IEEE, vol. 3, pp. 1651–1656. Wang, Q.-G., Hang, C.-C., Bi, Q., 1999. 基于机电反馈的频率响应辨识技术. *IEEE Trans. Control Syst. Technol.* 7, 122–128. Wolpert, D.H., Macready, W.G., et al., 1997. 无免费午餐优化定理. *IEEE Trans. Evol. Comput.* 1, 67–82. Wong, D., Leong, H.W., Liu, H., 2012. 模拟退火在VLSI设计中的应用, vol. 42. Springer Science & Business Media. Wu, H., Xu, D., Yang, M., Zhang, X., 2016. 基于模拟退火分子动力学模拟的羟基磷灰石表面结构. *Langmuir* 32, 4643–4652. Wu, X., Zhou, Y., Lu, Y., 2017. 精英反向水波优化算法用于全局优化. *Math. Probl. Eng.* 2017. Wu, Z., Zhao, X., Ma, Y., Zhao, X., 2019. 基于改进多目标布谷鸟搜索算法的短期负荷预测混合模型. *Appl. Energy* 237, 896–909. Xia, S., Li, L., 2017. 基于PSO-RBF神经网络的温室温度预测应用. *Comput. Eng. Des.* 38, 744–748. Xu, Y., Fan, P., Yuan, L., 2013. 一种简单高效的人工蜂群算法. *Math. Probl. Eng. Yachen, Z., Yueming, H., 2008. 基于模拟退火算法的PID控制器研究. In: 2008 27th Chinese Control Conference. IEEE, pp. 225–228. Yadav, S., Nagar, S.K., Mishra, A., 2020. 使用灰狼优化器整定PID控制器参数. Available at SSRN 3575432. Yadav, S., Verma, S.K., Nagar, S.K., 2016. 优化磁悬浮系统的PID控制器. *IFAC-PapersOnLine* 49, 778–782. Yadav, S., Verma, S.K., Nagar, S.K., 2018. 使用教与学优化提升磁悬浮系统性能. *Alex. Eng. J.* 57, 2427–2433. Yaghoobi, S., Mojjallali, H., 2016. 基于改进混沌磷虾群算法的PID控制器整定. *Optik* 127, 4803–4807. Yang, B., Wang, J., Zhang, X., Yu, T., Yao, W., Shu, H., Zeng, F., Sun, L., 2020. 元启发式算法在光伏电池参数识别中的应用综述. *Energy Convers. Manag.* 208, 112595. Yang, X.-S., 2009. 多模态优化的萤火虫算法. In: International Sym- posium on Stochastic Algorithms. Springer, pp. 169–178.*

Yang, X.-S., 2010. A new metaheuristic bat-inspired algorithm. In: Nature Inspired Cooperative Strategies for Optimization. NICSO 2010. Springer, pp. 65–74.

Yang, X.-S., 2011. Metaheuristic optimization: algorithm analysis and open problems. In: International Symposium on Experimental Algorithms. Springer, pp. 21–32.

Yang, X.-S., 2012a. Bat algorithm for multi-objective optimisation. arXiv preprint. arXiv: 1203.6571.

Yang, X.-S., 2012b. Flower pollination algorithm for global optimization. In: International Conference on Unconventional Computing and Natural Computation. Springer, pp. 240–249.

Yang, X.-S., Deb, S., 2009. Cuckoo search via Lévy flights. In: 2009 World Congress on Nature & Biologically Inspired Computing. NaBIC. IEEE, pp. 210–214.

Yang, X.-S., Deb, S., 2010. Engineering optimisation by cuckoo search. arXiv preprint. arXiv:1005.2908.

Yang, X.-S., Gandomi, A.H., 2012. Bat algorithm: a novel approach for global engineering optimization. Eng. Comput.

Yapici, H., Cetinkaya, N., 2019. A new meta-heuristic optimizer: pathfinder algorithm. Appl. Soft Comput. 78, 545–568.

Yildız, B.S., Yıldız, A.R., 2018. Comparison of grey wolf, whale, water cycle, ant lion and sine-cosine algorithms for the optimization of a vehicle engine connecting rod. Mater. Test. 60, 311–315.

Younesi, A., Tohidi, S., 2015. Design of a sensorless controller for pmsm using krill herd algorithm. In: The 6th Power Electronics, Drive Systems & Technologies Conference. PEDSTC2015. IEEE, pp. 418–423.

Yu, H., Chen, Y., Hassan, S.G., Li, D., 2016a. Prediction of the temperature in a Chinese solar greenhouse based on LSSVM optimized by improved PSO. Comput. Electron. Agric. 122, 94–102.

Yu, K., Wang, X., Wang, Z., 2016b. Constrained optimization based on improved teaching–learning-based optimization algorithm. Inf. Sci. 352, 61–78.

Yunus, A.S., Djalal, M.R., 2019. Optimal tuning of PID control on single machine infinite bus using ant colony optimization. In: 2019 International Conference on Technologies and Policies in Electric Power & Energy. IEEE, pp. 1–6.

Zhang, Y., Wang, S., Ji, G., 2015. A comprehensive survey on particle swarm optimization algorithm and its applications. Math. Probl. Eng.

Zheng, Y.-J., 2015. Water wave optimization: a new nature-inspired metaheuristic. Comput. Oper. Res. 55, 1–11.

Zhi, D., 2019. Optimization of PID controller for single phase inverter based on ABC. In: 2019 5th International Conference on Control, Automation and Robotics. ICCAR. IEEE, pp. 501–504.

Zhong, S., Dong, Y.-F., Sarosh, A., 2014. Artificial bee colony algorithm for parametric optimization of spacecraft attitude tracking controller. In: Foundations and Practical Applications of Cognitive Systems and Information Processing. Springer, pp. 501–510.

Zhuang, M., Atherton, D., 1993. Automatic tuning of optimum PID controllers. In: IEE Proceedings D (Control Theory and Applications), vol. 140. IET, pp. 216–224.

Ziegler, J., Nichols, N., 1993. Optimum settings for automatic controllers.

Ziegler, J.G., Nichols, N.B., 1942. Optimum settings for automatic controllers. Trans. Am. Soc. Mech. Eng. 64.

Zulu, A., John, S., 2014. A review of control algorithms for autonomous quadrotors. Open J. Appl. Sci. 4, 547–556.

Yang, X.-S., 2010. 一种新的元启发式蝙蝠算法。In: Nature Inspired Cooperative Strategies for Optimization. NICSO 2010. Springer, pp. 65–74.

Yang, X.-S., 2011. 元启发式优化：算法分析与开放问题。In: International Symposium on Experimental Algorithms. Springer, pp. 21–32.

Yang, X.-S., 2012a. 多目标优化的蝙蝠算法。arXiv preprint. arXiv : 1203 .6571.

Yang, X.-S., 2012b. 用于全局优化的花朵授粉算法。In: International Conference on Unconventional Computing and Natural Computation. Springer, pp. 240–249.

Yang, X.-S., Deb, S., 2009. 基于莱维飞行的布谷鸟搜索。In: 2009 World Congress on Nature & Biologically Inspired Computing. NaBIC. IEEE, pp. 210–214.

Yang, X.-S., Deb, S., 2010. 基于布谷鸟搜索的工程优化。arXiv preprint. arXiv :1005 .2908.

Yang, X.-S., Gandomi, A.H., 2012. 蝙蝠算法：一种新的全局工程优化方法。Eng. Comput. Yapici, H., Cetinkaya, N., 2019. 一种新的元启发式优化器：探路者算法。Appl. Soft Comput. 78, 545–568.

Yıldız, B.S., Yıldız, A.R., 2018. 比较灰狼、鲸鱼、水循环、蚁狮和正弦余弦算法用于车辆发动机连杆优化。Mater. Test. 60, 311–315.

Younesi, A., Tohidi, S., 2015. 使用磷虾群算法设计永磁同步电机无传感器控制器。In: The 6th Power Electronics, Drive Systems & Technologies Conference. PEDSTC2015. IEEE, pp. 418–423.

Yu, H., Chen, Y., Hassan, S.G., Li, D., 2016a. 基于改进粒子群优化LSSVM的中国日光温室温度预测。Comput. Electron. Agric. 122, 94–102.

Yu, K., Wang, X., Wang, Z., 2016b. 基于改进教与学优化算法的约束优化. Inf. Sci. 352, 61–78.

Yunus, A.S., Djalal, M.R., 2019. 基于蚁群优化的单机无穷大母线PID控制最优整定. 见: 2019年国际电力能源技术与政策会议. IEEE, 第1–6页.

Zhang, Y., Wang, S., Ji, G., 2015. 粒子群优化算法及其应用综合综述. Math. Probl. Eng. Zheng, Y.-J., 2015. 水波优化：一种新的自然启发式元启发方法. Comput. Oper. Res. 55, 1–11.

Zhi, D., 2019. 基于人工蜂群算法的单相逆变器PID控制器优化. 见: 2019年第五届国际控制、自动化与机器人会议. ICCAR. IEEE, 第501–504页.

Zhong, S., Dong, Y.-F., Sarosh, A., 2014. 用于航天器姿态跟踪控制器参数优化的人工蜂群算法. 见: 认知系统与信息处理的基础与应用. Springer, 第501–510页.

Zhuang, M., Atherton, D., 1993. 最优PID控制器的自动整定. 见: IEE Proceedings D (控制理论与应用), 第140卷. IET, 第216–224页.

Ziegler, J., Nichols, N., 1993. 自动控制器最优设置. Ziegler, J.G., Nichols, N.B., 1942. 自动控制器最优设置. Trans. Am. Soc. Mech. Eng. 64.

Zulu, A., John, S., 2014. 自主四旋翼飞行器控制算法综述. Open J. Appl. Sci. 4, 547–556.